Advanced Data Analysis

Lecture Notes (Ch. 1–13)

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Chapter 1. Prediction Problems and Evaluation Design

Predictive Modeling, Lecture 1 (Is the score trustworthy? Splits, CV, leakage, OLS, logistic regression)

Roadmap of the predictive-modeling part

Lecture	Key question	Models and concepts
1	Can the score be trusted?	splits, CV, leakage, OLS, logistic regression
2	Many correlated variables?	Ridge, Lasso, Elastic Net
3	How to learn nonlinear relations?	decision trees, bagging, random forest
4	Strong models for tabular data?	GBM, XGBoost, LightGBM, CatBoost
5	When are neural nets useful?	MLP, minibatch, Adam, regularization, TabPFN/TabICL
6	How to combine many models?	AutoGluon, OOF, bagging, stacking

Models may change; the prediction time, split, pipeline and final-test rule must not.

1 Prediction as a Rule for New Cases

1.1 The prediction problem

Definition: prediction rule

From observed data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^p$ we build a rule

$$\hat{f} : \mathcal{X} \to \mathcal{Y}, \qquad \hat{y}_{\text{new}} = \hat{f}(x_{\text{new}}).$$

x_i is the feature vector already observed at the **prediction time**, y_i the target observed later. The object of interest is the **prediction error on new cases**, not the fit to $\mathcal{D}$.

1.2 The prediction time decides which variables are usable

To predict next month's churn at the end of this month, nothing after the end of the month may be a feature.

Role of a column	Question at prediction time	Churn example
Feature X_j	Is it already known at that moment?	recent usage, contract length, # complaints
Target Y	What do we predict?	churn next month
Identifier (ID)	Does it carry anything beyond row identity?	customer ID, account serial number
Post-event variable	Does it arise only after the outcome?	reason for cancellation, refund after cancellation

Post-event variables raise the score but do not exist at prediction time: never use them.

1.3 Regression and classification

Task	Target	Direct model output	Focus of evaluation
Regression	continuous Y	$\hat{y}(x)$, conditional mean	average error and large errors
Binary classification	$Y \in \{0, 1\}$	$\hat{p}(x) = P(Y = 1 \mid X = x)$	ranking, probability, threshold errors
Multiclass	$Y \in \{1, \dots, C\}$	$(\hat{p}_1, \dots, \hat{p}_C)$	per-class errors and probability quality

A classifier outputs a **probability, not a label**; the action is a separate decision (§7).

1.4 Prediction versus causal inference

Prediction: $\hat{y} = \hat{f}(x)$	Causal inference: $\mathbb{E}\{Y(1) - Y(0)\}$
Who is likely to churn?	How much did a coupon reduce churn?
Small loss on new cases is the goal.	Identification assumptions and bias control are central.
Correlated (non-causal) variables can be useful features.	Confounders and post-treatment variables need care.

A good predictive coefficient never justifies a causal claim.

2 Loss, Risk and Generalization

Motivation. Risk is the expected loss on unseen data, which training loss underestimates.

2.1 Loss translates decision costs into numbers

Definition: loss and risk

A loss $L(y, f(x)) \geq 0$ measures the cost of predicting $f(x)$ when the truth is y. The **risk** in the deployment environment is

$$R(f) = \mathbb{E}_{(X,Y)\sim P_{\text{deploy}}}[L\{Y, f(X)\}].$$

$R(f)$ cannot be observed directly. The **training risk** is $\widehat{R}_{\text{train}}(f) = \frac{1}{n}\sum_{i=1}^{n} L\{y_i, f(x_i)\}$, and the **generalization gap** is $\text{Gap}(f) = R(f) - \widehat{R}_{\text{train}}(f)$.

Fix the loss *before* the model (overstock vs. stock-out, missed defaulter vs. rejected good customer); accuracy is only one loss.

Theorem: what each loss targets

Let $f^\star = \arg\min_f R(f)$ over all measurable f. Pointwise in x:

1. squared loss $(y - f)^2$: $f^\star(x) = \mathbb{E}[Y \mid X = x]$ (conditional mean);
2. absolute loss $|y - f|$: $f^\star(x) = \text{median}(Y \mid X = x)$;
3. 0–1 loss for $Y \in \{0, 1\}$: $f^\star(x) = \mathbb{I}\{P(Y = 1 \mid x) \geq 1/2\}$ (Bayes classifier);
4. log loss $-[y \log q + (1 - y)\log(1 - q)]$ over $q \in (0, 1)$: $q^\star(x) = P(Y = 1 \mid X = x)$ (a *proper* scoring rule).

Proof [Supplement]. $R(f) = \mathbb{E}_X\big[\mathbb{E}\{L(Y, f(X)) \mid X\}\big]$, so minimize $g(c) = \mathbb{E}[L(Y, c) \mid X = x]$ for each x.

(1) $g(c) = \mathrm{Var}(Y \mid x) + (\mathbb{E}[Y \mid x] - c)^2$.

(2) $g'(c) = 2P(Y \le c \mid x) - 1$ (right derivative): < 0 below the conditional median, ≥ 0 from it on.

(3) $g(1) = 1 - \pi(x)$, $g(0) = \pi(x)$ with $\pi(x) = P(Y = 1 \mid x)$; choose 1 iff $\pi \ge \frac{1}{2}$.

(4) $g(q) = -\pi \log q - (1-\pi)\log(1-q)$, $g'(q) = -\pi/q + (1-\pi)/(1-q) = 0 \iff q = \pi$, and $g'' > 0$. ■

2.2 Training score versus deployment risk

Fitting minimizes $\widehat{R}_{\text{train}}$, which complex models drive near zero; $R(f)$ needs an **independent validation procedure**. The gap becomes unstable with large model searches, splits ignoring time/groups, full-data preprocessing, and distribution shift.

Theorem: bias–variance decomposition

Let $Y = f(x) + \varepsilon$ with $\mathbb{E}\varepsilon = 0$, $\mathrm{Var}\varepsilon = \sigma^2$, and let $\hat{f}$ be fitted on a random training set independent of the new (x, Y). Then

$$\mathbb{E}[(Y - \hat{f}(x))^2] = \underbrace{\big(f(x) - \mathbb{E}\hat{f}(x)\big)^2}_{\text{Bias}^2} + \underbrace{\mathrm{Var}\hat{f}(x)}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{irreducible}}.$$

Proof [Supplement]. $Y - \hat{f} = \varepsilon + (f - \mathbb{E}\hat{f}) + (\mathbb{E}\hat{f} - \hat{f})$; the terms are pairwise uncorrelated ($\varepsilon \perp \hat{f}$ with mean 0; $f - \mathbb{E}\hat{f}$ constant; $\mathbb{E}(\mathbb{E}\hat{f} - \hat{f}) = 0$). Square and take expectations. ■

Training error falls monotonically with complexity; test error is U-shaped (**underfitting**: high bias; **overfitting**: high variance).

3 Splitting the Data

3.1 Three roles

Part	Used for	Never used for
Training part	fitting model parameters	—
Internal validation (holdout or CV)	choosing candidates and tuning	—
Test set	reporting final generalization once	choosing models

3.2 The split must preserve the data-generating structure

Data structure	Recommended split	Danger of a random split
independent rows, class imbalance	stratified split	minority-class proportion may differ by chance
clustered (customers, hospitals, households)	group split	the same group appears in training and test
time order	forward (time-ordered) split	future information leaks into training
duplicates / near-duplicates	deduplicate, then split	nearly identical rows on both sides

The validation split follows from the *unit of a new case*.

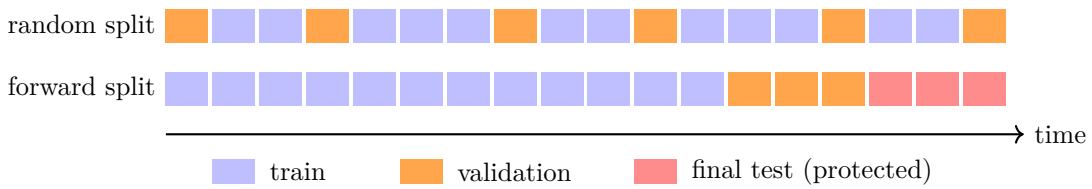


Figure 1: A random split trains on the future of its validation cases; a forward split trains on the past and protects the latest period as the final test.

3.3 Reproducibility

Fix: split rule and random seed; feature list and target definition; handling of missing values, categories, scaling; code computing metrics and thresholds.

Record: extraction time, data version, number of rows; class ratios and number of groups in each set; candidate models and hyperparameter grids; fold-by-fold variability, not only the CV mean.

4 Information Leakage

Definition: leakage

Leakage is any channel through which information unavailable at prediction time, or information from the validation/test part, enters the fitting procedure.

Type	Example	Why it is a problem
Target leakage	cancellation reason or default-handling result used as a feature	does not exist at prediction time
Preprocessing leakage	mean imputation / standardization on the full data	peeks at the validation/test distribution
Temporal leakage	future averages, after-the-fact aggregates	cannot reproduce a past decision
Duplicate / group leakage	same customer, document or product on both sides	the "new" case was effectively already seen

Example: why preprocessing must be learned inside each fold

Screening on the full data ("keep the 20 features most correlated with y") lets a model on pure noise score high in CV, since screening has seen the validation targets. Fix: re-fit every data-dependent step (imputation, scaling, screening, encoding) inside each training fold.

5 K-fold Cross-Validation

Definition: K-fold CV

Split the development data into folds $V_1, \dots, V_K$. For each k fit $\hat{f}^{(-k)}$ on the data without V_k and compute

$$\widehat{R}_k(f) = \frac{1}{|V_k|} \sum_{i \in V_k} L\{y_i, \hat{f}^{(-k)}(x_i)\}, \qquad \widehat{R}_{\mathrm{CV}}(f) = \frac{1}{K} \sum_{k=1}^{K} \widehat{R}_k(f).$$

Each observation is validated once and trained on $K-1$ times.

Remark: fold variability and its standard error

Report $\widehat{R}_{\mathrm{CV}}$ with $\mathrm{SE} = \mathrm{sd}(\widehat{R}_1, \dots, \widehat{R}_K)/\sqrt{K}$ (rough, usually too small; used by the 1-SE rule of Ch. 3).

Algorithm: selection with CV, reporting with the test set

1. Split off the test set first and lock it away.
2. On the remaining development data, define the pipeline (preprocessing + candidate model).
3. Choose candidates and hyperparameters using CV scores and fold variability.
4. Refit the chosen pipeline on the whole development data.
5. Report final performance, error patterns and operating cost on the test set.

6 Metrics are the Language of Decisions

A metric should track operational loss and be fixed across candidates. Ask first *which errors are more expensive.*

6.1 Regression metrics

Metric	Definition	Interpretation
MAE	$\frac{1}{n}\sum_i \lvert y_i - \hat{y}_i \rvert$	all errors linearly; linked to median prediction
MSE	$\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2$	squared penalty on large errors; linked to mean prediction
RMSE	$\sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}$	same unit as y; emphasizes large errors
R^2	$1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$	improvement in squared error over the mean baseline

Theorem: $\mathrm{MAE} \le \mathrm{RMSE} \le \sqrt{n}\,\mathrm{MAE}$

For residuals $e_i = y_i - \hat{y}_i$, $\frac{1}{n}\sum |e_i| \le \left(\frac{1}{n}\sum e_i^2\right)^{1/2}$, with equality iff all $|e_i|$ are equal.

Proof [Supplement]. Cauchy–Schwarz: $\left(\sum |e_i| \cdot 1\right)^2 \le n \sum e_i^2$. The upper bound follows from $\sum e_i^2 \le (\sum |e_i|)^2$. ■

RMSE − MAE grows with the dispersion of $|e_i|$. Prefer RMSE when large errors are costly, MAE to limit outlying targets.

7 Classification: Probabilities, Thresholds, Metrics

7.1 Probability first, action second

A classifier outputs $\hat{p}(x) = P(Y = 1 \mid X = x)$; one $\hat{p} = 0.18$ can mean different actions under different costs. Calibration asks whether cases predicted at 0.18 churn at about 18%.

7.2 Threshold and confusion matrix

$$\hat{y}_t(x) = \mathbb{I}\{\hat{p}(x) \ge t\},$$

	actual +	actual −
predicted +	TP	FP
predicted −	FN	TN

$$\text{precision} = \frac{TP}{TP+FP}, \qquad \text{recall} = \frac{TP}{TP+FN}, \qquad \text{accuracy} = \frac{TP+TN}{n}.$$

Lowering t raises recall and false positives.

Example: accuracy under imbalance

50 churners among 1,000: predicting "no churn" for all gives accuracy 95%, recall 0. Use precision, recall and costs.

7.3 Three questions about one probability

Ranking	Reliability of the probability	Action
Whom to look at first? ROC-AUC, PR-AUC	Does 0.30 mean about 30%? Brier score, log loss, calibration plot	Where to place the threshold? precision/recall at t, cost

Theorem: ROC-AUC is a ranking probability

For a score s, $\mathrm{AUC} = P(s(X^+) > s(X^-)) + \frac{1}{2}P(s(X^+) = s(X^-))$, where X^+, X^- are independent draws from the positive and negative classes. Empirically AUC equals the proportion of (positive, negative) pairs ranked correctly (Mann–Whitney statistic).

Proof [Supplement]. ROC plots $\mathrm{TPR}(t) = P(s^+ > t)$ against $\mathrm{FPR}(t) = P(s^- > t)$, so $\int \mathrm{TPR}\, d\mathrm{FPR} = \int P(s^+ > t)\, dF^-(t) = P(s^+ > s^-)$ for continuous scores (ties contribute half). ■

PR-AUC suits rare positives. Brier $\frac{1}{n}\sum (y_i - \hat{p}_i)^2$; log loss punishes confident errors. AUC is invariant to monotone transforms, so it ignores calibration.

7.4 Choosing the threshold from costs

Theorem: cost-optimal threshold

Let a false negative cost C_{FN} and a false positive cost C_{FP} (correct decisions cost 0). If $\hat{p}(x)$ is calibrated, acting positive minimizes expected cost iff

$$\hat{p}(x)\, C_{FN} > \{1 - \hat{p}(x)\}\, C_{FP} \iff \hat{p}(x) > t^\star = \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

Proof [Supplement]. Positive: cost C_{FP} w.p. $1 - \hat{p}$; negative: cost C_{FN} w.p. $\hat{p}$. Compare and rearrange. ■

Larger C_{FN} lowers $t^\star$; $t^\star = 1/2$ only for equal costs.

8 The Linear Baseline: OLS

8.1 Least squares

$$\hat{y}_i = \beta_0 + x_i^\top \beta, \qquad \hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - \beta_0 - x_i^\top \beta)^2.$$

With the design matrix X including the intercept column,

Derivation: normal equations

$Q(\beta) = \|y - X\beta\|^2 = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta$, so $\nabla Q = -2X^\top y + 2X^\top X\beta = 0$ gives $X^\top X\hat{\beta} = X^\top y$, $\hat{\beta} = (X^\top X)^{-1} X^\top y$ if invertible; Hessian $2X^\top X \succeq 0$ gives a global minimum. $\hat{y} = Hy$, $H = X(X^\top X)^{-1}X^\top$, is the orthogonal projection onto the column space: $X^\top(y - \hat{y}) = 0$.

Nearly singular $X^\top X$ means unstable coefficients; see regularization (Ch. 3).

8.2 Predictive versus causal reading of coefficients

Predictively, β_j is the change in prediction per unit of X_j with others fixed; under multicollinearity it wobbles. A causal reading also needs confounder control, no post-treatment variables, and no selection or measurement bias.

8.3 Residual diagnostics point to the next model

Observed phenomenon	Meaning	Next candidates
curved residual pattern	missed nonlinearity / interaction	transformations, splines, trees (Ch. 4)
variance changing with $\hat{y}$	heteroscedasticity, scale issue	transformations, robust loss, segment-wise evaluation
a few huge residuals	outliers, label errors	Huber loss, data checks (Ch. 2)
unstable coefficients with correlated features	multicollinearity, high dimension	Ridge, Lasso, Elastic Net (Ch. 3)

9 Logistic Regression

9.1 From linear score to probability

$$\eta(x) = \beta_0 + x^\top \beta, \qquad \hat{p}(x) = \sigma\{\eta(x)\} = \frac{1}{1+\exp\{-\eta(x)\}}, \qquad \log\frac{p(x)}{1-p(x)} = \beta_0 + x^\top\beta.$$

β_j is the change of the **log-odds** per unit of X_j, $\exp(\beta_j)$ the **odds ratio**, and $\partial p/\partial x_j = \beta_j p(1-p)$.

9.2 Log loss

$$\ell(y,p) = -[y\log p + (1-y)\log(1-p)], \qquad \hat{\beta} = \arg\min_\beta \sum_{i=1}^n \ell\{y_i, \sigma(\beta_0 + x_i^\top\beta)\}.$$

Confident wrong probabilities ($p \approx 0$ for a positive) get huge penalties.

Theorem: gradient and convexity of the logistic log loss

With $p_i = \sigma(x_i^\top\beta)$ (intercept absorbed), $J(\beta) = \sum_i \ell(y_i, p_i)$ satisfies

$$\nabla J(\beta) = X^\top(p-y), \qquad \nabla^2 J(\beta) = X^\top W X, \quad W = \mathrm{diag}\{p_i(1-p_i)\} \succeq 0,$$

so J is convex and minimizing log loss equals maximum likelihood for the Bernoulli model. Newton's method is iteratively reweighted least squares.

Proof [Supplement]. $\sigma' = \sigma(1-\sigma)$, so $\partial\ell/\partial\eta = -y(1-p) + (1-y)p = p - y$; hence $\nabla J = \sum_i (p_i - y_i)x_i$ and $\nabla^2 J = \sum_i p_i(1-p_i)x_i x_i^\top$. The Bernoulli log-likelihood is $\sum_i\{y_i \log p_i + (1-y_i)\log(1-p_i)\} = -J$. ■

Remark: separation

Under perfect separation J decreases as $\|\beta\| \to \infty$ and no MLE exists; penalization (Ch. 3) fixes this.

10 Comparing Candidates Fairly

Criterion	Conditions to control
same split	all candidates use the same training/CV/test structure
same feature definition	same prediction time, missing-value and category handling
same metric	primary and secondary metrics for selection fixed in advance
same compute budget	record search range, time and number of iterations
same reporting	show CV mean, variability, test score, complexity and operating constraints together

Summary

- Define the target and the prediction time first.

- The goal is low generalization risk in deployment, not low training error.
- The split must preserve the data-generating structure: time, groups, duplicates.
- CV is for selection, the test set for the final report; preprocessing is learned inside each training fold.
- Metrics, thresholds and baselines are chosen from decision costs before algorithms.

Chapter 2. Outliers and Robust Prediction

Predictive Modeling, Section 2 (MCD, LOF, Isolation Forest, leverage, Cook's distance, Huber, RANSAC, Theil–Sen)

1 An Outlier Is Not the Same as a Data Error

What the observation is	Example	First response
input / measurement error	wrong unit, mis-recorded target	correct from the raw source; remove only if uncorrectable
a genuine rare case	a very large transaction, a rare high-risk patient	keep it, and check that group's prediction error separately
a case from a new environment	a policy or market condition absent in training	check distribution shift and the need to retrain

An outlier is relative to a reference; the action depends on why it is unusual.

1.1 Three kinds of outliers

Kind	Meaning	Simple diagnostic
univariate	one variable is far from most observations	histogram, boxplot, IQR rule
multivariate	each variable is ordinary but the *combination* is rare	scatterplot, multivariate distance, density
conditional	given x_i, the target y_i differs greatly from what is expected	residual $e_i = y_i - \hat{y}_i$

IQR rule: flag $x < Q_1 - 1.5\,\mathrm{IQR}$ or $x > Q_3 + 1.5\,\mathrm{IQR}$, $\mathrm{IQR} = Q_3 - Q_1$; it misses outliers in combinations.

Method	What it checks	Observations it looks for
MCD	global distance in X	multivariate outliers far from the central distribution
LOF	local density in X	points less dense than their neighbors
Isolation Forest	isolation path in X	points isolated by few random splits
studentised residual	y given X	cases far from the regression prediction
leverage, Cook's distance	influence on the regression fit	cases that change coefficients and predictions a lot

"X is unusual" and "y is unusual given X" are different problems.

2 MCD: Robust Mahalanobis Distance

2.1 Classical Mahalanobis distance and masking

With the sample mean $\bar{x}$ and sample covariance S,

$$D_i^2 = (x_i - \bar{x})^\top S^{-1} (x_i - \bar{x}).$$

Theorem: null distribution of the squared distance

If $X \sim N_p(\mu, \Sigma)$ then $(X-\mu)^\top \Sigma^{-1}(X-\mu) \sim \chi^2_p$. Hence, approximately, a point with $D_i^2 > \chi^2_{p,1-\alpha}$ is an outlier candidate.

Proof [Supplement]. $Z = \Sigma^{-1/2}(X-\mu) \sim N_p(0, I_p)$ and $(X-\mu)^\top \Sigma^{-1}(X-\mu) = Z^\top Z = \sum_{j=1}^p Z_j^2$. ■

Masking. Outliers pull $\bar{x}$ and inflate S, shrinking their own distances; estimate location and scatter from the bulk only.

2.2 Minimum Covariance Determinant

Definition: MCD

For a fixed h with $n/2 \le h < n$ (typically $h \approx 0.75n$), find the subset H of h observations whose sample covariance has the smallest determinant:

$$H^\star = \arg\min_{|H|=h} \det S_H, \qquad \hat{\mu}_{\text{MCD}} = \bar{x}_{H^\star}, \quad \widehat{\Sigma}_{\text{MCD}} = c_{h,n}\, S_{H^\star},$$

($c_{h,n}$ is a consistency factor), and use the robust distance

$$D_i^2 = (x_i - \hat{\mu}_{\text{MCD}})^\top \widehat{\Sigma}_{\text{MCD}}^{-1} (x_i - \hat{\mu}_{\text{MCD}}), \qquad \text{flag if } D_i^2 > \chi^2_{p,1-\alpha}.$$

$\det S_H \propto$ squared ellipsoid volume, so MCD finds the *tightest* h points; breakdown point $\approx (n-h)/n$. FAST-MCD iterates C-steps (replace H by the h points of smallest distance), which never increase $\det S_H$ (Exercise 2.2).

Remark: when MCD fits

Continuous data, $p < n$, one elliptical bulk; otherwise use LOF or Isolation Forest.

3 LOF: Lower Density Than the Neighbors

Definition: local outlier factor

Let $N_k(i)$ be the k nearest neighbors of i, $d_k(j)$ the distance from j to its kth neighbor, and the reachability distance $\text{rd}_k(i,j) = \max\{d_k(j), d(i,j)\}$. The local reachability density and LOF are

$$\text{lrd}_k(i) = \Big(\frac{1}{|N_k(i)|}\sum_{j\in N_k(i)} \text{rd}_k(i,j)\Big)^{-1}, \qquad \text{LOF}_k(i) = \frac{1}{|N_k(i)|}\sum_{j\in N_k(i)} \frac{\text{lrd}_k(j)}{\text{lrd}_k(i)}.$$

LOF ≈ 1: density like the neighbors; LOF $\gg 1$: local outlier. The comparison is *relative to the neighborhood*, so sparse genuine clusters are not penalized; rd_k stabilizes very close pairs.

4 Isolation Forest: Quickly Isolated Points

Algorithm: Isolation Forest

1. On a subsample, choose a feature uniformly at random.
2. Choose a split point uniformly between the feature's minimum and maximum.
3. Split recursively until each observation is alone (or a depth limit).
4. Grow many such trees and average the path lengths.

Anomaly score:

$$s(x) = 2^{-\mathbb{E}[h(x)]/c(n)}, \qquad c(n) = 2H(n-1) - \frac{2(n-1)}{n}, \quad H(m) \approx \ln m + 0.5772,$$

where $h(x)$ is the path length isolating x and $c(n)$ the average unsuccessful-search path length in a binary search tree of n points.

$s \to 1$: anomalous; $s \approx 0.5$ everywhere: no clear anomaly; $s \ll 0.5$: normal. `contamination` only sets the share labeled anomalous; the score cannot tell an error from a valid rare case.

5 Regression: Residual, Leverage and Influence

Motivation. Residuals flag bad fits, leverage flags unusual inputs, and influence combines both.

5.1 Three different notions

Suppose most data follow $y \approx 2x$.

Observation	Feature x	Residual	Interpretation
$(3, 20)$	ordinary range	large	outlier in the target
$(10, 20)$	far from the rest	small	high-leverage point that follows the trend
$(10, 0)$	far from the rest	large	influential point likely to change the line

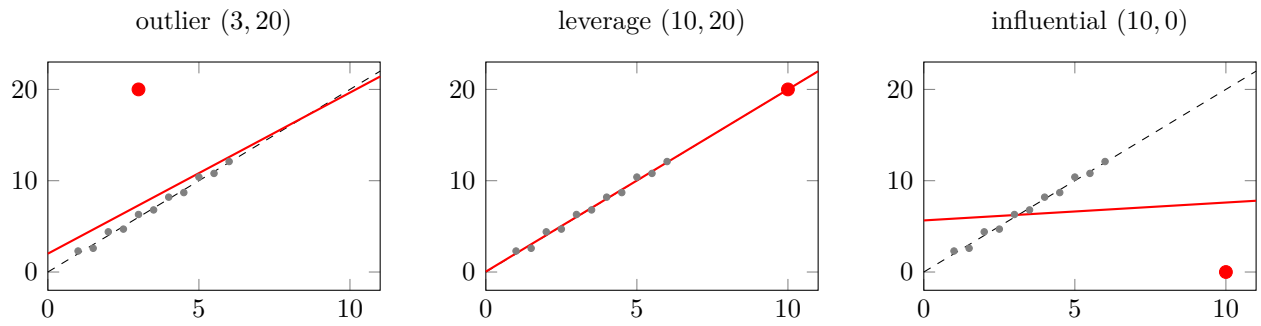


Figure 2: Dashed: OLS on the gray points; red: refit with the red point. Only the high-leverage point off the trend drags the slope (large Cook's distance).

5.2 Hat matrix and leverage

With $X \in \mathbb{R}^{n\times(p+1)}$ (intercept included) and $H = X(X^\top X)^{-1}X^\top$, $\hat{y} = Hy$ and $e = (I-H)y$. The leverage is $h_{ii} = x_i^\top (X^\top X)^{-1} x_i$.

Theorem: properties of leverage and residuals

1. $0 \le h_{ii} \le 1$ and $\sum_i h_{ii} = p+1$, so the average leverage is $(p+1)/n$.
2. $\hat{y}_i = \sum_j h_{ij} y_j$, so $h_{ii} = \partial \hat{y}_i / \partial y_i$.
3. Under $\mathrm{Var}(\varepsilon) = \sigma^2 I$, $\mathrm{Var}(e_i) = \sigma^2(1-h_{ii})$.

Proof [Supplement]. $H = H^\top = H^2$: $h_{ii} = \sum_j h_{ij}^2 \ge h_{ii}^2 \Rightarrow 0 \le h_{ii} \le 1$; $\sum_i h_{ii} = \mathrm{tr}\{(X^\top X)^{-1}X^\top X\} = p+1$. Item 2 is $\hat{y} = Hy$. Item 3: $(I-H)X = 0 \Rightarrow e = (I-H)\varepsilon$, $\mathrm{Var}(e) = \sigma^2(I-H)$. ■

High-leverage points have small *raw* residuals (item 3), so standardize.

5.3 Studentised residual and Cook's distance

Diagnostic	Computation	Cases it reveals
studentised residual	$r_i = \dfrac{e_i}{\hat{\sigma}\sqrt{1-h_{ii}}}$	y_i far from what x_i predicts
leverage	h_{ii}, diagonal of H	far from the other observations in feature space
Cook's distance	$D_i = \dfrac{r_i^2}{p+1}\dfrac{h_{ii}}{1-h_{ii}}$	removing it changes coefficients and predictions much

Screening rules $|r_i| > 2$, $h_{ii} > 2(p+1)/n$, $D_i > 4/n$ trigger *inspection*, not deletion.

Theorem: leave-one-out update and Cook's formula

Let $\hat{\beta}_{(i)}$ be the OLS estimate without case i. Then

$$\hat{\beta} - \hat{\beta}_{(i)} = \frac{(X^\top X)^{-1} x_i e_i}{1-h_{ii}},$$

and Cook's distance, defined as $D_i = \dfrac{\|X\hat{\beta} - X\hat{\beta}_{(i)}\|^2}{(p+1)\hat{\sigma}^2}$, equals $\dfrac{r_i^2}{p+1}\dfrac{h_{ii}}{1-h_{ii}}$.

Proof [Supplement]. $A = X^\top X$, $A_{(i)} = A - x_i x_i^\top$, $X_{(i)}^\top y_{(i)} = X^\top y - x_i y_i$; Sherman–Morrison: $A_{(i)}^{-1} = A^{-1} + \frac{A^{-1}x_i x_i^\top A^{-1}}{1-h_{ii}}$. Then

$$\hat{\beta}_{(i)} = \left(A^{-1} + \tfrac{A^{-1}x_i x_i^\top A^{-1}}{1-h_{ii}}\right)(X^\top y - x_i y_i) = \hat{\beta} - A^{-1}x_i y_i + \frac{A^{-1}x_i(\hat{y}_i - h_{ii}y_i)}{1-h_{ii}} = \hat{\beta} - \frac{A^{-1}x_i(y_i - \hat{y}_i)}{1-h_{ii}}.$$

So $\|X(\hat{\beta}-\hat{\beta}_{(i)})\|^2 = \frac{e_i^2}{(1-h_{ii})^2} x_i^\top A^{-1}AA^{-1}x_i = \frac{e_i^2 h_{ii}}{(1-h_{ii})^2}$; divide by $(p+1)\hat{\sigma}^2$ and use $e_i^2 = r_i^2\hat{\sigma}^2(1-h_{ii})$. ■

D_i is large only if *both* r_i and $h_{ii}/(1-h_{ii})$ are, as for $(10, 0)$.

Remark: leave-one-out residuals for free

$y_i - x_i^\top \hat{\beta}_{(i)} = e_i/(1-h_{ii})$, so OLS LOOCV error is $\frac{1}{n}\sum_i \{e_i/(1-h_{ii})\}^2$ without refitting.

6 What to Do With Detected Cases

Finding	Action	Caveat
input / measurement error	correct or remove in raw data	remove only if it cannot be corrected and cannot be used
heavily skewed variable	log or Box–Cox transform	keeps the order, compresses gaps among large values
extremes must be capped	winsorisation	cap both tails at quantiles *estimated on training data*
valid rare case	keep	use robust learning and segment-wise evaluation

Never delete or alter observations on a detection score alone.

7 Robust Learning

Method	Core principle	Suitable when
Huber regression	caps the growth rate of loss for large residuals	a few large target outliers
RANSAC	fits many small subsamples; keeps the model supported by most points	clear inlier vs. contaminated groups
Theil–Sen	median of pairwise slopes	linear regression with few features

7.1 Huber loss

With residual $r = y - \hat{y}$:

Loss	Definition	Large residual	Target of prediction
squared	r^2	grows quadratically	conditional mean
absolute	$\lvert r\rvert$	grows linearly	conditional median
Huber	quadratic for small, linear for large $\lvert r\rvert$	linear beyond δ	compromise between center fit and tail robustness

$$\rho_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta, \\ \delta(|r| - \frac{1}{2}\delta), & |r| > \delta, \end{cases} \qquad \psi_\delta(r) = \rho'_\delta(r) = \begin{cases} r, & |r| \le \delta, \\ \delta\,\mathrm{sign}(r), & |r| > \delta. \end{cases}$$

Derivation: why Huber limits outliers — IRLS view

Estimating equation $\sum_i \psi_\delta(y_i - x_i^\top\beta)x_i = 0$; with $\psi_\delta(r) = w(r)\,r$, $w(r) = \min\{1, \delta/|r|\}$, iterate $\beta \leftarrow (X^\top W X)^{-1} X^\top W y$, $W = \mathrm{diag}\{w(r_i)\}$. A residual 10δ gets weight 0.1; influence is bounded by δ. ρ_δ is convex and C^1; $\delta \to \infty$ gives OLS, $\delta \to 0$ (rescaled) LAD.

Usually $\delta = 1.345\hat{s}$, $\hat{s} = \mathrm{MAD}/0.6745$ (95% Gaussian efficiency).

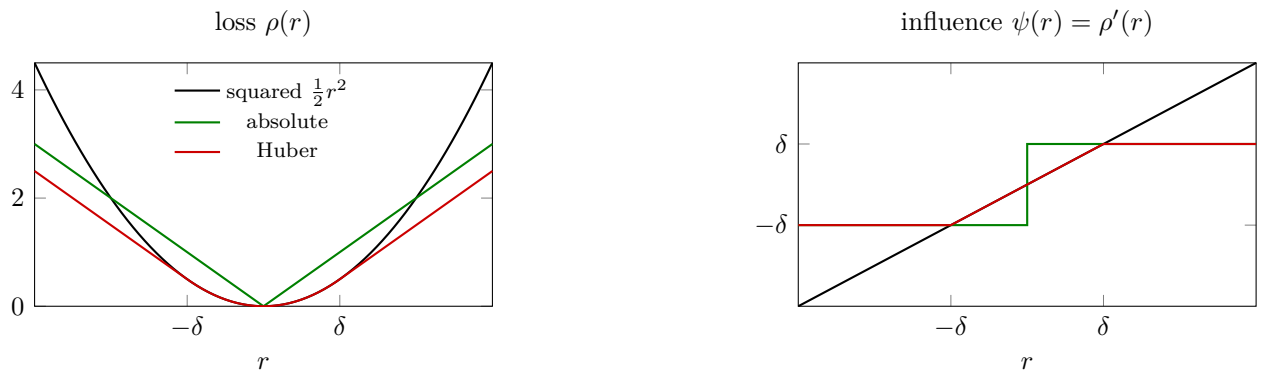


Figure 3: Losses (left) and influence functions (right): unbounded for squared loss, capped at $\pm\delta$ for Huber.

7.2 RANSAC

Algorithm: RANSAC

1. Fit the regression on a small random subsample (minimal size s).
2. Classify observations with residual below a threshold as inliers.
3. Repeat many times.
4. Refit the model on the largest inlier set.

With inlier fraction w, success at least once with probability q needs $N \geq \log(1-q)/\log(1-w^s)$ trials.

7.3 Theil–Sen

In simple regression compute all pairwise slopes:

$$\hat{\beta}_1 = \underset{i<j}{\text{median}} \frac{y_j - y_i}{x_j - x_i}, \qquad \hat{\beta}_0 = \underset{i}{\text{median}}(y_i - \hat{\beta}_1 x_i).$$

Breakdown point $\approx 1 - 1/\sqrt{2} \approx 29\%$.

8 Robust Prediction Is Also Chosen by CV

Algorithm: protocol

1. Split off the final test set first.
2. Inside each CV training fold, fit scaling and outlier-detection steps.
3. Set the detection threshold, LOF's k, Huber's δ, RANSAC's residual threshold from training folds only.
4. Compare OLS and robust regressions on the same validation folds.
5. Check errors of outlier candidates and of rare groups along with the overall error.

Result to check	Purpose
MAE and RMSE	evaluate typical errors and rare large errors together
sensitivity to the detection rule	does model selection change with the threshold?
error of rare groups	did robustification ignore important cases?

Summary

- Outliers are relative to a reference; errors, valid rare cases and new environments call for different actions.
- In X: MCD (global, elliptical), LOF (local density), Isolation Forest (isolation depth).
- In regression: residual (y given x), leverage (x unusual), Cook's distance (both together).
- Robust losses (Huber, LAD) and robust fitting (RANSAC, Theil–Sen) limit the dominance of a few points; their tuning constants are chosen inside CV.

9 Selected exercises

Motivation. Three facts used above: affine invariance, C-step monotonicity, and Huber as shifted least squares.

Exercise 2.1 (Affine invariance of Mahalanobis distances)

Let $x_1, \dots, x_n \in \mathbb{R}^p$ have sample covariance $S \succ 0$, and let $y_i = Ax_i + b$ with A an invertible $p \times p$ matrix and $b \in \mathbb{R}^p$.
(a) Show that the squared Mahalanobis distances D_i^2 computed from the y_i equal those computed from the x_i.
(b) Show that the MCD subset $H^\star$ is the same for both data sets, and so are the robust distances.

Solution [Supplement].

(a)

Step 1. $\bar{y} = A\bar{x} + b$ and $y_i - \bar{y} = A(x_i - \bar{x})$, so $S_y = AS_xA^\top$, which is invertible with $S_y^{-1} = A^{-\top}S_x^{-1}A^{-1}$.

Step 2. Hence

$$(y_i - \bar{y})^\top S_y^{-1}(y_i - \bar{y}) = (x_i - \bar{x})^\top A^\top A^{-\top} S_x^{-1} A^{-1} A(x_i - \bar{x}) = (x_i - \bar{x})^\top S_x^{-1}(x_i - \bar{x}).$$

(b)

Step 3. For any H, $S_H^{(y)} = AS_H^{(x)}A^\top$, so $\det S_H^{(y)} = (\det A)^2 \det S_H^{(x)}$ with a common factor $(\det A)^2 > 0$: same minimizing subsets.

Step 4. Then $\hat{\mu}^{(y)} = A\hat{\mu}^{(x)} + b$, $\widehat{\Sigma}^{(y)} = A\widehat{\Sigma}^{(x)}A^\top$ ($c_{h,n}$ common), and Step 2 gives equal robust distances.

LOF and Isolation Forest lack this invariance, so scale their inputs deliberately. ■

Exercise 2.2 (A C-step never increases the determinant)

For a subset H with $|H| = h$ put $T_H = \frac{1}{h}\sum_{i\in H} x_i$ and $S_H = \frac{1}{h}\sum_{i\in H}(x_i - T_H)(x_i - T_H)^\top$. Let $|H_1| = h$ with $S_1 = S_{H_1} \succ 0$, $T_1 = T_{H_1}$, and $d_1(i) = (x_i - T_1)^\top S_1^{-1}(x_i - T_1)$. Let H_2 consist of the h indices with the smallest $d_1(i)$, and $S_2 = S_{H_2}$. Show that $\det S_2 \le \det S_1$.

Solution [Supplement].

Step 1. Using $a^\top M a = \operatorname{tr}(Maa^\top)$,

$$\sum_{i\in H_1} d_1(i) = \operatorname{tr}\Big(S_1^{-1}\sum_{i\in H_1}(x_i - T_1)(x_i - T_1)^\top\Big) = \operatorname{tr}(S_1^{-1} h S_1) = hp.$$

Step 2. $\sum_{i\in H_2} d_1(i) \le \sum_{i\in H_1} d_1(i) = hp$ by the choice of H_2.

Step 3. For $M \succ 0$ and any c, since $\sum_{i\in H_2}(x_i - T_2) = 0$ with $T_2 = T_{H_2}$,

$$\sum_{i\in H_2}(x_i - c)^\top M(x_i - c) = \sum_{i\in H_2}(x_i - T_2)^\top M(x_i - T_2) + h(T_2 - c)^\top M(T_2 - c).$$

With $M = S_1^{-1}$, $c = T_1$: left side $= \sum_{i\in H_2} d_1(i)$, first right term $= h\operatorname{tr}(S_1^{-1}S_2)$, so $\operatorname{tr}(S_1^{-1}S_2) \le p$ by Step 2.

Step 4. $S_1^{-1/2} S_2 S_1^{-1/2} \succeq 0$ has eigenvalues $\lambda_j \ge 0$, trace $\le p$, determinant $\det S_2/\det S_1$. By AM–GM,

$$\Big(\frac{\det S_2}{\det S_1}\Big)^{1/p} = \Big(\prod_j \lambda_j\Big)^{1/p} \le \frac{1}{p}\sum_j \lambda_j \le 1.$$

With finitely many h-subsets the non-increasing sequence stabilizes, where FAST-MCD stops. (Normalizing by $h - 1$ rescales all determinants equally.) ■

Exercise 2.3 (Huber regression is least squares with outlier shifts)

(a) For $r \in \mathbb{R}$ and $\delta > 0$ show that

$$\rho_\delta(r) = \min_{\gamma\in\mathbb{R}}\Big\{\tfrac{1}{2}(r - \gamma)^2 + \delta|\gamma|\Big\},$$

with the unique minimizer $\gamma^\star(r) = \operatorname{sign}(r)(|r| - \delta)_+$.
(b) Let $F(\boldsymbol{\beta}, \gamma) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta} - \gamma\|_2^2 + \delta\|\gamma\|_1$ with $\gamma \in \mathbb{R}^n$. Show that $(\hat{\boldsymbol{\beta}}, \hat{\gamma})$ minimizes F if and only if $\hat{\boldsymbol{\beta}}$ minimizes $\sum_i \rho_\delta(y_i - x_i^\top\boldsymbol{\beta})$ and $\hat{\gamma}_i = \gamma^\star(y_i - x_i^\top\hat{\boldsymbol{\beta}})$ for all i.

Solution [Supplement].

(a)

Step 1. $\phi(\gamma) = \frac{1}{2}(r - \gamma)^2 + \delta|\gamma|$ is strictly convex and coercive; its unique minimizer solves $0 \in \gamma - r + \delta\,\partial|\gamma|$ ($\partial|\gamma| = \{\operatorname{sign}\gamma\}$ for $\gamma \ne 0$, $[-1, 1]$ at 0).

Step 2. $|r| \le \delta$: $\gamma = 0$ ($r \in \delta[-1, 1]$), $\phi(0) = \frac{1}{2}r^2$. $|r| > \delta$: $\gamma = r - \delta\operatorname{sign}(r)$ satisfies $\gamma - r + \delta\operatorname{sign}\gamma = 0$, value $\frac{1}{2}\delta^2 + \delta(|r| - \delta) = \delta(|r| - \frac{1}{2}\delta)$. Both give $\rho_\delta(r)$ at $\gamma^\star(r)$.

(b)

Step 3. With $r_i = y_i - x_i^\top\boldsymbol{\beta}$, $F = \sum_i\{\frac{1}{2}(r_i - \gamma_i)^2 + \delta|\gamma_i|\}$ separates, so by (a) $\min_\gamma F(\boldsymbol{\beta}, \gamma) = \sum_i \rho_\delta(r_i) =: \mathcal{H}(\boldsymbol{\beta})$, attained only at $\gamma_i = \gamma^\star(r_i)$.

Step 4. If $(\hat{\boldsymbol{\beta}}, \hat{\gamma})$ minimizes F, then for every $\boldsymbol{\beta}$

$$\mathcal{H}(\hat{\boldsymbol{\beta}}) \le F(\hat{\boldsymbol{\beta}}, \hat{\gamma}) \le \min_\gamma F(\boldsymbol{\beta}, \gamma) = \mathcal{H}(\boldsymbol{\beta}),$$

so $\hat{\boldsymbol{\beta}}$ minimizes $\mathcal{H}$; $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ gives $F(\hat{\boldsymbol{\beta}}, \hat{\gamma}) = \mathcal{H}(\hat{\boldsymbol{\beta}})$, and uniqueness forces $\hat{\gamma}_i = \gamma^\star(y_i - x_i^\top \hat{\boldsymbol{\beta}})$. Conversely, $F(\hat{\boldsymbol{\beta}}, \hat{\gamma}) = \mathcal{H}(\hat{\boldsymbol{\beta}}) \leq \mathcal{H}(\boldsymbol{\beta}) \leq F(\boldsymbol{\beta}, \gamma)$ for all $(\boldsymbol{\beta}, \gamma)$.

Step 5. Given $\hat{\gamma}$, F is quadratic in $\boldsymbol{\beta}$: $X^\top(\mathbf{y} - \hat{\gamma} - X\hat{\boldsymbol{\beta}}) = 0$, OLS on $\mathbf{y} - \hat{\gamma}$.

Cases with $|r_i| \leq \delta$ get no shift; others are shifted by $|r_i| - \delta$, so only $\pm\delta$ reaches the fit (bounded influence). ■

Chapter 3. High-Dimensional Data and Regularization

Predictive Modeling, Lecture 2 (OLS instability, Ridge, Lasso, Elastic Net, choosing λ)

1 Linear Prediction and Why It Becomes Unstable

1.1 Setting

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad X \in \mathbb{R}^{n\times p}, \qquad \hat{y} = x_{\text{new}}^{\top}\hat{\boldsymbol{\beta}}.$$

Regularization picks, among $\hat{\boldsymbol{\beta}}$ that fit $\mathbf{y}$ well, one *less sensitive* on new x; stability is not causal identification.

1.2 High dimension is about p relative to n

Situation	Example	Problem for prediction
$p \ll n$	100,000 customers, 20 variables	fairly stable, but collinearity is still possible
p close to n	500 patients, 400 variables	small data changes move coefficients and selection a lot
$p \geq n$	1,000 documents, 50,000 words	OLS is not unique or cannot be computed
many derived features	raw variables, interactions, one-hot encodings	search degrees of freedom grow faster than signal

1.3 OLS and what it assumes

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 = (X^{\top}X)^{-1}X^{\top}\mathbf{y} \quad \text{(if } X^{\top}X \text{ is invertible)}.$$

It needs a well-conditioned $X^{\top}X$; otherwise this baseline *becomes unstable*.

1.4 When $p \geq n$: non-uniqueness

Theorem: interpolation is not unique

If $\text{rank}(X) < p$ (always the case when $p > n$), there is $v \neq 0$ with $Xv = 0$, and then $X(\hat{\boldsymbol{\beta}} + v) = X\hat{\boldsymbol{\beta}}$ for every solution $\hat{\boldsymbol{\beta}}$. In particular, when $\text{rank}X = n$, infinitely many $\boldsymbol{\beta}$ give $X\boldsymbol{\beta} = \mathbf{y}$ with zero training error.

$x_{\text{new}}^{\top}v$ need not be 0, so zero training error does not pick a good solution. (The minimum-norm interpolator $X^{\top}(XX^{\top})^{-1}\mathbf{y}$ is the $\lambda \to 0$ limit of Ridge.)

1.5 Collinearity and the variance of coefficients

Income, credit limit and card spending may measure one dimension: coefficients take opposite large signs and are reallocated across samples.

Theorem: variance along eigen-directions

Under $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I$, $\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^\top X)^{-1}$. With the SVD $X = UDV^\top$, $D = \mathrm{diag}(d_1, \ldots, d_p)$,

$$\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 \sum_{j=1}^{p} \frac{v_j v_j^\top}{d_j^2}, \qquad \mathrm{Var}(v_j^\top \hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{d_j^2}.$$

Proof [Supplement]. $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon} \Rightarrow \mathrm{Var} = \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1} = \sigma^2 V D^{-2} V^\top$. ■

$d_j \approx 0$ makes the variance along v_j explode, and $\Delta\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} = (X^\top X)^{-1} X^\top \Delta \mathbf{y}$ amplifies perturbations. Check fold-wise stability of coefficients and selections, not only CV means.

2 Regularization: Buying Variance With Bias

$$\mathbb{E}[(Y - \hat{f}(X))^2] = \mathrm{Bias}^2 + \mathrm{Var} + \sigma^2.$$

In high dimensions OLS variance dominates; regularization trades bias for lower variance, judged by CV, not training loss. Too little tracks noise; too much shrinks the signal.

2.1 One template, three models

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \left[\frac{1}{n} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda P(\boldsymbol{\beta}) \right], \qquad P(\boldsymbol{\beta}) = \begin{cases} \|\boldsymbol{\beta}\|_2^2 & \text{Ridge} \\ \|\boldsymbol{\beta}\|_1 & \text{Lasso} \\ \alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2 & \text{Elastic Net} \end{cases}$$

$\lambda = 0$ gives OLS; larger λ shrinks more. Choose λ by loss on new data.

2.2 Standardization comes first

$$x_{ij}^{\mathrm{std}} = \frac{x_{ij} - \mu_j^{(\mathrm{train})}}{s_j^{(\mathrm{train})}}.$$

Rescaling X_j by c rescales β_j by $1/c$, changing its penalty; standardization makes it unit-free. The intercept is not penalized (center first); one-hot and sparse inputs need judgement. μ_j, s_j come from the **training part only**.

3 Ridge

Motivation. Ridge shrinks most along poorly determined directions, trading a little bias for stability.

3.1 Closed form and constrained form

$$\hat{\boldsymbol{\beta}}_\lambda^{\mathrm{Ridge}} = \arg\min_{\boldsymbol{\beta}} \left[\frac{1}{n} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right] = (X^\top X + n\lambda I)^{-1} X^\top \mathbf{y}.$$

Proof [Supplement]. Gradient $-\frac{2}{n} X^\top (\mathbf{y} - X\boldsymbol{\beta}) + 2\lambda\boldsymbol{\beta} = 0 \iff (X^\top X + n\lambda I)\boldsymbol{\beta} = X^\top \mathbf{y}$; the matrix is positive definite for $\lambda > 0$, so the strictly convex objective has a unique minimizer even if $p \geq n$. ■

Coefficients shrink toward but not to 0; suited to spread-out signal. Equivalent (Lagrange) to $\min \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2$ s.t. $\|\boldsymbol{\beta}\|_2^2 \le c(\lambda)$.

3.2 Ridge shrinks small singular directions most

Theorem: SVD form of Ridge

With $X = UDV^\top$ and $\tilde{\lambda} = n\lambda$ (the slides absorb n into λ),

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = V\,\text{diag}\Big(\frac{d_j}{d_j^2 + \tilde{\lambda}}\Big)U^\top\mathbf{y}, \qquad X\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \sum_j u_j \frac{d_j^2}{d_j^2 + \tilde{\lambda}} u_j^\top \mathbf{y}.$$

Proof [Supplement]. $X^\top X + \tilde{\lambda} I = V(D^2 + \tilde{\lambda} I)V^\top$ and $X^\top\mathbf{y} = VDU^\top\mathbf{y}$, so $(X^\top X + \tilde{\lambda} I)^{-1}X^\top\mathbf{y} = V(D^2 + \tilde{\lambda} I)^{-1}DU^\top\mathbf{y}$. ■

OLS multiplies direction j by $1/d_j$, Ridge by $d_j/(d_j^2 + \tilde{\lambda}) \le 1/(2\sqrt{\tilde{\lambda}})$: small-$d_j$ directions shrink most. **Effective degrees of freedom** $\text{df}(\lambda) = \sum_j d_j^2/(d_j^2 + \tilde{\lambda})$ fall from p to 0.

3.3 Ridge shares signal among correlated variables

Theorem: identical columns get identical Ridge coefficients

If $x_{(1)} = x_{(2)}$, then $\hat{\beta}_1^{\text{Ridge}} = \hat{\beta}_2^{\text{Ridge}}$ for every $\lambda > 0$.

Proof [Supplement]. The fit depends only on $\beta_1 + \beta_2$, so swapping $\beta_1 \leftrightarrow \beta_2$ preserves the strictly convex objective; its unique minimizer is symmetric. ■

OLS may give $\hat{y} \approx 12x_1 - 11x_2$; Ridge $\approx 0.8x_1 + 0.7x_2$ (illustrative): similar variables shrink together.

Data / goal	Why Ridge fits	What to check
many weak signals	keeps all coefficients a little	does it beat OLS?
strong multicollinearity	stabilizes unstable directions	standardization and coefficient paths
$p \ge n$ or sparse matrices	unique, computable solution	sparse implementation, memory
prediction over interpretation	targets generalization, not zeros	final metric and operating cost

4 Lasso

4.1 The L_1 penalty sets some coefficients exactly to zero

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1\Big], \qquad \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^{p} |\beta_j|.$$

Small coefficients become exactly 0 (**sparse** model); unselected does not mean irrelevant.

Theorem: soft-thresholding under an orthogonal design

If $X^\top X = nI$, let $z = X^\top \mathbf{y}/n$ (the OLS estimate). Then

$$\hat{\beta}_j^{\text{Lasso}} = S(z_j, \lambda/2) = \text{sign}(z_j)(|z_j| - \lambda/2)_+, \qquad \hat{\beta}_j^{\text{Ridge}} = \frac{z_j}{1+\lambda}.$$

Proof [Supplement]. $\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \frac{1}{n}\|\mathbf{y}\|^2 - 2\boldsymbol{\beta}^\top z + \|\boldsymbol{\beta}\|^2$, so the objective is $\sum_j\{(\beta_j - z_j)^2 + \lambda|\beta_j|\}$ + const. Subgradient: $2(\beta_j - z_j) + \lambda s_j = 0$, $s_j = \text{sign}(\beta_j)$ if $\beta_j \neq 0$, $s_j \in [-1, 1]$ if $\beta_j = 0$. $|z_j| \leq \lambda/2 \Rightarrow \beta_j = 0$; else $\beta_j = z_j - \frac{\lambda}{2}\text{sign}(z_j)$. Ridge: $(\beta_j - z_j)^2 + \lambda\beta_j^2$ is minimized at $z_j/(1+\lambda)$. ■

Ridge shrinks proportionally; Lasso subtracts a constant, zeroing small coefficients. Coordinate descent applies this soft-threshold to partial residuals.

4.2 Geometry: the ball and the diamond

Loss contours meet the smooth Ridge ball $\|\boldsymbol{\beta}\|_2^2 \leq c$ off the axes, but the Lasso diamond $\|\boldsymbol{\beta}\|_1 \leq c$ often at a corner, where some $\beta_j = 0$.

4.3 Lasso is unstable among correlated variables

Among variables measuring one factor, Lasso picks one, and a new sample may pick another. Check selection frequencies across folds; keep Ridge or Elastic Net as candidates.

Remark: a limit on Lasso selection

When $p > n$, the Lasso selects at most n variables. For two identical columns the Lasso solution is not unique: any split (β_1, β_2) with the same sum and signs is optimal.

Situation	Why Lasso is attractive	Caveat
variables costly to measure or deploy	simpler prediction pipeline	compare selection frequency and loss in performance
very many variables, sparse signal suspected	shrinks unnecessary coefficients to 0	may keep only one of correlated signals
an explainable operating model is needed	easy to present a variable list	do not turn selection into causal claims
screening for later analysis	starting point of exploratory reduction	needs separate validation and domain review

5 Elastic Net

$$\hat{\boldsymbol{\beta}}^{\text{EN}} = \arg\min_{\boldsymbol{\beta}} \left[\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\{\alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2\}\right], \qquad \alpha \in [0, 1].$$

$\alpha = 1$: Lasso; $\alpha = 0$: Ridge. The L_2 part gives strict convexity (unique solution) and a **grouping effect** (Exercise 3.3); the L_1 part keeps sparsity. Orthogonal design: $S(z_j, \lambda\alpha/2)/\{1 + \lambda(1-\alpha)\}$. Tune λ, α by CV.

6 Choosing λ Inside the Folds

Standardization and fitting per λ use only each fold's training part. Two rules:

- **minimum rule**: $\lambda_{\min}$ with the smallest mean CV loss;
- **1-SE rule**: $\lambda_{1\mathrm{SE}} = \max\{\lambda : \widehat{R}_{\mathrm{CV}}(\lambda) \le \widehat{R}_{\min} + \mathrm{SE}_{\min}\}$, the most regularized model within one SE.

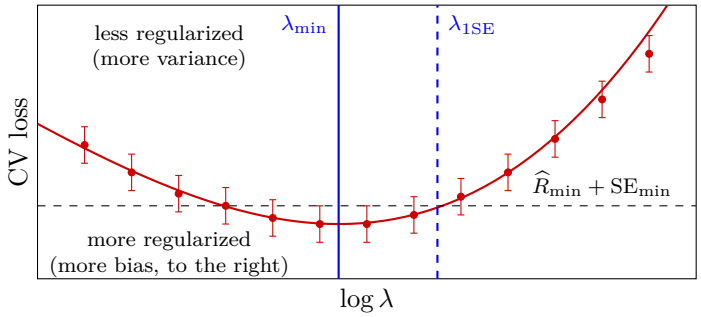


Figure 4: Schematic CV curve with ±1 SE bars: $\lambda_{1\mathrm{SE}}$ is the largest λ below the dashed line.

Fix the rule *before* the analysis; differences within CV noise are not meaningful.

7 Regularized Logistic Regression

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\sum_{i=1}^{n} \ell\{y_i, \sigma(\beta_0 + x_i^\top \boldsymbol{\beta})\} + \lambda P(\boldsymbol{\beta})\Big], \qquad \ell(y,p) = -[y\log p + (1-y)\log(1-p)].$$

Only the loss changes. With $\lambda > 0$ the objective is coercive, so a finite minimizer exists even under separation (Ch. 1).

8 Regularization Versus Robust Regression

Regularization	Robust regression
reduces coefficient variance and overfitting caused by many variables, collinearity and large degrees of freedom	mitigates loss domination caused by a few large residuals, label errors and extreme cases
Ridge, Lasso, Elastic Net	Huber, LAD
shrinkage, selection, generalization risk	influence of outliers, conditional location

Different failure causes; combinable (Huber loss + Elastic Net penalty).

9 Practical Selection and Reporting

Within one SE, prefer the simpler, more stable, faster candidate; one Lasso fit does not settle importance. Then refit on all training data and evaluate the test set once.

Example: reading a results table (illustrative)

Candidate	CV primary metric	Complexity	Interpretation
OLS	0.842 ± 0.031	180 variables	baseline; unstable under collinearity
Ridge	0.801 ± 0.019	180 shrunk coefficients	better performance and stability
Lasso	0.808 ± 0.026	24 selected	small loss in performance for simplicity
Elastic Net	0.799 ± 0.020	47 selected	compromise for correlated blocks

(Loss: smaller is better.)

Example: customer churn case

Situation. 8,000 customers, 600 derived variables; strong correlations among usage, payment and inquiry variables; churn rate 8%. **Candidates.** Standardized logistic baseline, Ridge, Lasso, Elastic Net. Primary metrics PR-AUC and log loss; operating metric recall@budget. **Split.** Protect the test period by time order; stratified CV within the training period. **Report.** CV mean and variability, chosen λ, number of selected variables, probability calibration and test performance in one table.

Summary

1. Many variables and collinearity make *coefficient variance and generalization risk*, rather than training fit, the main problem of OLS.
2. Ridge shrinks, Lasso selects, Elastic Net compromises; the right choice depends on the signal structure and the operating goal.
3. Standardization and λ selection are done inside CV folds; regularization does not automatically fix leakage or outliers.

10 Selected exercises

Motivation. Ridge risk, the Lasso path start $\lambda_{\max}$, and the Elastic Net grouping effect.

Exercise 3.1 (Some Ridge penalty always beats OLS)

Let X be fixed with $X^\top X = nI$, $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I$ and $\boldsymbol{\beta} \neq \mathbf{0}$. With the Ridge estimate $\hat{\boldsymbol{\beta}}_\lambda = z/(1+\lambda)$, $z = X^\top \mathbf{y}/n$, of the orthogonal-design theorem:

(a) show that $\mathbb{E}\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2 = \dfrac{\lambda^2\|\boldsymbol{\beta}\|^2 + p\sigma^2/n}{(1+\lambda)^2}$;

(b) show that this risk decreases on $[0, \lambda^\star]$ and increases afterwards, where $\lambda^\star = p\sigma^2/(n\|\boldsymbol{\beta}\|^2)$, and that its minimum is strictly smaller than the OLS risk $p\sigma^2/n$.

Solution [Supplement].

(a)

Step 1. $z = X^\top(X\boldsymbol{\beta} + \boldsymbol{\varepsilon})/n = \boldsymbol{\beta} + X^\top\boldsymbol{\varepsilon}/n$, so $\mathbb{E}z = \boldsymbol{\beta}$ and $\mathrm{Var}(z) = \sigma^2 X^\top X/n^2 = (\sigma^2/n)I$.

Step 2. $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta} = -\frac{\lambda}{1+\lambda}\boldsymbol{\beta}$, $\mathrm{Var}(\hat{\boldsymbol{\beta}}_\lambda) = \frac{\sigma^2}{n(1+\lambda)^2} I$; MSE = squared bias + trace:

$$\mathbb{E}\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2 = \frac{\lambda^2\|\boldsymbol{\beta}\|^2}{(1+\lambda)^2} + \frac{p\sigma^2}{n(1+\lambda)^2}.$$

(b)

Step 3. Write $a = \|\boldsymbol{\beta}\|^2 > 0$, $c = p\sigma^2/n$ and $f(\lambda) = (\lambda^2 a + c)/(1+\lambda)^2$. Then

$$f'(\lambda) = \frac{2\lambda a(1+\lambda) - 2(\lambda^2 a + c)}{(1+\lambda)^3} = \frac{2(\lambda a - c)}{(1+\lambda)^3},$$

negative for $\lambda < c/a = \lambda^\star$ and positive for $\lambda > \lambda^\star$.

Step 4. Substituting $\lambda^\star = c/a$,

$$f(\lambda^\star) = \frac{c^2/a + c}{(1 + c/a)^2} = \frac{ac}{a+c} < c = f(0).$$

The relative gain $c/(a+c)$ is largest at low signal-to-noise a/c. ■

Exercise 3.2 (When the Lasso selects nothing)

For a general design X and $\lambda > 0$, consider the Lasso objective $Q(\boldsymbol{\beta}) = \frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1$.
(a) Show that $\mathbf{0}$ minimizes Q if and only if $\lambda \geq \lambda_{\max} := \frac{2}{n}\|X^\top \mathbf{y}\|_\infty$.
(b) Show that all minimizers of Q have the same fitted values $X\hat{\boldsymbol{\beta}}$, and deduce that for $\lambda \geq \lambda_{\max}$ the minimizer $\mathbf{0}$ is unique.

Solution [Supplement].

(a)

Step 1. Q is convex, so $\boldsymbol{\beta}$ is a minimizer iff $\mathbf{0} \in \partial Q(\boldsymbol{\beta}) = -\frac{2}{n}X^\top(\mathbf{y} - X\boldsymbol{\beta}) + \lambda\,\partial\|\boldsymbol{\beta}\|_1$.

Step 2. $\partial\|\mathbf{0}\|_1 = [-1,1]^p$: $\frac{2}{n}X^\top\mathbf{y} \in \lambda[-1,1]^p \iff \frac{2}{n}|x_{(j)}^\top \mathbf{y}| \leq \lambda\ \forall j \iff \lambda \geq \lambda_{\max}$.

(b)

Step 3. If minimizers have $X\hat{\boldsymbol{\beta}}_1 \neq X\hat{\boldsymbol{\beta}}_2$, put $\bar{\boldsymbol{\beta}} = \frac{1}{2}(\hat{\boldsymbol{\beta}}_1 + \hat{\boldsymbol{\beta}}_2)$. Strict convexity of $u \mapsto \|\mathbf{y} - u\|^2$ and $\|\bar{\boldsymbol{\beta}}\|_1 \leq \frac{1}{2}(\|\hat{\boldsymbol{\beta}}_1\|_1 + \|\hat{\boldsymbol{\beta}}_2\|_1)$ give $Q(\bar{\boldsymbol{\beta}}) < \min Q$, a contradiction.

Step 4. For $\lambda \geq \lambda_{\max}$, $\mathbf{0}$ minimizes, so every minimizer has $X\hat{\boldsymbol{\beta}} = \mathbf{0}$ and $\frac{1}{n}\|\mathbf{y}\|^2 + \lambda\|\hat{\boldsymbol{\beta}}\|_1 = \min Q = \frac{1}{n}\|\mathbf{y}\|^2$, i.e. $\hat{\boldsymbol{\beta}} = \mathbf{0}$.

Hence the λ grid starts at $\lambda_{\max}$ (orthogonal design: $|z_j| \leq \lambda/2$ for all j). ■

Exercise 3.3 (Grouping effect of the Elastic Net)

Let $\mathbf{y}$ be centered and the columns of X standardized ($\mathbf{1}^\top x_{(j)} = 0$, $\|x_{(j)}\|^2 = n$), with sample correlation $\rho_{ij} = x_{(i)}^\top x_{(j)}/n$. Let $0 \leq \alpha < 1$, $\lambda > 0$, and $\hat{\boldsymbol{\beta}}$ the Elastic Net estimate. Show that if $\hat{\beta}_i\hat{\beta}_j > 0$, then

$$|\hat{\beta}_i - \hat{\beta}_j| \leq \frac{\|\mathbf{y}\|}{\sqrt{n}} \cdot \frac{\sqrt{2(1-\rho_{ij})}}{\lambda(1-\alpha)}.$$

Solution [Supplement].

Step 1. With $\hat{r} = \mathbf{y} - X\hat{\boldsymbol{\beta}}$, for $\hat{\beta}_k \neq 0$ the objective is differentiable in β_k:

$$-\frac{2}{n}x_{(k)}^\top \hat{r} + \lambda\alpha\,\mathrm{sign}(\hat{\beta}_k) + 2\lambda(1-\alpha)\hat{\beta}_k = 0.$$

Step 2. Equal signs for $k = i, j$; subtract:

$$2\lambda(1-\alpha)(\hat\beta_i - \hat\beta_j) = \frac{2}{n}(x_{(i)} - x_{(j)})^\top \hat r.$$

Step 3. By Cauchy–Schwarz, $|\hat\beta_i - \hat\beta_j| \le \|x_{(i)} - x_{(j)}\|\,\|\hat r\| / \{n\lambda(1-\alpha)\}$, and $\|x_{(i)} - x_{(j)}\|^2 = 2n - 2n\rho_{ij}$.

Step 4. Comparing with $\mathbf{0}$: $\frac{1}{n}\|\hat r\|^2 \le \frac{1}{n}\|\hat r\|^2 + \lambda P(\hat{\boldsymbol\beta}) \le \frac{1}{n}\|\mathbf{y}\|^2$, so $\|\hat r\| \le \|\mathbf{y}\|$ and

$$|\hat\beta_i - \hat\beta_j| \le \frac{\sqrt{2n(1-\rho_{ij})}\,\|\mathbf{y}\|}{n\lambda(1-\alpha)} = \frac{\|\mathbf{y}\|}{\sqrt{n}} \cdot \frac{\sqrt{2(1-\rho_{ij})}}{\lambda(1-\alpha)}.$$

For the Lasso ($\alpha = 1$) the bound is void. ■

Chapter 4. Decision Trees and Random Forests

Predictive Modeling, Lecture 3 (CART, impurity, pruning, bagging, OOB, random forest, importance)

1 After the Linear Baseline: Where Does the Relation Change?

Linear models need hand-designed interactions; trees split the input space by if–then conditions with a constant per region. **A tree discovers interactions and thresholds; a random forest stabilizes it.**

2 The Tree as a Piecewise-Constant Function

Motivation. Rectangles with constants give interactions for free but high variance.

Definition: regression tree

$$\hat{f}(x) = \sum_{m=1}^{M} c_m \, \mathbb{I}(x \in R_m),$$

where $R_1, \ldots, R_M$ are the leaf regions (a partition of $\mathcal{X}$), c_m is the prediction in region m, and the boundaries are axis-aligned splits by a threshold on one variable.

Each leaf is a rectangle and each root-to-leaf path an if–then rule. One changed split changes everything below it: high variance.

2.1 Nested conditions create interactions automatically

Two splits already give

$$\hat{f}(x) = \begin{cases} c_1, & x_1 < s_1, \\ c_2, & x_1 \geq s_1 \ \wedge \ x_2 < s_2, \\ c_3, & x_1 \geq s_1 \ \wedge \ x_2 \geq s_2. \end{cases}$$

x_2 matters only when $x_1 \geq s_1$: an interaction, which a linear model needs as a hand-made $x_1 x_2$.

2.2 Leaf predictions

Theorem: optimal leaf constants

For a fixed partition, (i) under squared loss the optimal c_m is the leaf mean $\hat{c}_m = \frac{1}{N_m} \sum_{i: x_i \in R_m} y_i$;
(ii) under absolute loss it is the leaf median;
(iii) for classification, the leaf class proportions $\hat{p}_{mk} = \frac{1}{N_m} \sum_{i: x_i \in R_m} \mathbb{I}(y_i = k)$ minimize the log loss, and $\hat{y}(x) = \arg\max_k \hat{p}_{mk}$ minimizes misclassification.

Proof [Supplement]. The loss separates over leaves; in each, apply the constant-prediction result of Ch. 1 to the leaf's empirical distribution. ■

Predictions jump like a staircase at leaf boundaries. $\hat{p}_{mk}$ is a probability estimate, unstable in small leaves; check calibration before acting.

3 Growing the Tree: CART

3.1 One variable, one threshold per node

$$R_1(j,s) = \{x : x_j < s\}, \qquad R_2(j,s) = \{x : x_j \geq s\}.$$

Search all (j,s), take the best split, recurse: **greedy**, not globally optimal.

3.2 Regression split criterion

$$\Delta_{\text{RSS}}(j,s) = \sum_{i \in R} (y_i - \bar{y}_R)^2 - \sum_{r \in \{1,2\}} \sum_{i \in R_r(j,s)} (y_i - \bar{y}_{R_r})^2.$$

Choose (j,s) maximizing Δ_{RSS}.

Derivation: efficient search and the ANOVA form of the gain

For node R with $N = N_1 + N_2$, the within/between decomposition gives

$$\Delta_{\text{RSS}} = N_1(\bar{y}_{R_1} - \bar{y}_R)^2 + N_2(\bar{y}_{R_2} - \bar{y}_R)^2 = \frac{N_1 N_2}{N}(\bar{y}_{R_1} - \bar{y}_{R_2})^2 \ \geq 0.$$

Continuous x_j: after sorting, running sums update in $O(1)$ per cut, $O(N \log N)$ in total. Categorical with K levels: $2^{K-1} - 1$ partitions, but sorting levels by mean target leaves $K - 1$ (Exercise 4.1).

3.3 Classification split criterion

$$G(R) = 1 - \sum_{k=1}^{K} p_{Rk}^2 \quad \text{(Gini)}, \qquad H(R) = -\sum_{k=1}^{K} p_{Rk} \log p_{Rk} \quad \text{(entropy)}.$$

Pure leaves have impurity 0; minimize $\frac{N_1}{N} I(R_1) + \frac{N_2}{N} I(R_2)$. Gini and entropy give similar trees.

Theorem: interpretation and monotonicity of impurity

(i) $G(R)$ is the probability that two labels drawn independently from the leaf distribution differ, equivalently the misclassification rate of a rule that labels at random according to p_R. (ii) Gini and entropy are strictly concave functions of p; hence any split satisfies $\frac{N_1}{N} I(R_1) + \frac{N_2}{N} I(R_2) \leq I(R)$, with equality iff both children have the parent's class proportions.

Proof [Supplement]. (i) $P(\text{two draws differ}) = 1 - \sum_k p_k^2$.

(ii) $p_R = \frac{N_1}{N} p_{R_1} + \frac{N_2}{N} p_{R_2}$; apply Jensen to the concave impurity. ■

$1 - \max_k p_k$ is only piecewise linear, so many splits gain zero (Exercise 4.2). The split criterion is not the evaluation metric.

3.4 Many candidate splits cause selection bias

Variables with many cut points or levels get more chances to split, and the best training split in a small node may not generalize. Remedies: minimum leaf size, pruning, validation; check high-cardinality IDs for leakage.

3.5 Greedy CART: fast but not globally optimal

Early splits are never revisited and lock in later structure; a tree's structure is not "the discovered truth".

3.6 Axis-aligned splits approximate oblique boundaries by staircases

Threshold rules suit trees; $x_1 + x_2 > c$ or smooth functions need many small, high-variance leaves. Compare with linear baselines under one protocol.

4 Controlling Complexity

Depth always lowers training error; test error can rise again.

Approach	Example	Pros and cons
pre-pruning	maximum depth, minimum leaf size, minimum split gain	fast, but stopping too early can miss good structure
post-pruning	grow large, then remove branches	sees a wider set of candidates, but requires a separate complexity choice
validation-based choice	choose depth / leaf size / pruning strength by CV	targets generalization; must be done inside the pipeline

Choose depth by generalization performance.

4.1 Minimum leaf size is a statistical regularizer

$$\hat{p}_m = \frac{n_{m,+}}{N_m}, \qquad \mathrm{SE}(\hat{p}_m) \approx \sqrt{\frac{\hat{p}_m(1-\hat{p}_m)}{N_m}}.$$

Small leaves are noisy ($N_m = 5$, $\hat{p} = 0.4$: SE ≈ 0.22); large leaves mix groups (bias). Minimum leaf size is a regularizer, not a speed option.

4.2 Cost-complexity pruning

$$R_\alpha(T) = R(T) + \alpha|T|,$$

where $R(T)$ is the training risk, $|T|$ the number of leaves and α the price per leaf, chosen by CV.

Derivation: weakest-link pruning

For an internal node t with subtree T_t, collapsing T_t to a leaf changes R_α by $\{R(t) - R(T_t)\} - \alpha(|T_t| - 1)$. Collapsing pays off as soon as

$$\alpha \geq g(t) = \frac{R(t) - R(T_t)}{|T_t| - 1}.$$

Collapsing the smallest-$g(t)$ node repeatedly gives nested $T_0 \supset T_1 \supset \cdots \supset \{\text{root}\}$ at $\alpha_1 < \alpha_2 < \cdots$, containing the optimal subtree for every α (Breiman et al.); CV chooses among finitely many trees.

5 From One Tree to Many: Bagging

5.1 A single tree has high variance

A few changed observations can flip the root split and everything below. Averaging slightly different trees cancels this variance: **trees are strong but unstable**.

5.2 Bagging

Algorithm: bagging (bootstrap aggregating)

For $b = 1, \dots, B$: draw a bootstrap sample of size n with replacement; grow a deep tree $\hat{f}_b$. Aggregate by averaging (regression) or by averaging probabilities / majority vote (classification).

A single tree is pruned; ensemble trees are grown deep (low bias) and averaging lowers variance, though leaf size and depth still matter for small samples, rare classes and probabilities.

5.3 A bootstrap sample contains about 63.2% of the distinct observations

Theorem: inclusion probability

The probability that a given observation never appears in a bootstrap sample of size n is $(1 - 1/n)^n \to e^{-1} \approx 0.368$. Hence on average about $1 - 0.368 = 0.632$ of the distinct observations are used to grow one tree.

Proof [Supplement]. Each of n independent draws misses it w.p. $1 - 1/n$, and $n \log(1 - 1/n) = -1 - \frac{1}{2n} + O(n^{-2}) \to -1$. ■

Left-out (out-of-bag, OOB) observations allow internal evaluation.

5.4 Averages are sensitive to correlation between trees

Theorem: variance of an average of correlated predictors

If each $\hat{f}_b(x)$ has variance σ^2 and every pair has correlation $\rho \geq 0$, then

$$\operatorname{Var}\Big(\frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)\Big) = \sigma^2\Big[\rho + \frac{1-\rho}{B}\Big].$$

Proof [Supplement]. $\operatorname{Var}(\sum_b \hat{f}_b) = B\sigma^2 + B(B-1)\rho\sigma^2$. Divide by B^2: $\frac{\sigma^2}{B} + \frac{B-1}{B}\rho\sigma^2 = \sigma^2\{\rho + (1-\rho)/B\}$. ■

The floor is $\rho\sigma^2$; bias is unchanged ($\mathbb{E}[\frac{1}{B}\sum_b \hat{f}_b] = \mathbb{E}\hat{f}_1$). Random forests lower ρ.

Task	Output of one tree	Ensemble output
regression	$\hat{f}_b(x)$	$\frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)$
binary classification	$\hat{p}_b(x)$ or class	average the probabilities then threshold, or majority vote
multiclass	class probability vector	argmax of the averaged class probabilities
ranking / policy	risk score, probability	combine with top-k, a threshold, a cost rule

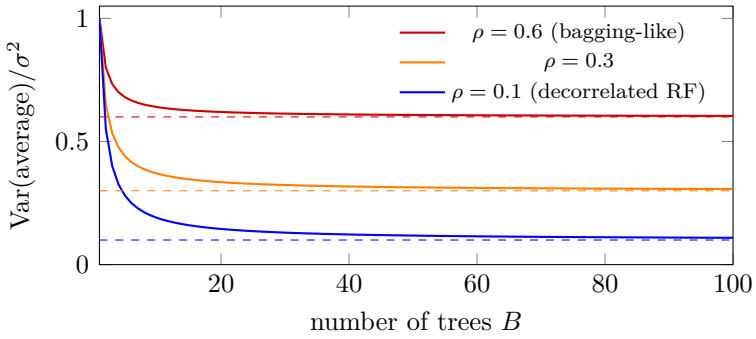


Figure 5: $\rho + (1-\rho)/B$ levels off at the floor ρ (dashed); decorrelation lowers the floor.

6 Random Forests

6.1 Bagging alone can let every tree use the same strong variable

If every tree splits first on the same strong variable, trees are alike and high ρ limits the gain for any B.

6.2 Randomness over both observations and features

Algorithm: random forest

For $b = 1, \ldots, B$: draw a bootstrap sample; grow a tree in which, **at each node**, only m_{try} randomly chosen variables are allowed as split candidates; grow deep (minimum leaf size small). Aggregate as in bagging.

Bootstrap varies the data; feature subsampling varies the structure, lowering ρ.

6.3 m_{try}

$m_{\text{try}} = p$ is bagging; smaller means lower correlation but weaker trees. Start at $\sqrt{p}$ (classification) or $p/3$ (regression); tune by OOB or CV.

6.4 OOB scores are convenient but do not always replace CV

For observation i, average only the trees for which i was out of bag: $\hat{f}^{\text{OOB}}(x_i) = \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} \hat{f}_b(x_i)$, with $|\mathcal{B}_i| \approx 0.368B$; the OOB error is the average loss of these predictions.

Method	Strength	Caveat
OOB	fast internal estimate without a separate validation set	does not respect time/group structure; not independent validation
CV	compares preprocessing and candidates under the same rule	expensive; the pipeline must be repeated inside folds
final test	reports generalization of a fixed procedure	used once; not material for re-tuning

OOB does not replace the time/group/leakage rules of Ch. 1.

7 Variable Importance Is Not Causal Importance

Impurity-decrease importance: total impurity reduction from splits on a variable. **Permutation importance**: drop in validation/OOB score after shuffling one column:

$$I_j = \text{Score}(D_{\text{eval}}) - \text{Score}(D_{\text{eval}}^{\pi(j)}),$$

with only column j permuted (sign chosen so a loss increase is positive); use OOB/validation data, not the test set, and repeat permutations.

Importance means the model *relied on* a variable, not that changing it changes the outcome. Correlated substitutes mask each other; impurity importance favors high-cardinality variables.

8 Comparison and Summary

Model	Randomness	Main strength	Main limitation
single tree	none or limited	rules are directly readable	high variance, unstable boundaries
bagging	bootstrap of rows	reduces variance of deep trees	gain limited if trees are correlated
random forest	rows + feature subsets	lower correlation, more stable prediction	hard to explain by one rule
boosting (next)	sequential residual / weight updates	reduces bias, strong prediction	more sensitive to tuning, overfitting, time structure

What matters is **how dissimilar the trees' errors are**.

Summary

1. Trees split the input space by conditions and capture nonlinearity and interactions.
2. A deep single tree is expressive but has high variance and overfits.
3. Bagging averages bootstrap trees to lower variance; random forests add feature subsampling to lower correlation between trees.
4. OOB, importance and partial dependence are useful diagnostics, but do not replace independent validation or causal interpretation.
5. Trees must be compared with the linear baseline under the same split, CV and final-test rules.

9 Selected exercises

Motivation. Level sorting, misclassification as a split criterion, and when aggregation helps.

Exercise 4.1 (Ordering the levels of a categorical variable)

A node R contains N observations of a categorical feature with levels $1, \dots, K$, level k having $N_k \geq 1$ observations with mean target $\bar{y}_k$. A binary split sends a nonempty proper subset L of the levels left and the rest right. Show that some split maximizing Δ_{RSS} has the form $L = \{k : \bar{y}_k \leq \tau\}$ for a threshold τ, so only the $K-1$ splits along the sorted level means need

to be examined. Show also that for $y_i \in \{0,1\}$ the Gini decrease equals $\frac{2}{N}\Delta_{\mathrm{RSS}}$, so the same holds for binary classification with Gini.

Solution [Supplement].

Step 1. With $W(L)$ the within-children SS, $\Delta_{\mathrm{RSS}}(L) = \mathrm{SS}_{\mathrm{tot}} - W(L)$; minimize W. Means minimize SS, so

$$W(L) = \min_{c_1,c_2} C(L,c_1,c_2), \qquad C(L,c_1,c_2) = \sum_{i:\,\mathrm{lev}(i)\in L} (y_i - c_1)^2 + \sum_{i:\,\mathrm{lev}(i)\notin L} (y_i - c_2)^2.$$

Step 2. Within level k, $\sum_{i:\,\mathrm{lev}(i)=k}(y_i - c)^2 = Q_k + N_k(\bar{y}_k - c)^2$, where Q_k does not depend on c.

Step 3. Let $L^\star$ be optimal with child means $c_1 \le c_2$ (else take the complement). If $c_1 = c_2$, $\Delta_{\mathrm{RSS}}(L^\star) = 0 \le \Delta_{\mathrm{RSS}}$ of any split, so all are optimal. Else put $\tau = \frac{1}{2}(c_1 + c_2)$, $L' = \{k : \bar{y}_k \le \tau\}$; by Step 2 level k is cheaper on the c_1 side iff $(\bar{y}_k - c_1)^2 \le (\bar{y}_k - c_2)^2$ iff $\bar{y}_k \le \tau$. Hence

$$W(L') \le C(L', c_1, c_2) \le C(L^\star, c_1, c_2) = W(L^\star).$$

Step 4. L' is proper: otherwise $C(L', c_1, c_2)$ is an SS about one constant, $\ge \mathrm{SS}_{\mathrm{tot}}$, so $\Delta_{\mathrm{RSS}}(L^\star) \le 0$, contradicting $\Delta_{\mathrm{RSS}}(L^\star) = \frac{N_1 N_2}{N}(c_1 - c_2)^2 > 0$. So L' is optimal.

Step 5. For $y_i \in \{0,1\}$, $G = 2p(1-p)$ and $\mathrm{SS} = Np(1-p)$, so $N\,G(R) = 2\,\mathrm{SS}(R)$ for every node and

$$G(R) - \frac{N_1}{N}G(R_1) - \frac{N_2}{N}G(R_2) = \frac{2}{N}\{\mathrm{SS}(R) - \mathrm{SS}(R_1) - \mathrm{SS}(R_2)\} = \frac{2}{N}\Delta_{\mathrm{RSS}}.$$

■

Exercise 4.2 (Misclassification error ignores useful splits)

Let $E(R) = 1 - \max_k p_{Rk}$ be the misclassification impurity.
(a) Show that if a class $k^\star$ is a majority class of the parent and of both children, then $N_1 E(R_1) + N_2 E(R_2) = N\,E(R)$: the split has zero gain.
(b) A node has 400 cases of each of two classes. Split A gives children $(300, 100)$ and $(100, 300)$; split B gives $(200, 400)$ and $(200, 0)$. Compare the weighted impurities of the two splits under E and under Gini.

Solution [Supplement].

(a) $N\,E(R) = N - n_{k^\star}$, $N_r E(R_r) = N_r - n_{r,k^\star}$, and $N_1 + N_2 - (n_{1,k^\star} + n_{2,k^\star}) = N - n_{k^\star}$.

(b)

Step 1. Under E: A makes $100 + 100$ errors, B $200 + 0$; both $200/800 = 0.25$ (parent 0.5).

Step 2. Gini, A: children $(\frac{3}{4}, \frac{1}{4})$, $G = 1 - \frac{9}{16} - \frac{1}{16} = 0.375$. B: $(\frac{1}{3}, \frac{2}{3})$ with $G = 1 - \frac{1}{9} - \frac{4}{9} = \frac{4}{9}$, weight $\frac{600}{800}$, plus a pure child: $\frac{3}{4} \cdot \frac{4}{9} = \frac{1}{3} < 0.375$.

Gini (and entropy: 0.562 vs. 0.477 nats) prefers B with its pure leaf; E cannot tell. ■

Exercise 4.3 (Aggregation helps under squared loss, not always under a vote)

Let $\varphi(x, \mathcal{D})$ be a predictor fitted on a random training sample $\mathcal{D}$, and $\varphi_A(x) = \mathbb{E}_{\mathcal{D}}\,\varphi(x, \mathcal{D})$ the ideal aggregated predictor (assume $\mathbb{E}_{\mathcal{D}}\varphi(x,\mathcal{D})^2 < \infty$).
(a) Show that for every (x, y), $\mathbb{E}_{\mathcal{D}}\{y - \varphi(x,\mathcal{D})\}^2 = \{y - \varphi_A(x)\}^2 + \mathrm{Var}_{\mathcal{D}}\varphi(x,\mathcal{D})$, and conclude that for a test pair (X, Y) independent of $\mathcal{D}$ the risk of φ_A is at most the expected risk of φ.

(b) For classification let $\varphi(x, \mathcal{D}) \in \{0, 1\}$ and $\varphi_A(x) = \mathbb{I}\{P_{\mathcal{D}}(\varphi(x, \mathcal{D}) = 1) > \frac{1}{2}\}$ (the majority vote over training samples). Give a point x where the vote has a larger error than φ.

Solution [Supplement].

(a)

Step 1. $y - \varphi = (y - \varphi_A) + (\varphi_A - \varphi)$; the cross term $2(y - \varphi_A)\mathbb{E}_{\mathcal{D}}(\varphi_A - \varphi) = 0$.

Step 2. $(X, Y) \perp \mathcal{D}$, so by Fubini

$$\mathbb{E}_{X,Y,\mathcal{D}}\{Y - \varphi(X, \mathcal{D})\}^2 = \mathbb{E}_{X,Y}\{Y - \varphi_A(X)\}^2 + \mathbb{E}_X \mathrm{Var}_{\mathcal{D}}\varphi(X, \mathcal{D}) \;\geq\; \mathbb{E}_{X,Y}\{Y - \varphi_A(X)\}^2.$$

The gain is the variance of the base learner, largest for unstable learners.

(b) $P(Y = 1 \mid x) = 1$, $P_{\mathcal{D}}(\varphi(x, \mathcal{D}) = 1) = 0.4$: φ errs w.p. 0.6, but $\varphi_A(x) = 0$ always errs. A vote amplifies the more frequent answer. ■

Chapter 5. Boosting Methods

Predictive Modeling, Lecture 4 (AdaBoost, gradient boosting, XGBoost, LightGBM, CatBoost)

1 From Averaging to Sequential Correction

Random forest (Ch. 4)	Gradient boosting (this chapter)
$\hat{f}_{\mathrm{RF}}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x)$	$F_M(x) = F_0(x) + \sum_{m=1}^{M} \eta\, h_m(x)$
trees learned **independently** on different bootstrap samples	h_m: a shallow tree learning the **error (residual) direction** left by the previous model
averaging reduces the instability of individual trees	η: the **learning rate**, shrinking each correction to prevent overfitting
every tree looks at the same target Y	small corrections ηh_m accumulate **sequentially**, not independently

Random forests mainly reduce variance; boosting mainly reduces bias with many shallow learners.

2 AdaBoost

Motivation. AdaBoost reweights toward current errors, which minimizes the exponential loss stagewise.

2.1 Algorithm

Algorithm: AdaBoost (labels $y_i \in \{-1, +1\}$)

Initialize $D_1(i) = 1/N$. For $b = 1, \ldots, B$:

1. **Train**: fit a classifier $h_b(x) \in \{-1, +1\}$ using weights D_b.
2. **Evaluate**: weighted error $\epsilon_b = \sum_{i=1}^{N} D_b(i)\, \mathbb{I}\{h_b(x_i) \neq y_i\}$.
3. **Weight**: $\alpha_b = \frac{1}{2} \log \dfrac{1 - \epsilon_b}{\epsilon_b}$.
4. **Update**: $D_{b+1}(i) = \dfrac{D_b(i) \exp\{-\alpha_b y_i h_b(x_i)\}}{Z_b}$, Z_b the normalizer.

Output: $H(x) = \operatorname{sign}\Big(\sum_{b=1}^{B} \alpha_b h_b(x)\Big)$.

Errors are multiplied by e^{α_b}, correct cases by $e^{-\alpha_b}$; smaller ϵ_b gives larger α_b ($\epsilon_b = \frac{1}{2} \Rightarrow \alpha_b = 0$; $\epsilon_b > \frac{1}{2}$ flips the learner).

2.2 AdaBoost is forward stagewise fitting of the exponential loss

Theorem: derivation of α_b and the weight update

Let $F_{b-1} = \sum_{s<b} \alpha_s h_s$ and consider the exponential loss $L(y, F) = e^{-yF}$. Choosing (α, h) to minimize $\sum_i \exp\{-y_i(F_{b-1}(x_i) + \alpha h(x_i))\}$ gives: h_b minimizes the weighted error with weights

$D_b(i) \propto e^{-y_i F_{b-1}(x_i)}$, and $\alpha_b = \frac{1}{2}\log\frac{1-\epsilon_b}{\epsilon_b}$. The weights evolve exactly by the AdaBoost update.

Proof [Supplement]. With $w_i = e^{-y_i F_{b-1}(x_i)}$, normalized to $D_b(i)$, and $y_i h(x_i) = \pm 1$,

$$\sum_i D_b(i) e^{-\alpha y_i h(x_i)} = e^{-\alpha}(1-\epsilon) + e^{\alpha}\epsilon = e^{-\alpha} + (e^{\alpha} - e^{-\alpha})\epsilon.$$

For $\alpha > 0$ this increases in ϵ, so h_b minimizes weighted error; ∂_α: $-e^{-\alpha}(1-\epsilon) + e^{\alpha}\epsilon = 0 \iff e^{2\alpha} = (1-\epsilon)/\epsilon$; and $w_i^{\text{new}} = w_i e^{-\alpha_b y_i h_b(x_i)}$ normalizes to the update. ■

Theorem: training-error bound

$Z_b = 2\sqrt{\epsilon_b(1-\epsilon_b)}$ and

$$\frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\{H(x_i) \neq y_i\} \leq \prod_{b=1}^{B} Z_b \leq \exp\Big(-2\sum_{b=1}^{B}\gamma_b^2\Big), \qquad \gamma_b = \tfrac{1}{2} - \epsilon_b.$$

Proof [Supplement]. $Z_b = e^{-\alpha_b}(1-\epsilon_b) + e^{\alpha_b}\epsilon_b = 2\sqrt{\epsilon_b(1-\epsilon_b)}$. Unrolling, $D_{B+1}(i) = \frac{e^{-y_i F_B(x_i)}}{N\prod_b Z_b}$ sums to 1, so $\frac{1}{N}\sum_i e^{-y_i F_B(x_i)} = \prod_b Z_b$; and $\mathbb{I}\{H(x_i) \neq y_i\} \leq e^{-y_i F_B(x_i)}$. Finally $2\sqrt{\epsilon(1-\epsilon)} = \sqrt{1-4\gamma^2} \leq e^{-2\gamma^2}$. ■

With $\gamma_b \geq \gamma > 0$ training error falls exponentially. Minimizing $pe^{-f} + (1-p)e^{f}$ gives $f^\star(x) = \frac{1}{2}\log\frac{P(Y=1|x)}{P(Y=-1|x)}$: half the log-odds.

Example: AdaBoost on a toy case (classic illustration)

Ten points, decision stumps. Round 1: vertical stump, three + errors, $\epsilon_1 = 0.30$, $\alpha_1 = \frac{1}{2}\log(0.7/0.3) = 0.42$. Round 2: vertical, $\epsilon_2 = 0.21$, $\alpha_2 \approx 0.65$. Round 3: horizontal, $\epsilon_3 = 0.14$, $\alpha_3 \approx 0.92$:

$$H_{\text{final}} = \text{sign}(0.42\, h_1 + 0.65\, h_2 + 0.92\, h_3)$$

is a staircase no single stump represents. (The formula gives 0.66, 0.91; the figures round ϵ_b.)

3 Gradient Boosting Machines

3.1 Algorithm

Algorithm: GBM

Initialize $F_0(x) = \arg\min_\gamma \sum_{i=1}^{N} L(y_i, \gamma)$. For $b = 1, \ldots, B$:

1. **Gradient**: pseudo-residuals $r_{ib} = -\Big[\dfrac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\Big]_{F=F_{b-1}}$.
2. **Fit**: fit a regression tree $h_b(x)$ (leaves R_{jb}) to the targets r_{ib}; optionally set each leaf value by a line search $\gamma_{jb} = \arg\min_\gamma \sum_{x_i \in R_{jb}} L(y_i, F_{b-1}(x_i) + \gamma)$.
3. **Update**: $F_b(x) = F_{b-1}(x) + \eta\, h_b(x)$.

Output: $F_B(x) = F_0(x) + \sum_{b=1}^{B} \eta\, h_b(x)$.

For squared error $r_{ib} = y_i - F_{b-1}(x_i)$; **any differentiable loss** works. This is **gradient descent in function space**: $(F(x_1), \ldots, F(x_N))$ moves along the negative gradient, approximated by a tree that extends to new x.

3.2 Loss functions and their pseudo-residuals

Loss	$\Psi(y, f)$	$-\partial\Psi/\partial f$	F_0
squared (L_2)	$\frac{1}{2}(y-f)^2$	$y - f$	mean
absolute (L_1)	$\lvert y - f\rvert$	$\operatorname{sign}(y-f)$	median
Huber	$\frac{1}{2}(y-f)^2$ if $\lvert y-f\rvert \le \delta$; $\delta(\lvert y-f\rvert - \delta/2)$ otherwise	$y - f$ or $\delta\operatorname{sign}(y-f)$	robust center
Bernoulli	$\log(1+e^{-2\tilde{y}f})$, $\tilde{y}\in\{-1,1\}$	$\frac{2\tilde{y}}{1+e^{2\tilde{y}f}}$	$\frac{1}{2}\log\frac{\bar{p}}{1-\bar{p}}$
exponential (AdaBoost)	$e^{-\tilde{y}f}$	$\tilde{y}e^{-\tilde{y}f}$	$\frac{1}{2}\log\frac{\bar{p}}{1-\bar{p}}$

Bernoulli: f is half the log-odds ($p = 1/(1+e^{-2f})$); on the log-odds scale the pseudo-residual is $y - p$, $y \in \{0, 1\}$. Absolute/Huber losses limit large residuals (Ch. 2).

3.3 Shrinkage, subsampling and early stopping

Small η (0.01–0.1) needs more trees but generalizes better. **Stochastic gradient boosting** subsamples rows/columns per tree. B is set by **early stopping** on validation loss. η, B, depth (interaction order) and subsampling control the *same* overfitting.

3.4 Limitations of the standard GBM

Standard GBM overfits with too many trees, has no explicit complexity penalty, and is slow.

4 XGBoost: Regularized Second-Order Boosting

Motivation. A second-order expansion gives closed-form leaf values and a regularized split gain.

- **Regularization**: L_1/L_2 penalties on leaf weights and a per-leaf cost.
- **Second-order optimization**: uses the Hessian, not only the gradient.

Theorem: optimal leaf weights and split gain

At round t the new tree $f_t(x) = w_{q(x)}$ (leaf index $q(x)$, T leaves) minimizes

$$\mathcal{L}^{(t)} = \sum_i L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \gamma T + \tfrac{1}{2}\lambda \sum_{j=1}^{T} w_j^2.$$

With $g_i = \partial_{\hat{y}} L(y_i, \hat{y}_i^{(t-1)})$, $h_i = \partial^2_{\hat{y}} L(y_i, \hat{y}_i^{(t-1)})$, $G_j = \sum_{i\in I_j} g_i$, $H_j = \sum_{i\in I_j} h_i$, the second-order approximation gives

$$w_j^\star = -\frac{G_j}{H_j + \lambda}, \qquad \tilde{\mathcal{L}}^\star = -\frac{1}{2}\sum_{j=1}^{T} \frac{G_j^2}{H_j + \lambda} + \gamma T,$$

and the gain of splitting a leaf into L and R is

$$\text{Gain} = \frac{1}{2}\Big[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\Big] - \gamma.$$

Proof [Supplement]. $L(y_i, \hat{y}+f) \approx L(y_i,\hat{y}) + g_i f + \frac{1}{2}h_i f^2$; grouping by leaves, $\tilde{\mathcal{L}} = \sum_j\{G_j w_j + \frac{1}{2}(H_j+\lambda)w_j^2\} + \gamma T$, minimized at $w_j = -G_j/(H_j+\lambda)$ with value $-\frac{1}{2}G_j^2/(H_j+\lambda)$. Gain = one-leaf objective − two-leaf objective $-\gamma$. ■

Example: what $w^\star$ is for familiar losses

Squared loss: $g_i = \hat{y}_i - y_i$, $h_i = 1$, $w_j^\star = \sum_{i\in I_j}(y_i-\hat{y}_i)/(n_j+\lambda)$, a *shrunken* mean residual. Logistic: $g_i = p_i - y_i$, $h_i = p_i(1-p_i)$, one Newton step. `min_child_weight` bounds H_j below; γ prunes weak splits.

System side: learned default direction for missing values, quantile-sketch candidates, cache-aware parallelism.

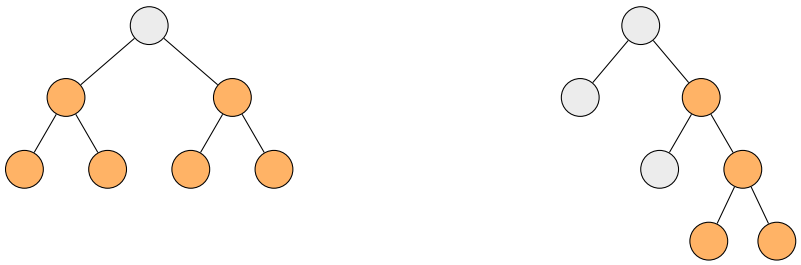
5 LightGBM

- **Histogram-based learning**: continuous variables bucketed into few bins.
- **Leaf-wise tree growth**: expand only the leaf with the largest loss reduction.

Aspect	Standard GBM	LightGBM
split search	pre-sorts every continuous value and scans all of them; bottleneck on large data	integer bins (histogram) cut computation and memory
tree growth	level-wise: grows all nodes of a level symmetrically, even unnecessary ones	leaf-wise: expands only the node that reduces the error most

Derivation: why histograms are cheap

With k bins (default 255), a node's (G,H) histogram costs $O(\#\text{rows})$ per feature and the split search $O(k)$. A child's histogram is parent minus sibling (*histogram subtraction*). Coarse bins mildly regularize.



level-wise (GBM, XGBoost default) leaf-wise (LightGBM)

Figure 6: Orange: most recent splits. Leaf-wise growth gives deep, unbalanced trees for the same number of leaves.

Remark: leaf-wise growth and overfitting

Lower training loss per leaf, but overfits small data: control `num_leaves`, `max_depth`, `min_data_in_leaf`. Also GOSS (keep large $|g_i|$, reweighted subsample of the rest) and EFB (bundle rarely co-nonzero sparse features).

6 CatBoost

Motivation. CatBoost computes everything for a row from earlier rows only, so its own label never leaks.

6.1 The curse of dimensionality of categorical variables

- GBM: one-hot encoding gives hundreds of dummy columns.
- CatBoost: one column of target = 1 proportions (target statistics).

Example: one-hot (GBM) versus target statistics (CatBoost)

Category	Share of target = 1 in that category	CatBoost value (one column)
A	80 of 100 (80%)	$\to 0.80$
B	10 of 40 (25%)	$\to 0.25$
C	100 of 200 (50%)	$\to 0.50$
D	5 of 50 (10%)	$\to 0.10$

One column replaces four dummies.

6.2 Naive target statistics leak the target

Theorem: target leakage of in-sample target encoding

If each row's encoding uses its own target, $\hat{x}_i = \frac{1}{n_{c_i}} \sum_{j:c_j=c_i} y_j$, then for a category observed once the encoded value equals y_i. In the extreme of all categories unique (e.g. an ID), the encoded feature equals the target on the training data and a single split fits it perfectly, while on new data the feature carries no information.

Generally $\mathbb{E}[\hat{x}_i \mid y_i]$ depends on y_i via y_i/n_{c_i}, so training and test encodings differ (*target shift*). CatBoost's fix:

- **smoothing with a prior**: $\hat{x}_i = \dfrac{\sum_{j\in\mathcal{P}_i} \mathbb{I}(c_j = c_i)\, y_j + a\,\bar{p}}{\sum_{j\in\mathcal{P}_i} \mathbb{I}(c_j = c_i) + a}$, shrinking rare categories toward the global rate $\bar{p}$;
- **ordered target statistics**: for a random permutation σ, $\mathcal{P}_i = \{j : \sigma(j) < \sigma(i)\}$, so only *earlier* rows encode row i, as at test time.

Ordered boosting likewise computes a row's residual with a model that has not seen it (no "prediction shift"). Symmetric (oblivious) trees use one split per level: fast and regularizing.

7 Summary and Practical Rules

Algorithm	Direction of development
XGBoost	explicit regularization controls overfitting; second-order information (Hessian) gives precise optimization
LightGBM	histograms and leaf-wise growth drastically reduce computation on large numerical/sparse data
CatBoost	ordered target statistics handle leakage and instability of categorical features

Rules for using boosting.

- Boosting adds shallow trees along the steepest loss decrease.
- Learning rate, iterations, tree complexity, subsampling and early stopping control one overfitting problem.
- Choose by repeated performance under a fixed validation procedure (Ch. 1 rules; early stopping on a validation fold, never the test set).

8 Selected exercises

Motivation. AdaBoost's reset to chance, L_2 boosting as shrinkage, and the XGBoost gain versus CART.

Exercise 5.1 (The last weak learner becomes a coin flip)

In AdaBoost with $0 < \epsilon_b < 1$, show that the weighted error of h_b under the updated weights is exactly one half:

$$\sum_{i=1}^{N} D_{b+1}(i)\, \mathbb{I}\{h_b(x_i) \neq y_i\} = \tfrac{1}{2}.$$

Solution [Supplement].

Step 1. Errors ($y_i h_b = -1$) get $e^{\alpha_b} = \sqrt{(1-\epsilon_b)/\epsilon_b}$, correct cases $e^{-\alpha_b}$: unnormalized totals $\epsilon_b e^{\alpha_b} = \sqrt{\epsilon_b(1-\epsilon_b)} = (1-\epsilon_b)e^{-\alpha_b}$.

Step 2. So $Z_b = 2\sqrt{\epsilon_b(1-\epsilon_b)}$ and the errors' normalized weight is $\frac{1}{2}$. ■

Exercise 5.2 (L_2 boosting with a linear smoother)

Use squared loss, $F_0 = \mathbf{0}$ (centered $\mathbf{y}$), and a base learner that maps a target vector r to the fitted vector Sr, where S is a symmetric $N \times N$ matrix with eigenvalues $\mu_1, \ldots, \mu_N \in [0, 1]$ (e.g. the OLS hat matrix, or a Ridge smoother). The fitted vector is updated by $F_b = F_{b-1} + \nu S(\mathbf{y} - F_{b-1})$ with $0 < \nu \leq 1$.
(a) Show that $F_B = \{I - (I - \nu S)^B\}\mathbf{y}$.
(b) With $S = \sum_k \mu_k u_k u_k^\top$ (orthonormal u_k), show that $F_B = \sum_k \{1 - (1 - \nu\mu_k)^B\}(u_k^\top \mathbf{y})\, u_k$, that each factor lies in $[0, 1]$ and increases to $\mathbb{I}\{\mu_k > 0\}$ as $B \to \infty$, and that the degrees of freedom $\mathrm{tr}\{I - (I - \nu S)^B\}$ increase in B.
(c) For $S = H$ a projection, show that $F_B = \{1 - (1 - \nu)^B\}H\mathbf{y}$.

Solution [Supplement].

(a) $R_b = \mathbf{y} - F_b = (I - \nu S)R_{b-1}$, so $R_B = (I - \nu S)^B \mathbf{y}$ and $F_B = \mathbf{y} - R_B$.

(b)

Step 1. $(I-\nu S)^B = \sum_k (1-\nu\mu_k)^B u_k u_k^\top$ and $I = \sum_k u_k u_k^\top$; apply (a).

Step 2. $0 \le 1-\nu\mu_k \le 1$, <1 iff $\mu_k > 0$: $(1-\nu\mu_k)^B$ is non-increasing, $\to 0$ if $\mu_k > 0$, $\equiv 1$ if $\mu_k = 0$.

Step 3. $\mathrm{tr}\{I-(I-\nu S)^B\} = \sum_k \{1-(1-\nu\mu_k)^B\}$ is a sum of non-decreasing terms.

(c) Eigenvalues $0, 1$: factor $1-(1-\nu)^B$ on the range of H, 0 off it.

B acts like $1/\lambda$ in Ridge; real boosting gains come from base learners that adapt to the residuals. ■

Exercise 5.3 (The XGBoost gain and the CART criterion)

(a) For $L = \frac{1}{2}(y-\hat{y})^2$ with current residuals $r_i = y_i - \hat{y}_i^{(t-1)}$, show that $w_j^\star = n_j \bar{r}_j/(n_j+\lambda)$ and that for $\lambda = \gamma = 0$ the gain equals $\frac{1}{2}\Delta_{\mathrm{RSS}}$ of Ch. 4 computed on the residuals.
(b) Show that for $\lambda = \gamma = 0$ and $h_i > 0$ the gain is never negative, but that for $\lambda > 0$ splitting a leaf into two halves with equal (G, H) has negative gain.

Solution [Supplement].

(a)

Step 1. $g_i = -r_i$, $h_i = 1$: $G_j = -n_j\bar{r}_j$, $H_j = n_j$, $w_j^\star = n_j\bar{r}_j/(n_j+\lambda)$.

Step 2. $\lambda = 0$: $G_j^2/H_j = n_j\bar{r}_j^2$ and $n\bar{r} = n_L\bar{r}_L + n_R\bar{r}_R$, so

$$\text{Gain} = \tfrac{1}{2}(n_L\bar{r}_L^2 + n_R\bar{r}_R^2 - n\bar{r}^2) = \tfrac{1}{2}\{n_L(\bar{r}_L-\bar{r})^2 + n_R(\bar{r}_R-\bar{r})^2\},$$

which is half the ANOVA form of Δ_{RSS} for the targets r_i.

(b)

Step 3. By Cauchy–Schwarz,

$$(G_L+G_R)^2 = \Big(\sqrt{H_L}\,\frac{G_L}{\sqrt{H_L}} + \sqrt{H_R}\,\frac{G_R}{\sqrt{H_R}}\Big)^2 \le (H_L+H_R)\Big(\frac{G_L^2}{H_L} + \frac{G_R^2}{H_R}\Big),$$

which is the claim Gain ≥ 0.

Step 4. With $G_L = G_R = g \neq 0$, $H_L = H_R = h$ and $\gamma = 0$,

$$\text{Gain} = \frac{1}{2}\Big[\frac{2g^2}{h+\lambda} - \frac{4g^2}{2h+\lambda}\Big] = -\frac{\lambda g^2}{(h+\lambda)(2h+\lambda)} < 0.$$

Each half has half the curvature and is shrunk more, so λ alone discourages uninformative splits. ■

Chapter 6. Neural Networks and In-Context Learning for Tabular Data

Predictive Modeling, Lecture 5 (MLP, embeddings, backpropagation, Adam, regularization, TabPFN, TabICL)

1 When Tree Boosting Is Not Enough

Limitation of tree boosting		What neural networks offer
finds splits on given features but does not learn the representation of features	$\longrightarrow$	learns representations needed for prediction through **embeddings** and **hidden layers**
smooth relations may need many splits and trees	$\longrightarrow$	combines several features continuously to represent complex relations
text, image, time series must first be summarized into tabular features	$\longrightarrow$	different input types can be learned in one model
usually re-learns a model from scratch for each new data set	$\longrightarrow$	pre-training and in-context learning reuse information from other problems

$$\underbrace{\text{fixed features} \longrightarrow \text{split learning}}_{\text{tree boosting}} \qquad \underbrace{\text{representation learning} + \text{prediction learning}}_{\text{neural network}}$$

2 The MLP as a Composite Nonlinear Function

2.1 One hidden layer

$$z^{(1)} = W^{(1)}x + b^{(1)} \quad \text{weighted sums of the raw features,}$$
$$a^{(1)} = \mathrm{ReLU}(z^{(1)}) \quad \text{hidden representation after activation,}$$
$$\hat{y} = g(W^{(2)}a^{(1)} + b^{(2)}) \quad \text{final prediction from the hidden representation.}$$

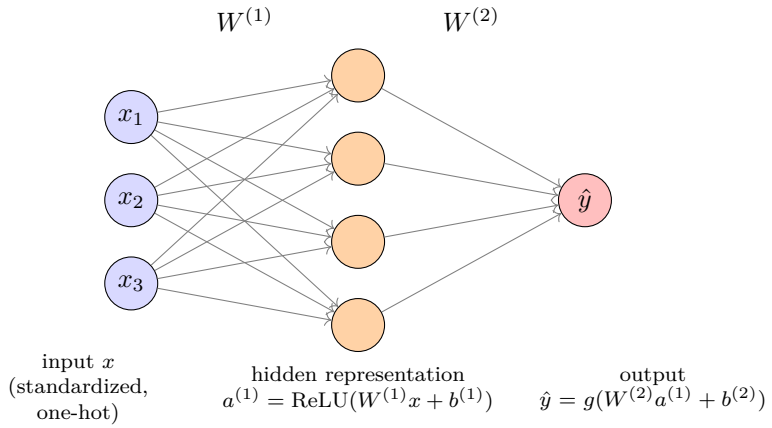


Figure 7: A one-hidden-layer MLP; g is identity, sigmoid or softmax. Backpropagation sends $\hat{y} - y$ back along the edges.

Hidden layer: a one-hidden-layer ReLU network is continuous piecewise linear, cut by the hyperplanes $\{w_k^{(1)\top}x + b_k^{(1)} = 0\}$; with enough units it approximates any continuous function on a compact set (universal approximation; Exercise 6.2).

Output layer by target:

Target	Output g	Loss	Gradient w.r.t. the output score
regression	identity $g(u) = u$	$\frac{1}{2}(y - \hat{y})^2$	$\hat{y} - y$
binary	sigmoid	log loss	$\hat{p} - y$
multiclass	softmax $p_k = e^{u_k} / \sum_l e^{u_l}$	cross-entropy $-\sum_k y_k \log p_k$	$p - y$

Proof of the softmax gradient [Supplement]. $\partial \log p_k / \partial u_m = \mathbb{I}(k = m) - p_m$, so $\partial(-\sum_k y_k \log p_k)/\partial u_m = -y_m + p_m \sum_k y_k = p_m - y_m$. ■

3 Neural-Network Inputs Are Numbers

One customer record:

Raw data	Conversion to numbers	Input
age 52	standardize with mean and SD (training data)	0.8
occupation: teacher	one-hot vector	$(0, 1, 0)$
income: missing	median imputation and missing indicator	$(40, 1)$

Input $(0.8, 0, 1, 0, 40, 1)$: the unscaled 40 dominates the weighted sums, so standardize (training statistics only).

4 Categorical Embeddings Are Learned From Prediction Error

For K occupations learn $E \in \mathbb{R}^{K \times d}$, $d \ll K$; teacher $= 1$ fetches row 1:

$$e_{\text{teacher}} = E[1, :] = (0.7, -0.2, 0.4), \qquad o_{\text{teacher}}^\top E = E[1, :],$$

a lookup equal to one-hot times a weight matrix.

Algorithm: how an embedding is learned

1. Initialize the entries of E with small random numbers.
2. Use the row of the observed category as network input.
3. Compute the error between truth and prediction.
4. Update, by backpropagation, only the row that was used:

$$E[1, :] \leftarrow E[1, :] - \eta \frac{\partial \ell}{\partial E[1, :]}.$$

Theorem: the gradient touches only the used row

If $e = o_k^\top E$ enters the network, then $\dfrac{\partial \ell}{\partial E} = o_k \left(\dfrac{\partial \ell}{\partial e}\right)^\top$, a $K \times d$ matrix whose only nonzero row is row k.

Proof [Supplement]. $e_m = \sum_l o_{k,l} E_{lm} = E_{km}$, so $\partial e_m / \partial E_{lm'} = \mathbb{I}(l = k)\mathbb{I}(m = m')$; apply the chain rule. ■

E is learned with the other weights; similar categories get nearby rows, rare ones stay noisy.

5 Training an MLP: Forward, Backward, Update

Algorithm: minibatch training

Initialize the weights θ. For each epoch and each minibatch B_t:

1. **Forward**: compute predictions and the average loss $\widehat{R}_{B_t}(\theta) = \frac{1}{|B_t|}\sum_{i\in B_t} L\{y_i, f_\theta(x_i)\}$.
2. **Backward**: compute the gradient $g_t = \nabla_\theta \widehat{R}_{B_t}(\theta)$.
3. **Update**: move the weights in the opposite direction, $\theta \leftarrow \theta - \eta g_t$.

At the end of each epoch record the validation loss; at the end, restore the weights with the lowest validation loss.

Backward pass: chain rule $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial w}$.

Derivation: backpropagation for one hidden layer

For $z^{(1)} = W^{(1)}x + b^{(1)}$, $a^{(1)} = \mathrm{ReLU}(z^{(1)})$, $\hat{y} = W^{(2)}a^{(1)} + b^{(2)}$, $L = \frac{1}{2}\|y - \hat{y}\|^2$:

$$\delta^{(2)} = \hat{y} - y, \quad \frac{\partial L}{\partial W^{(2)}} = \delta^{(2)}a^{(1)\top}, \quad \frac{\partial L}{\partial b^{(2)}} = \delta^{(2)},$$

$$\delta^{(1)} = \left(W^{(2)\top}\delta^{(2)}\right) \odot \mathbb{I}\{z^{(1)} > 0\}, \quad \frac{\partial L}{\partial W^{(1)}} = \delta^{(1)}x^\top, \quad \frac{\partial L}{\partial b^{(1)}} = \delta^{(1)}.$$

Reusing δ makes all gradients cost about one forward pass; units with $z \le 0$ pass no gradient ("dead" ReLU).

5.1 Minibatches and Adam

For uniform batches $\mathbb{E}[g_t] = \nabla\widehat{R}(\theta)$ at cost $O(|B_t|)$. Adam adapts per-parameter steps:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t, \quad v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2,$$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \ \hat{v}_t = \frac{v_t}{1-\beta_2^t}, \quad \theta \leftarrow \theta - \eta\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon},$$

($\beta_1 = 0.9$, $\beta_2 = 0.999$). Bias correction undoes $m_0 = v_0 = 0$: constant mean μ gives $\mathbb{E}m_t = (1-\beta_1^t)\mu$. $\hat{m}/\sqrt{\hat{v}}$ is roughly scale-free. Minibatch, Adam: *optimization* devices.

6 The Core Difficulty of Tabular MLPs: Overfitting

training loss keeps decreasing + validation loss rises again $\Longrightarrow$ overfitting.

Device	Role	Signal to check
small model	limit depth and width, fewer parameters	does even the training loss stay high (underfitting)?
weight decay	penalize large weights	does the train–validation gap shrink?
dropout	randomly switch off some hidden units during training	does validation improve, or does it overfit?
early stopping	restore the weights of the epoch with minimal validation loss	from when does validation loss rise again?

Theorem: weight decay and dropout as regularizers

(i) Adding $\frac{\lambda}{2}\|\theta\|^2$ to the loss changes the gradient step to $\theta \leftarrow (1-\eta\lambda)\theta - \eta g_t$: every step multiplicatively shrinks the weights (Ridge, Ch. 3).
(ii) Inverted dropout replaces a unit a by $\tilde{a} = a\,m/(1-q)$ with $m \sim \text{Bernoulli}(1-q)$, so $\mathbb{E}\tilde{a} = a$. For a linear model $\hat{y} = \beta^\top \tilde{x}$ with dropout on the inputs,

$$\mathbb{E}_m[(y - \beta^\top \tilde{x})^2] = (y - \beta^\top x)^2 + \frac{q}{1-q}\sum_j \beta_j^2 x_j^2,$$

i.e. dropout behaves like a data-dependent Ridge penalty.

Proof [Supplement]. (i) $\nabla(\widehat{R} + \frac{\lambda}{2}\|\theta\|^2) = g_t + \lambda\theta$.

(ii) $\mathbb{E}[\beta^\top \tilde{x}] = \beta^\top x$, so $\mathbb{E}(y - \beta^\top \tilde{x})^2 = (y - \beta^\top x)^2 + \text{Var}(\beta^\top \tilde{x})$, and independent m_j with $\text{Var}\{m_j/(1-q)\} = q/(1-q)$ give $\text{Var}(\beta^\top \tilde{x}) = \sum_j \beta_j^2 x_j^2\, q/(1-q)$. ■

Weight decay, dropout, early stopping: *generalization* devices; early stopping $\approx L_2$ (Exercise 6.3).

7 Trees and MLPs Have Different Inductive Biases

	Tree boosting	MLP
representation	piecewise functions built from stacked feature thresholds	continuous hidden representation from weighted sums and activations
preprocessing	relatively insensitive to the scale of numerical features	sensitive to scaling, missing values, categorical encoding
data size	strong baseline on small/medium tabular data	benefits from sufficient data
computation	many efficient CPU implementations	suited to minibatches and GPU matrix operations

On small tabular data MLPs have many parameters per observation, must build thresholds from many neurons, and depend on encoding; heterogeneous, irregular, partly uninformative columns favor trees.

8 Another Route: In-Context Learning

Motivation. A model pretrained on synthetic tables predicts from labeled examples given as input.

8.1 Refitting versus conditioning

MLP and GBM: fit anew	TabPFN, TabICL: use as context
$\mathcal{D}_{\text{train}} \xrightarrow{\text{fit}} \theta_{\mathcal{D}} \xrightarrow{x_q} \hat{y}_q$	$\Big(\underbrace{\{(x_i, y_i)\}_{i \in C}}_{\text{labeled cases}}, \underbrace{x_q}_{\text{new case}}\Big) \xrightarrow{\text{fixed pre-trained model}} \hat{y}_q$
a new model is learned for each new data set	uses a model pre-trained on many tabular tasks
weights or trees are updated from the training data	predicts usually without re-training weights
estimates can be unstable with small data	

In-context learning keeps the model fixed and feeds the data as input.

8.2 TabPFN pre-trains the learning algorithm itself

Algorithm: TabPFN

Pre-training. (1) Generate many synthetic tables from diverse generating structures (prior over structural causal models, MLPs, trees, noise, ...).
(2) Split each table into rows with visible labels and rows with hidden labels.
(3) Train to predict the hidden labels from the visible rows.
(4) Repeat over many tabular tasks to learn Θ.

Prediction on a new data set. $(X_{\text{train}}, y_{\text{train}}, x_q) \xrightarrow{\text{fixed } \Phi_\Theta} \hat{p}(y_q)$: a Transformer reads the whole table with unchanged weights.

Theorem: what prior-fitted training approximates

Let tasks $\phi \sim \pi$, and given ϕ draw a data set $\mathcal{D}$ and a query (x_q, y_q). If q_Θ ranges over all conditional distributions, the minimizer of the pre-training loss

$$\mathbb{E}_{\phi,\mathcal{D},x_q,y_q}\big[-\log q_\Theta(y_q \mid x_q, \mathcal{D})\big]$$

is the **posterior predictive distribution** $p(y_q \mid x_q, \mathcal{D}) = \int p(y_q \mid x_q, \phi)\, p(\phi \mid \mathcal{D})\, d\phi$.

Proof [Supplement]. Conditioning on $(\mathcal{D}, x_q)$, the loss is $\mathbb{E}_{\mathcal{D},x_q}[\mathrm{KL}(p(\cdot \mid x_q, \mathcal{D}) \,\|\, q_\Theta(\cdot \mid x_q, \mathcal{D}))] + \mathbb{E}[H(p(\cdot \mid x_q, \mathcal{D}))]$; the entropy is free of Θ and $\mathrm{KL} \geq 0$ with equality iff $q_\Theta = p(\cdot \mid x_q, \mathcal{D})$. Finally $p(y_q \mid x_q, \mathcal{D}) = \int p(y_q \mid x_q, \phi) p(\phi \mid x_q, \mathcal{D}) d\phi$ with $p(\phi \mid x_q, \mathcal{D}) = p(\phi \mid \mathcal{D})$. ■

TabPFN approximates Bayesian prediction under its synthetic prior, so quality depends on prior coverage.

8.3 TabICL summarizes rows first, then learns in context

Let the input table be $X = (x_{ij}) \in \mathbb{R}^{n \times m}$ with n rows and m features.

1. **Column-wise**: $(x_{1j}, \ldots, x_{nj}) \longrightarrow (e_{1j}, \ldots, e_{nj})$ (50 means different things in age and income).
2. **Row-wise**: $(e_{i1}, \ldots, e_{im}) \longrightarrow h_i \in \mathbb{R}^d$, fixed length for any m.
3. **In-context**: $(\{(h_i, y_i)\}_{i=1}^n, h_q) \xrightarrow{\text{pre-trained TabICL}} \hat{y}_q$.

Row compression lets the in-context step scale to more rows.

Remark: cautions for in-context models

Context = training rows of the current split only (else Ch. 1 leakage). Limits: context length, prior mismatch, prediction-time compute.

9 Summary

Data condition	Candidates to compare first	What to check
small/medium mixed-type data	regularized regression, CatBoost, LightGBM	is performance stable across splits?
enough data or useful categorical embeddings	tree boosting and a small MLP	were preprocessing and overfitting controlled?
pre-training and ICL usable	add TabPFN/TabICL to the existing baselines	do problem type and data size fit, within the compute budget?

MLP: preprocessing and embeddings to a hidden representation; minibatch/Adam optimize, weight decay/dropout/early stopping generalize. **TabPFN, TabICL**: labeled cases as context; sensitive to context length, prior and leakage.

10 Selected exercises

Motivation. Random initialization, ReLU representation in one dimension, early stopping as Ridge.

Exercise 6.1 (Identical hidden units stay identical)

Consider the one-hidden-layer network of the backpropagation box with scalar output $\hat{y} = w^{(2)\top} a^{(1)} + b^{(2)}$ and any differentiable loss $L(y, \hat{y})$, trained by (minibatch) gradient descent; for ReLU use the convention $\mathbb{I}\{z > 0\}$ for its derivative. Suppose that at $t = 0$ hidden units j and k have the same incoming weights and bias, $W^{(1)}_{j\cdot} = W^{(1)}_{k\cdot}$, $b^{(1)}_j = b^{(1)}_k$, and the same outgoing weight, $w^{(2)}_j = w^{(2)}_k$. Show that these equalities hold at every step t.

Solution [Supplement].

Step 1. If the equalities hold at step t, then for every input x, $z_j = z_k$ and $a_j = a_k$.

Step 2. With $\delta^{(2)} = \partial L/\partial \hat{y}$: $\partial L/\partial w^{(2)}_j = \delta^{(2)} a_j = \partial L/\partial w^{(2)}_k$; $\delta^{(1)}_j = w^{(2)}_j \delta^{(2)} \mathbb{I}\{z_j > 0\} = \delta^{(1)}_k$, so $\partial L/\partial W^{(1)}_{j\cdot} = \delta^{(1)}_j x^\top = \partial L/\partial W^{(1)}_{k\cdot}$ and $\partial L/\partial b^{(1)}_j = \partial L/\partial b^{(1)}_k$; minibatch averaging preserves this.

Step 3. Equal parameters get equal updates (also under Adam, which is coordinate-wise); induct on t.

Equal starts collapse the width to one unit; random initialization breaks the symmetry. ■

Exercise 6.2 (One hidden ReLU layer and piecewise-linear functions)

(a) Let $f : [a, b] \to \mathbb{R}$ be continuous and linear on each $[t_{k-1}, t_k]$, $a = t_0 < t_1 < \cdots < t_K = b$, with slope m_k there. Show that for $x \in [a, b]$

$$f(x) = f(a) + m_1 \,\mathrm{ReLU}(x - a) + \sum_{k=1}^{K-1} (m_{k+1} - m_k) \,\mathrm{ReLU}(x - t_k),$$

so a network with K hidden ReLU units represents f exactly on $[a, b]$.
(b) Deduce that every continuous f on $[a, b]$ is a uniform limit of such networks.

Solution [Supplement].

(a)

Step 1. Let g be the right side. On $[t_{k-1}, t_k]$, $\mathrm{ReLU}(x - t_l) = 0$ for $l \geq k$, $= x - t_l$ for $l \leq k-1$, and $\mathrm{ReLU}(x - a) = x - a$: g is continuous and piecewise linear.

Step 2. Slope on (t_{k-1}, t_k): $m_1 + \sum_{l=1}^{k-1}(m_{l+1} - m_l) = m_k$; and $g(a) = f(a)$.

Step 3. $f - g$ is continuous with slope 0 on every piece and 0 at a, so $f = g$. Weights: input 1, biases $-a, -t_1, \ldots, -t_{K-1}$, outputs $m_1, m_2 - m_1, \ldots, m_K - m_{K-1}$, output bias $f(a)$.

(b)

Step 4. Given $\varepsilon > 0$, uniform continuity gives $h > 0$ with $|f(x) - f(x')| < \varepsilon$ for $|x - x'| \leq h$; let ℓ interpolate f linearly on knots of mesh $\leq h$.

Step 5. On $[t_{k-1}, t_k]$, $\ell(x)$ is a convex combination of $f(t_{k-1}), f(t_k)$, both within ε of $f(x)$, so $|\ell - f| < \varepsilon$; ℓ is a network by (a). ■

Exercise 6.3 (Early stopping and Ridge)

Let $L(\theta) = \frac{1}{2}\|\mathbf{y} - X\theta\|^2$ with thin SVD $X = UDV^\top$, $d_1 \geq \cdots \geq d_r > 0$, and run gradient descent $\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$ from $\theta_0 = \mathbf{0}$ with $0 < \eta < 1/d_1^2$.
(a) Show that $\theta_t = V\mathrm{diag}(\phi_t(d_j)/d_j)U^\top \mathbf{y}$ with $\phi_t(d) = 1 - (1 - \eta d^2)^t$.
(b) The Ridge solution of $\frac{1}{2}\|\mathbf{y} - X\theta\|^2 + \frac{\lambda}{2}\|\theta\|^2$ is $V\mathrm{diag}(\psi_\lambda(d_j)/d_j)U^\top \mathbf{y}$ with $\psi_\lambda(d) = d^2/(d^2 + \lambda)$ (SVD form of Ridge). Show that for $\lambda = 1/(\eta t)$

$$\psi_\lambda(d) \leq \phi_t(d) \leq 2\,\psi_\lambda(d) \qquad \text{for all } d \in (0, d_1].$$

Solution [Supplement].

(a)

Step 1. $\theta_{t+1} = (I - \eta X^\top X)\theta_t + \eta X^\top \mathbf{y}$ with $X^\top \mathbf{y} = VDU^\top \mathbf{y}$, $X^\top X = VD^2V^\top$ and $\theta_0 = \mathbf{0}$, so $\theta_t = Vc_t$.

Step 2. $c_{t+1,j} = (1 - \eta d_j^2)c_{t,j} + \eta d_j\, u_j^\top \mathbf{y}$ has fixed point $c_j^\star = u_j^\top \mathbf{y}/d_j$, and $c_{t,j} - c_j^\star = (1 - \eta d_j^2)^t(0 - c_j^\star)$, so $c_{t,j} = \phi_t(d_j)\, u_j^\top \mathbf{y}/d_j$.

(b)

Step 3. Put $u = \eta d^2 \in (0, 1)$. Then $\psi_\lambda(d) = \eta t d^2/(\eta t d^2 + 1) = ut/(1 + ut)$ and $\phi_t(d) = 1 - (1 - u)^t$.

Step 4. (Lower bound.) $(1 - u)^t \leq e^{-ut} \leq 1/(1 + ut)$, so $\phi_t \geq 1 - 1/(1 + ut) = \psi_\lambda$.

Step 5. (Upper bound.) Bernoulli: $(1 - u)^t \geq 1 - ut$, so $\phi_t \leq \min(1, ut)$. If $ut \leq 1$, $\psi_\lambda \geq ut/2$; if $ut > 1$, $\psi_\lambda > \frac{1}{2}$. Either way $\min(1, ut) \leq 2\psi_\lambda$.

More steps correspond to a smaller penalty. ■

Chapter 7. AutoML and AutoGluon

Predictive Modeling, Lecture 6 (model selection, OOF predictions, fold bagging, stacking, weighted ensembles)

1 When There Are Many Candidates, Model Selection Becomes a Problem

No tabular model is always best, so the question is *how to combine the candidates* of Lectures 1–5.

Stage	What AutoGluon does
fit candidates	trains several candidates: linear models, tree boosting, neural networks, ...
internal validation	non-bagged models on a holdout set, bagged models with OOF predictions
ensemble	combines candidates by fold bagging, stacking, weighted ensembles
leaderboard	compares internal validation score, fit time, prediction time, stack level
final prediction	predicts new cases with the selected single model or ensemble

AutoGluon **manages fitting, validation and combination**; its internal splits are random/stratified, so time/group structure and a protected test set must be imposed from outside (Ch. 1).

2 Stacking: Candidate Predictions Become New Inputs

Candidate predictions for x_i:

$$z_i = \big(h_{\text{linear}}(x_i),\, h_{\text{boost}}(x_i),\, h_{\text{MLP}}(x_i)\big), \qquad \text{e.g. } z_i = (0.30,\ 0.75,\ 0.60)$$

(probabilities) become features of a higher-level model:

$$\hat{y}_i = g(x_i, z_i),$$

Stacking **learns again from the candidates' predictions**, exploiting that they err on different cases.

2.1 Why the level-1 inputs must be out-of-fold

Theorem: in-sample stacking rewards overfitting

If g is trained on in-sample predictions $z_i = h(x_i)$ of models that were fitted on the same rows, a model that memorizes the training data (e.g. 1-nearest neighbor, whose in-sample prediction is exactly y_i) looks perfect, and g puts all its weight on it, although its out-of-sample error can be the worst of all candidates.

g must train on predictions for cases *the candidate did not see*, as at prediction time: OOF predictions.

3 K-fold Bagging and OOF Predictions

Definition: fold bagging

Split the training data into folds $F_1, \ldots, F_K$ and fit the same model K times, the kth time without F_k, giving $h^{(-1)}, \ldots, h^{(-K)}$ (same algorithm and settings, different training rows).

- **Training data: OOF predictions.** $i \in F_k \implies \hat{y}_i^{\text{OOF}} = h^{(-k)}(x_i)$: each case is predicted by the model that did not use it.
- **New cases: averaged prediction.** $x_{\text{new}} \mapsto \frac{1}{K}\sum_{k=1}^{K} h^{(-k)}(x_{\text{new}})$.

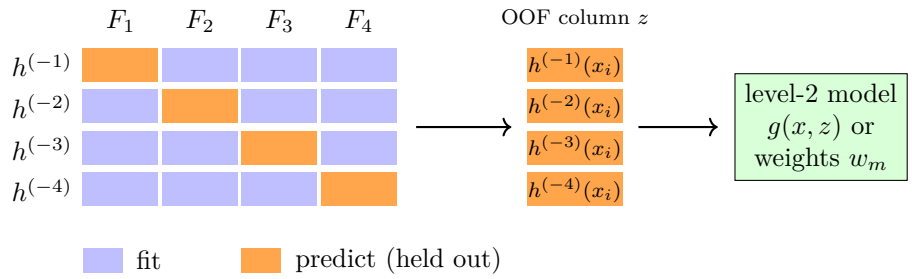


Figure 8: Fold bagging, $K = 4$: $h^{(-k)}$ predicts only F_k, giving an OOF column z on which the level-2 learner is trained.

The OOF vector gives (i) an honest internal score and (ii) leakage-free stacking inputs; averaging fold models lowers variance (Ch. 4). `num_bag_sets` repeats with new seeds.

Remark: a subtle mismatch

OOF predictions come from single fold models, test predictions from smoother averages; small for $K = 5$–10.

4 Weighted Ensembles

Motivation. Similar models often err differently, so weighted averages on held-out predictions help.

Example: three candidates

	linear	boosting	MLP
prediction	0.30	0.75	0.60
weight	0.10	0.60	0.30

$$\hat{y} = 0.10(0.30) + 0.60(0.75) + 0.30(0.60) = 0.66.$$

$\hat{y} = \sum_{m=1}^{M} w_m h_m(x)$ with fixed weights chosen on OOF performance.

Algorithm: greedy ensemble selection (Caruana et al.)

Start from the empty ensemble. Repeat R times: add (with replacement) the candidate whose inclusion gives the best OOF score of the *averaged* ensemble. The weight of model m is (number of times selected)$/R$.

Weights are convex; useless models get 0. This overfits OOF less than regressing y on z.

Theorem: ambiguity decomposition

For weights $w_m \geq 0$, $\sum_m w_m = 1$, and $\hat{y} = \sum_m w_m h_m$, under squared loss

$$(y - \hat{y})^2 = \sum_m w_m (y - h_m)^2 - \sum_m w_m (h_m - \hat{y})^2.$$

Hence the ensemble's error is never larger than the weighted average of the members' errors, and the improvement equals the weighted spread ("ambiguity") of the members' predictions.

Proof [Supplement]. $\sum_m w_m (y - h_m)^2 = \sum_m w_m \{(y - \hat{y}) + (\hat{y} - h_m)\}^2 = (y - \hat{y})^2 + 2(y - \hat{y}) \sum_m w_m (\hat{y} - h_m) + \sum_m w_m (\hat{y} - h_m)^2$, and the middle sum is $\hat{y} - \hat{y} = 0$. ■

Gains need good *and* disagreeing members. For any convex loss Jensen gives the inequality (Exercise 7.3).

5 How the Three Mechanisms Fit Together in AutoGluon

	Fold bagging	Stacking	Weighted ensemble
learning	same model on K folds	higher model g on candidates' OOF predictions z and original features x	repeatedly selects candidates that improve internal validation
new case	average of fold predictions	feeds candidates' predictions and original features to $g(x, z)$	sums candidates' predictions with fixed weights
prediction	$\frac{1}{K} \sum_{k=1}^{K} h^{(-k)}(x)$	$g(x, z)$	$\sum_{m=1}^{M} w_m h_m(x)$
core role	less prediction variance, OOF generation	learns new prediction rules from candidates' error patterns	combines the selected candidates into one final prediction

With `num_stack_levels`, level-2 models are fold-bagged too and a final weighted ensemble combines the top layer; x is passed to every level.

6 Setting the Search Space

```
predictor = TabularPredictor(label=label, eval_metric=metric)
predictor.fit(train_data, presets=..., time_limit=..., hyperparameters=...)
```

Setting	Role
`eval_metric`	metric used to rank candidates and to select the ensemble
`presets`	bundles the candidate portfolio, ensembling, refitting and cost settings (e.g. quality versus speed)
`time_limit`	approximate time budget for `fit()`
`hyperparameters`	the candidate portfolio: which model families and which settings of each
`num_bag_folds`	number of folds used to create OOF predictions
`num_stack_levels`	number of extra stacking layers that use lower models' predictions as inputs

Match the metric to the decision (Ch. 1); report the time limit with the leaderboard.

Summary

- No single tabular model is always best; combining candidates is a decision in its own right.
- Stacking needs out-of-fold predictions; in-sample predictions reward memorization.
- Fold bagging creates OOF predictions and averages fold models for new cases; stacking learns $g(x, z)$; the weighted ensemble combines selected candidates with fixed convex weights.
- A convex combination's loss is at most the average member loss; gains come from diverse, individually strong candidates.
- AutoGluon manages fitting, validation and combination; the validation design and the final test remain the analyst's responsibility.

7 Selected exercises

Motivation. Best two-model blend, winner's optimism, and log loss under convex combination.

Exercise 7.1 (The best blend of two models)

Let $e_m = Y - h_m(X)$, $m = 1, 2$, with $s_m^2 = \mathbb{E}e_m^2$, $c = \mathbb{E}(e_1 e_2)$ and $\Delta = s_1^2 + s_2^2 - 2c = \mathbb{E}(e_1 - e_2)^2 > 0$. For $w \in \mathbb{R}$ let $h_w = wh_1 + (1 - w)h_2$.
(a) Show that $\mathbb{E}\{Y - h_w(X)\}^2$ is minimized at $w^\star = (s_2^2 - c)/\Delta$, with minimum $M = (s_1^2 s_2^2 - c^2)/\Delta$.
(b) Show that $M \le \min(s_1^2, s_2^2)$, with equality iff $c = \min(s_1^2, s_2^2)$.
(c) Show that $w^\star \in [0, 1]$ iff $c \le \min(s_1^2, s_2^2)$.

Solution [Supplement].

(a) $Y - h_w = we_1 + (1 - w)e_2$, so the risk is

$$q(w) = w^2 s_1^2 + 2w(1 - w)c + (1 - w)^2 s_2^2 = \Delta w^2 - 2(s_2^2 - c)w + s_2^2,$$

with leading coefficient $\Delta > 0$, minimized at $w^\star = (s_2^2 - c)/\Delta$:

$$q(w^\star) = s_2^2 - \frac{(s_2^2 - c)^2}{\Delta} = \frac{s_2^2(s_1^2 + s_2^2 - 2c) - (s_2^2 - c)^2}{\Delta} = \frac{s_1^2 s_2^2 - c^2}{\Delta}.$$

(b) Suppose $s_1^2 \le s_2^2$ (the other case is symmetric). Then

$$s_1^2 - M = \frac{s_1^2(s_1^2 + s_2^2 - 2c) - s_1^2 s_2^2 + c^2}{\Delta} = \frac{(s_1^2 - c)^2}{\Delta} \ge 0,$$

with equality iff $c = s_1^2$.

(c) $w^\star \ge 0$ iff $c \le s_2^2$, and $w^\star \le 1$ iff $s_2^2 - c \le \Delta$ iff $c \le s_1^2$.

For $c > \min s_m^2$ the optimal weight extrapolates; convex weights trade this for stability. ■

Exercise 7.2 (The winner's internal score is optimistic)

Candidates $k = 1, \ldots, K$ have true risks R_k and internal validation estimates $\widehat{R}_k$ with $\mathbb{E}\widehat{R}_k = R_k$. Let $\hat{k} = \arg\min_k \widehat{R}_k$.
(a) Show that $\mathbb{E}\widehat{R}_{\hat{k}} \le \min_k R_k \le \mathbb{E}R_{\hat{k}}$.

(b) If $R_1 = \cdots = R_K = R$ and $\widehat{R}_k = R + \tau Z_k$ with $Z_k \overset{\text{iid}}{\sim} N(0,1)$, show that the optimism is $R - \mathbb{E}\widehat{R}_{\hat{k}} = \tau\,\mathbb{E}\max_k Z_k$, and that it equals $\tau/\sqrt{\pi}$ for $K = 2$ and does not decrease in K.

Solution [Supplement].

(a) $\widehat{R}_{\hat{k}} \le \widehat{R}_j\ \forall j \Rightarrow \mathbb{E}\widehat{R}_{\hat{k}} \le R_j$; minimize over j. And $R_{\hat{k}} \ge \min_k R_k$ pointwise.

(b)

Step 1. $\min_k(R + \tau Z_k) = R - \tau \max_k(-Z_k)$ and $(-Z_k) \overset{d}{=} (Z_k)$.

Step 2. For $K = 2$, $\max(Z_1, Z_2) = \frac{1}{2}(Z_1 + Z_2) + \frac{1}{2}|Z_1 - Z_2|$ and $Z_1 - Z_2 \sim N(0,2)$, so

$$\mathbb{E}\max(Z_1, Z_2) = \tfrac{1}{2}\mathbb{E}|Z_1 - Z_2| = \tfrac{1}{2}\sqrt{2}\sqrt{2/\pi} = 1/\sqrt{\pi} \approx 0.56.$$

Step 3. An extra candidate can only raise the maximum.

Hence the final number must come from a test set not used for selection. ■

Exercise 7.3 (Convex combinations under log loss)

Let $\ell(y,p) = -y\log p - (1-y)\log(1-p)$ for $y \in \{0,1\}$, $p \in (0,1)$, members $p_1, \ldots, p_M \in (0,1)$, weights $w_m \ge 0$ with $\sum_m w_m = 1$, and $\bar{p} = \sum_m w_m p_m$.
(a) Show that $\ell(y,\bar{p}) \le \sum_m w_m \ell(y,p_m)$, with equality iff all p_m with $w_m > 0$ are equal.
(b) Two cases, both with $y = 1$; member 1 predicts $(0.9, 0.1)$ and member 2 predicts $(0.1, 0.9)$. Compare the average log losses of the members and of their equal-weight average.

Solution [Supplement].

(a)

Step 1. $(-\log p)'' = 1/p^2 > 0$, $(-\log(1-p))'' = 1/(1-p)^2 > 0$: strictly convex.

Step 2. Strict Jensen gives the inequality, with equality iff the p_m with $w_m > 0$ coincide.

Summing over cases, the same holds for average log loss on any data set.

(b) Each member: $\frac{1}{2}(-\log 0.9 - \log 0.1) \approx \frac{1}{2}(0.105 + 2.303) = 1.204$. The average predicts 0.5: $\log 2 \approx 0.693$, beating *both*. ■

Chapter 8. Conditional Independence and Gaussian Graphical Models

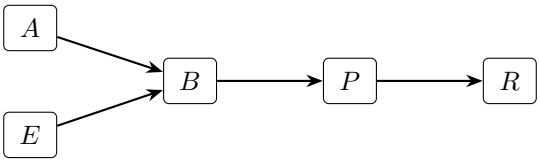
Graphical models for discovering relations, Lecture 1 (partial correlation, precision matrix, GGM, graphical lasso, neighborhood selection)

Roadmap of the causal-discovery part

Lecture	Condition on the data	Relation to be discovered
1 (Ch. 8)	Gaussian observational data	undirected graph of conditional relations
2 (Ch. 9)	linear SEM with given directions	conditional independences a graph implies; limits of direction
3 (Ch. 10)	all main variables observed	the equivalence class of DAGs that data can distinguish
4 (Ch. 11)	hidden common causes allowed	latent confounding, relations and directions compatible with it
5 (Ch. 12)	extra assumption on the error distribution	more specific directions inside the equivalence class
6 (Ch. 13)	further identifiable models	directions for which causal discovery is possible

A running toy health example

Used in all six lectures:



A	age	E	amount of exercise
B	body-mass index (BMI)	P	blood pressure
R	disease risk		

Lecture 4 adds a hidden lifestyle variable U; Lecture 6 adds measurement error.

1 Correlation Mixes Direct and Indirect Relations

$$X_1 \sim N(0,1), \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.8X_2 + \epsilon_3, \qquad \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0, 0.36),$$

with $X_1, \epsilon_2, \epsilon_3$ independent, so all variances equal 1.

Pair	Correlation	Why the relation arises
X_1, X_2	0.8	X_1 and X_2 are directly connected
X_2, X_3	0.8	X_2 and X_3 are directly connected
X_1, X_3	0.64	the relation is transmitted through X_2

$\mathrm{Cov}(X_1, X_3) = \mathrm{Cov}(X_1, 0.64X_1 + 0.8\epsilon_2 + \epsilon_3) = 0.64$. The question is: **does a relation remain after accounting for the other variables?**

2 Conditional Independence

Definition: conditional independence

$X \perp\!\!\!\perp Y \mid Z$ if $p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$ for all z, equivalently $p(x \mid y, z) = p(x \mid z)$.

In the chain, $X_1 \not\perp\!\!\!\perp X_3$ (Corr $= 0.64$) but $X_1 \perp\!\!\!\perp X_3 \mid X_2$:

$$p(x_1, x_3 \mid x_2) = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Conditional independence does not fix direction, but it **narrows the candidate edges**.

3 Partial Correlation

Motivation. Partial correlation asks whether a direct association remains once the other variables are removed linearly.

3.1 Definition by residuals

$$X_1 = \alpha_1 + \beta_1 X_2 + r_1, \qquad X_3 = \alpha_3 + \beta_3 X_2 + r_3.$$

Quantity	Meaning
r_1	the part of X_1 not explained by X_2
r_3	the part of X_3 not explained by X_2
$\mathrm{Corr}(r_1, r_3)$	the partial correlation of X_1, X_3 given X_2

$$\rho_{13\cdot 2} = \mathrm{Corr}(r_1, r_3),$$

with population (linear-projection) residuals; $\rho_{jk\cdot S}$ regresses both on X_S.

Theorem: first-order partial correlation

$$\rho_{13\cdot 2} = \frac{\rho_{13} - \rho_{12}\rho_{23}}{\sqrt{(1-\rho_{12}^2)(1-\rho_{23}^2)}}.$$

Proof [Supplement]. Standardize. Then $r_1 = X_1 - \rho_{12} X_2$, $r_3 = X_3 - \rho_{23} X_2$, so $\mathrm{Cov}(r_1, r_3) = \rho_{13} - \rho_{12}\rho_{23} - \rho_{23}\rho_{12} + \rho_{12}\rho_{23} = \rho_{13} - \rho_{12}\rho_{23}$, $\mathrm{Var}(r_1) = 1 - \rho_{12}^2$, $\mathrm{Var}(r_3) = 1 - \rho_{23}^2$. ■

Example: a correlation of 0.64 can disappear

With $\rho_{12} = \rho_{23} = 0.8$, $\rho_{13} = 0.64$,

$$\rho_{13\cdot 2} = \frac{0.64 - (0.8)(0.8)}{\sqrt{(1-0.8^2)(1-0.8^2)}} = 0,$$

so under Gaussianity $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

4 The Precision Matrix

4.1 Zeros of the precision matrix are conditional independences

Let $X \sim N_p(\mu, \Sigma)$ and $\Omega = \Sigma^{-1}$ (the **precision matrix**).

Precision matrix	Interpretation
$\Omega_{jk} = 0$	$X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$
$\Omega_{jk} \neq 0$	a relation between X_j, X_k remains given all the other variables

Theorem: precision matrix and full-order partial correlation

For $X \sim N_p(\mu, \Sigma)$ with $\Sigma \succ 0$ and $\Omega = \Sigma^{-1}$,

$$\rho_{jk\cdot -\{j,k\}} = -\frac{\Omega_{jk}}{\sqrt{\Omega_{jj}\Omega_{kk}}}, \qquad \text{so} \qquad \Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}.$$

Proof [Supplement]. Let $A = \{j,k\}$, B the rest. $\mathrm{Cov}(X_A \mid X_B) = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA} = (\Omega_{AA})^{-1}$ (block inversion), so

$$\mathrm{Cov}(X_A \mid X_B) = \begin{pmatrix} \Omega_{jj} & \Omega_{jk} \\ \Omega_{jk} & \Omega_{kk} \end{pmatrix}^{-1} = \frac{1}{\Omega_{jj}\Omega_{kk} - \Omega_{jk}^2} \begin{pmatrix} \Omega_{kk} & -\Omega_{jk} \\ -\Omega_{jk} & \Omega_{jj} \end{pmatrix},$$

with correlation $-\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$, free of x_B, hence the partial correlation. For a bivariate Gaussian, zero correlation $\iff$ independence. ■

4.2 Turning the precision matrix into a graph: the GGM

Definition: Gaussian graphical model

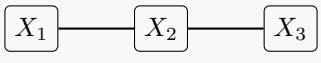
The GGM of $N_p(\mu, \Sigma)$ is the undirected graph on $\{1, \ldots, p\}$ with

$$(j,k) \text{ is an edge} \iff \Omega_{jk} \neq 0.$$

Example: the three-variable chain

$$R = \begin{pmatrix} 1 & 0.8 & 0.64 \\ 0.8 & 1 & 0.8 \\ 0.64 & 0.8 & 1 \end{pmatrix}, \qquad \Omega = R^{-1} \approx \begin{pmatrix} 2.78 & -2.22 & 0 \\ -2.22 & 4.56 & -2.22 \\ 0 & -2.22 & 2.78 \end{pmatrix}.$$

Exactly: $\Omega_{11} = \Omega_{33} = 1/0.36$, $\Omega_{22} = 1.64/0.36$, $\Omega_{12} = \Omega_{23} = -0.8/0.36$, $\Omega_{13} = 0$. GGM:



$\rho_{12\cdot 3} = 0.8/\sqrt{1.64} \approx 0.62 < 0.8$.

Remark: from pairwise to global statements

For a positive density the pairwise Markov property is equivalent to the global one: if S separates A and B in the graph, $X_A \perp\!\!\!\perp X_B \mid X_S$ (Hammersley–Clifford).

5 Estimating the Graph

5.1 Inverting the sample covariance produces spurious edges

The naive estimate is $\widehat{\Omega} = S^{-1}$.

Data condition	Problem
finite sample	because of sampling error, entries that are truly 0 are estimated as small nonzero values
many variables	S^{-1} is unstable and most entries are nonzero
$p \geq n$	S is singular and the inverse may not exist

We need a **sparse** estimate.

Method	Core idea	Suitable when
partial-correlation tests	compute partial correlations from S^{-1} and test $H_0 : \rho_{jk\cdot -\{j,k\}} = 0$	$p < n$ and sample large enough
graphical lasso	ℓ_1 penalty on the whole precision matrix, some entries estimated as 0	a sparse positive-definite precision matrix is needed
neighborhood selection	Lasso regression of each variable on the rest; choose neighbors variable by variable	many variables, node-wise relations
CLIME	directly estimate a sparse Ω satisfying $S\Omega \approx I$	estimating the precision matrix in high dimension

5.2 Testing a zero partial correlation

For Gaussian data, $X_j \perp\!\!\!\perp X_k \mid X_S \iff \rho_{jk\cdot S} = 0$. Let $\hat{\rho} = \hat{\rho}_{jk\cdot S}$, $q = |S|$.

Test	Statistic	Reference distribution
partial-correlation t-test	$T = \hat{\rho}\sqrt{\dfrac{n-q-2}{1-\hat{\rho}^2}}$	t_{n-q-2}
Fisher-z test	$Z = \sqrt{n-q-3}\,\operatorname{arctanh}(\hat{\rho})$	approximately $N(0,1)$
conditional distance correlation	$\mathrm{dCor}(X_j, X_k \mid X_{-\{j,k\}})$	p-value by resampling

Derivation: where the degrees of freedom come from

Regressing on q variables plus an intercept leaves a residual correlation like one from $m = n - q$ pairs. For m pairs and $\rho = 0$, $T = r\sqrt{(m-2)/(1-r^2)} \sim t_{m-2}$, and $\operatorname{arctanh}(r) \approx N(\operatorname{arctanh}\rho, 1/(m-3))$ by the delta method ($\mathrm{Var}(r) \approx (1-\rho^2)^2/m$, $\frac{d}{d\rho}\operatorname{arctanh}\rho = 1/(1-\rho^2)$).

Failing to reject H_0 does *not* prove conditional independence.

5.3 Graphical lasso

Derivation: Gaussian log-likelihood of Ω

For centered data, $\log p(x_i) = \frac{1}{2}\log\det\Omega - \frac{1}{2}x_i^\top \Omega x_i + c$ and $\sum_i x_i^\top \Omega x_i = n\,\mathrm{tr}(S\Omega)$, so the log-likelihood is $\frac{n}{2}\{\log\det\Omega - \mathrm{tr}(S\Omega)\} + c$, maximized at S^{-1} when $S \succ 0$.

$$\widehat{\Omega}_\lambda = \arg\min_{\Omega \succ 0} \Big[\mathrm{tr}(S\Omega) - \log\det(\Omega) + \lambda \sum_{j \neq k} |\Omega_{jk}|\Big].$$

Part of the objective	Role
$\operatorname{tr}(S\Omega) - \log\det(\Omega)$	matches the covariance structure of the observed data
$\lambda \sum_{j \neq k} \lvert\Omega_{jk}\rvert$	sets small off-diagonal entries exactly to 0

Convex; optimality: $\Omega^{-1} - S - \lambda\Gamma = 0$, $\Gamma_{jk} \in \partial|\Omega_{jk}|$, $\Gamma_{jj} = 0$. Hence $|(\widehat{\Omega}^{-1})_{jk} - S_{jk}| \leq \lambda$, with equality on edges. The solution is positive definite even for $p \geq n$; λ by CV log-likelihood, EBIC or stability selection.

5.4 Neighborhood selection

Theorem: conditional distribution of one node

For centered $X \sim N_p(0, \Sigma)$ and $\Omega = \Sigma^{-1}$,

$$X_j \mid X_{-j} \sim N\Big(-\frac{\Omega_{j,-j}}{\Omega_{jj}} X_{-j},\ \frac{1}{\Omega_{jj}} \Big),$$

so the population coefficients of the regression of X_j on the other variables are $\beta_k^{(j)} = -\Omega_{jk}/\Omega_{jj}$, and $\beta_k^{(j)} = 0 \iff \Omega_{jk} = 0$.

Proof [Supplement]. Schur complement with $A = \{j\}$: $\operatorname{Var}(X_j \mid X_{-j}) = (\Omega_{jj})^{-1}$, and $\Sigma_{j,-j}\Sigma_{-j,-j}^{-1} X_{-j} = -\Omega_{jj}^{-1}\Omega_{j,-j} X_{-j}$ by $\Omega_{j,-j} = -\Omega_{jj}\Sigma_{j,-j}\Sigma_{-j,-j}^{-1}$. ■

Estimate by the Lasso:

$$\hat{\beta}^{(j)} = \arg\min_{\beta} \Big\{ \frac{1}{2n} \|X_j - X_{-j}\beta\|_2^2 + \lambda \|\beta\|_1 \Big\}.$$

$\hat{\beta}_k^{(j)} \neq 0$ makes $j - k$ a candidate; the two regressions are combined by the **AND** rule (both nonzero) or the **OR** rule (at least one). It parallelises over nodes but gives a graph, not a positive-definite $\widehat{\Omega}$.

5.5 CLIME

$$\widehat{\Omega} = \arg\min_{\Omega} \|\Omega\|_1 \quad \text{subject to} \quad \|S\Omega - I\|_\infty \leq \lambda,$$

a column-wise linear program, symmetrized afterwards.

6 GGM Edges Are Conditional Associations, Not Causal Directions

Interpretable from a GGM	Not interpretable from a GGM alone
Does a relation remain after accounting for the other observed variables?	Which variable is the cause and which the effect?
What are the sign and size of the partial correlation?	Is an edge a direct causal effect?
Which variables are conditionally connected to many others?	Is there a hidden common cause?

$X_j - X_k$ means neither $X_j \to X_k$ nor $X_j \leftarrow X_k$; conditioning on a common *effect* can even create an edge between independent causes (Ch. 9). Directions need DAGs.

Summary

- Partial correlations are residual correlations; for Gaussians $\rho_{jk\cdot\text{rest}} = -\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$, so $\Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$.
- GGM edge $\iff$ $\Omega_{jk} \neq 0$; estimated by tests, graphical lasso, neighborhood selection or CLIME.
- GGM edges are undirected conditional associations.

7 Selected exercises

Motivation. These prove the block-inverse identity and Gaussian conditioning, the graphical-lasso $\lambda_{\max}$, and the link between partial correlations and node-wise regressions.

Exercise 8.1 (Block inverse and Gaussian conditioning)

Let $\Sigma \succ 0$ be partitioned into blocks A, B, $\Omega = \Sigma^{-1}$, and $M = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$.
(a) Show that $M \succ 0$, $\Omega_{AA} = M^{-1}$ and $\Omega_{AB} = -\Omega_{AA}\Sigma_{AB}\Sigma_{BB}^{-1}$.
(b) For $X \sim N_p(\mu, \Sigma)$ show that $X_A \mid X_B = x_B \sim N\big(\mu_A + \Sigma_{AB}\Sigma_{BB}^{-1}(x_B - \mu_B),\, M\big)$.

Solution [Supplement].

(a)

Step 1. For $v \neq 0$, $u = (v^\top, -(\Sigma_{BB}^{-1}\Sigma_{BA}v)^\top)^\top \neq 0$ gives $v^\top M v = u^\top \Sigma u > 0$, so $M \succ 0$.

Step 2. With $K = \Sigma_{AB}\Sigma_{BB}^{-1}$ let

$$\Omega' = \begin{pmatrix} M^{-1} & -M^{-1}K \\ -K^\top M^{-1} & \Sigma_{BB}^{-1} + K^\top M^{-1} K \end{pmatrix}.$$

Using $\Sigma_{AA} - K\Sigma_{BA} = M$ and $\Sigma_{BB}K^\top = \Sigma_{BA}$, the blocks of $\Sigma\Omega'$ are

$$\begin{aligned} \Sigma_{AA}M^{-1} - \Sigma_{AB}K^\top M^{-1} &= MM^{-1} = I, \\ -\Sigma_{AA}M^{-1}K + \Sigma_{AB}\Sigma_{BB}^{-1} + \Sigma_{AB}K^\top M^{-1}K &= -MM^{-1}K + K = 0, \\ \Sigma_{BA}M^{-1} - \Sigma_{BB}K^\top M^{-1} &= 0, \\ -\Sigma_{BA}M^{-1}K + I + \Sigma_{BB}K^\top M^{-1}K &= I. \end{aligned}$$

So $\Omega = \Omega'$: $\Omega_{AA} = M^{-1}$, $\Omega_{AB} = -M^{-1}K = -\Omega_{AA}\Sigma_{AB}\Sigma_{BB}^{-1}$.

(b)

Step 3. $R = X_A - \mu_A - K(X_B - \mu_B)$ has $\operatorname{Cov}(R, X_B) = \Sigma_{AB} - K\Sigma_{BB} = 0$ and

$$\operatorname{Var}(R) = \Sigma_{AA} - K\Sigma_{BA} - \Sigma_{AB}K^\top + K\Sigma_{BB}K^\top = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA} = M.$$

Step 4. (R, X_B) is jointly Gaussian and uncorrelated, hence independent: $R \sim N(0, M) \perp\!\!\!\perp X_B$, and $X_A = \mu_A + K(x_B - \mu_B) + R$ given $X_B = x_B$. $M = (\Omega_{AA})^{-1}$ is free of x_B. ■

Exercise 8.2 (When the graphical lasso returns no edge)

Let $S_{jj} > 0$ for all j and $\lambda \geq \max_{j \neq k} |S_{jk}|$, $\lambda > 0$. Show that $\widehat{\Omega}_\lambda = \operatorname{diag}(1/S_{11}, \ldots, 1/S_{pp})$, and that this minimizer is unique.

Solution [Supplement].

Step 1. (Strict convexity.) For $\Omega \succ 0$, symmetric $\Delta \neq 0$,

$$\frac{d^2}{dt^2}\{-\log\det(\Omega + t\Delta)\}\Big|_{t=0} = \operatorname{tr}(\Omega^{-1}\Delta\Omega^{-1}\Delta) = \|\Omega^{-1/2}\Delta\Omega^{-1/2}\|_F^2 > 0,$$

so $f(\Omega) = \operatorname{tr}(S\Omega) - \log\det\Omega + \lambda\sum_{j\neq k}|\Omega_{jk}|$ is strictly convex; a point with $0 \in \partial f$ is the unique minimizer.

Step 2. (Optimality.) $\nabla[\operatorname{tr}(S\Omega) - \log\det\Omega] = S - \Omega^{-1}$, so $0 \in \partial f(\Omega) \iff \Omega^{-1} - S - \lambda\Gamma = 0$ with $\Gamma_{jj} = 0$, $\Gamma_{jk} = \operatorname{sign}(\Omega_{jk})$ if $\Omega_{jk} \neq 0$, $\Gamma_{jk} \in [-1, 1]$ otherwise.

Step 3. (Check.) $\Omega_0 = \operatorname{diag}(1/S_{jj})$: diagonal $S_{jj} - S_{jj} = 0$; off-diagonal $\Gamma_{jk} = -S_{jk}/\lambda \in [-1, 1]$ gives $0 - S_{jk} - \lambda\Gamma_{jk} = 0$.

Thus $\lambda_{\max} = \max_{j\neq k}|S_{jk}|$ starts the path (the largest absolute correlation for standardized data). ■

Exercise 8.3 (Partial correlation from the two node-wise regressions)

With $\beta_k^{(j)}$ the population coefficient of X_k in the regression of X_j on X_{-j}, show that for $j \neq k$

$$\rho^2_{jk\cdot-\{j,k\}} = \beta_k^{(j)}\beta_j^{(k)}, \qquad \beta_k^{(j)} = \rho_{jk\cdot-\{j,k\}}\sqrt{\frac{\operatorname{Var}(X_j \mid X_{-j})}{\operatorname{Var}(X_k \mid X_{-k})}},$$

so $\beta_k^{(j)}$, $\beta_j^{(k)}$ and $\rho_{jk\cdot-\{j,k\}}$ have the same sign and vanish together. Conclude that the AND and OR rules agree in the population.

Solution [Supplement].

Step 1. $\beta_k^{(j)} = -\Omega_{jk}/\Omega_{jj}$, $\beta_j^{(k)} = -\Omega_{kj}/\Omega_{kk}$, with $\Omega_{jj}, \Omega_{kk} > 0$.

Step 2. Product $= \Omega_{jk}^2/(\Omega_{jj}\Omega_{kk}) = \rho^2_{jk\cdot-\{j,k\}}$.

Step 3. $-\Omega_{jk} = \rho_{jk\cdot-\{j,k\}}\sqrt{\Omega_{jj}\Omega_{kk}}$ gives $\beta_k^{(j)} = \rho_{jk\cdot-\{j,k\}}\sqrt{\Omega_{kk}/\Omega_{jj}}$, and $1/\Omega_{jj} = \operatorname{Var}(X_j \mid X_{-j})$.

In samples AND/OR differ only through estimation error. ■

Chapter 9. Gaussian Linear SEMs and Directed Relations

Graphical models, Lecture 2 (DAGs, chain/fork/collider, d-separation, faithfulness, Markov equivalence, CPDAG)

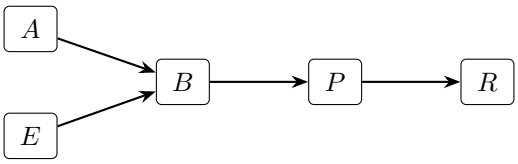
1 From Undirected to Directed Graphs

Lecture 1: conditional-dependence graph	Lecture 2: directed graph
$X_j - X_k \iff X_j \not\perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$ finds relations that remain after accounting for the others edges but no directions	$X_k \to X_j$ expresses each arrow by a generating equation per variable reads the conditional independences the graph creates; checks which directions data cannot distinguish

Question: **how far can directions be inferred from Gaussian observational data?**

2 A Gaussian Linear SEM Is a Bundle of Regressions on a DAG

2.1 The health example as equations



$$A = \epsilon_A, \quad E = \epsilon_E, \quad B = \beta_{AB}A + \beta_{EB}E + \epsilon_B, \quad P = \beta_{BP}B + \epsilon_P, \quad R = \beta_{PR}P + \epsilon_R,$$
$$\epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma_j^2).$$

Definition: DAG and Gaussian linear SEM

A directed acyclic graph (DAG) G has directed edges and no directed cycle. With $\text{pa}(j)$ the parents of node j,

$$X_j = \sum_{k \in \text{pa}(j)} \beta_{kj} X_k + \epsilon_j, \qquad \epsilon_1, \ldots, \epsilon_p \text{ independent}, \ \epsilon_j \sim N(0, \sigma_j^2).$$

Variable	Parents	Local regression
A, E	none	$A = \epsilon_A$, $E = \epsilon_E$
B	A, E	$B = \beta_{AB}A + \beta_{EB}E + \epsilon_B$
P	B	$P = \beta_{BP}B + \epsilon_P$
R	P	$R = \beta_{PR}P + \epsilon_R$

E is not in the equation of P but is related to P through $E \to B \to P$.

2.2 Matrix form and the implied covariance

Let $B_{jk} = \beta_{kj}$ and $D = \mathrm{diag}(\sigma_1^2, \dots, \sigma_p^2)$. Then $X = BX + \epsilon$; acyclicity makes B permutable to strictly lower triangular, so

$$X = (I - B)^{-1}\epsilon, \qquad \Sigma = (I - B)^{-1} D (I - B)^{-\top}, \qquad \Omega = \Sigma^{-1} = (I - B)^{\top} D^{-1} (I - B).$$

Markov factorization:

$$p(x_1, \dots, x_p) = \prod_{j=1}^{p} p(x_j \mid x_{\mathrm{pa}(j)}).$$

Theorem: the GGM of an SEM is its moral graph

For $j \neq k$,

$$\Omega_{jk} = -\frac{B_{jk}}{\sigma_j^2} - \frac{B_{kj}}{\sigma_k^2} + \sum_{l \neq j,k} \frac{B_{lj} B_{lk}}{\sigma_l^2}.$$

Hence $\Omega_{jk} \neq 0$ only if j, k are adjacent in G or have a common child; generically (no cancellation) exactly then. The GGM of Ch. 8 is the **moral graph**: the skeleton plus edges "marrying" the parents of each child.

Proof [Supplement]. $\Omega_{jk} = \sum_l (I - B)_{lj} (I - B)_{lk} / \sigma_l^2$: $l = j$ gives $-B_{jk}/\sigma_j^2$, $l = k$ gives $-B_{kj}/\sigma_k^2$, $l \neq j, k$ gives $B_{lj} B_{lk} / \sigma_l^2$, nonzero only for a common child l. ■

In the health example $A \perp\!\!\!\perp E$, yet the GGM has $A - E$ (common child B).

3 Chains, Forks and Colliders

Motivation. Chains and forks transmit association unless the middle node is conditioned on; colliders do the opposite.

Structure	Graph	Marginally	Conditioning on Z
chain	$X \to Z \to Y$	X and Y related	$X \perp\!\!\!\perp Y \mid Z$
fork	$X \leftarrow Z \to Y$	X and Y related	$X \perp\!\!\!\perp Y \mid Z$
collider	$X \to Z \leftarrow Y$	$X \perp\!\!\!\perp Y$	X and Y related

(Under Markov and faithfulness, §4.) Conditioning **closes** chains and forks and **opens** colliders.



chain: open given Z: blocked | fork: open given Z: blocked | collider: blocked given Z: open

Figure 9: The three elementary paths and the effect of conditioning on Z. Only the collider reverses, so v-structures are detectable; conditioning on a descendant of a collider also opens it.

Theorem: the three patterns in a Gaussian SEM

Let all errors be independent with variances 1 and coefficients $a, b \neq 0$.

1. Chain $Z = aX + \epsilon_Z$, $Y = bZ + \epsilon_Y$: $\mathrm{Cov}(X, Y) = ab \neq 0$, $\rho_{XY \cdot Z} = 0$.
2. Fork $X = aZ + \epsilon_X$, $Y = bZ + \epsilon_Y$: $\mathrm{Cov}(X, Y) = ab \neq 0$, $\rho_{XY \cdot Z} = 0$.

3. Collider $Z = aX + bY + \epsilon_Z$ with $X \perp\!\!\!\perp Y$: $\mathrm{Cov}(X,Y) = 0$ but $\rho_{XY\cdot Z} = -\frac{ab}{\sqrt{(1+b^2)(1+a^2)}} \neq 0$.

Proof [Supplement]. (1) $Y = abX + b\epsilon_Z + \epsilon_Y$, so $\mathrm{Cov}(X,Y) = ab$; the residual of Y on Z is $\epsilon_Y \perp\!\!\!\perp (X,Z)$.

(2) $\mathrm{Cov}(X,Y) = ab\mathrm{Var}Z$; residuals on Z are ϵ_X, ϵ_Y, independent.

(3) $\mathrm{Cov}(X,Y) = 0$. With $\mathrm{Var}Z = a^2 + b^2 + 1$, $\mathrm{Cov}(X,Z) = a$, $\mathrm{Cov}(Y,Z) = b$: residual covariance $-ab/(a^2+b^2+1)$, residual variances $(1+b^2)/(a^2+b^2+1)$ and $(1+a^2)/(a^2+b^2+1)$; divide. ■

Collider "explaining away": high BMI and low age suggest low exercise. **This change of conditional independence distinguishes a collider.**

4 d-Separation, the Markov Condition and Faithfulness

Definition: d-separation

A path between X and Y is **blocked** by a set S if it contains (i) a non-collider (chain or fork node) that is in S, or (ii) a collider that is not in S and has no descendant in S. X and Y are d-separated by S, written $X \perp_G Y \mid S$, if every path between them is blocked.

Definition: causal Markov condition and faithfulness

Markov: $X \perp_G Y \mid S \Rightarrow X \perp\!\!\!\perp Y \mid S$ in the distribution (equivalent to the Markov factorization). **Faithfulness**: the converse, $X \perp\!\!\!\perp Y \mid S \Rightarrow X \perp_G Y \mid S$: every conditional independence in the distribution is implied by the graph.

Under both, CIs of the data are *exactly* the d-separations of G (used in Ch. 10).

Example: how faithfulness fails — cancellation

$X \sim N(0,1)$, $Z = aX + \epsilon_Z$, $Y = cX + bZ + \epsilon_Y$: $\mathrm{Cov}(X,Y) = c + ab$. If $c = -ab$, $X \perp\!\!\!\perp Y$ despite the edge $X \to Y$. Such values have Lebesgue measure zero, but near-cancellation makes finite-sample tests unreliable.

Example: d-separation in the health example

$A \perp E$ (the only path $A \to B \leftarrow E$ has an unconditioned collider), but $A \not\perp E \mid P$ (P is a descendant of the collider B). $E \perp R \mid B$ and $E \perp R \mid P$ (the chain is blocked). $A \perp P \mid B$.

5 Markov Equivalence

5.1 Directions with the same conditional independences cannot be distinguished

$$X_1 \to X_2 \to X_3, \qquad X_1 \leftarrow X_2 \to X_3, \qquad X_1 \leftarrow X_2 \leftarrow X_3$$

all imply only $X_1 \perp\!\!\!\perp X_3 \mid X_2$. DAGs with the same CIs form a **Markov equivalence class**.

Theorem: the three DAGs define the same Gaussian model

A Gaussian distribution on (X_1, X_2, X_3) factorizes according to one of the three DAGs iff $X_1 \perp\!\!\!\perp X_3 \mid X_2$; so all three describe exactly the same set of covariance matrices.

Proof [Supplement]. If $X_1 \perp\!\!\!\perp X_3 \mid X_2$, $p = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_2)$ (chain); Bayes $p(x_1)p(x_2 \mid x_1) = p(x_2)p(x_1 \mid x_2)$ gives the fork, and $p(x_2)p(x_3 \mid x_2) = p(x_2 \mid x_3)p(x_3)$ the reversed chain. Conversely each factorization implies the CI. Gaussian conditionals are linear regressions, so each factorization is a Gaussian linear SEM. ■

The collider $X_1 \to X_2 \leftarrow X_3$ ($X_1 \perp\!\!\!\perp X_3$, $X_1 \not\perp\!\!\!\perp X_3 \mid X_2$) *is* distinguishable.

5.2 Skeleton and unshielded colliders

Definition: skeleton and v-structure

The **skeleton** is the adjacency structure left after deleting arrowheads: $X \to Z \to Y \mapsto X - Z - Y$. An **unshielded collider** (v-structure) is $X \to Z \leftarrow Y$ with X and Y non-adjacent.

Theorem: Verma–Pearl characterization

Two DAGs are Markov equivalent iff they have the same skeleton and the same unshielded colliders.

Idea [Supplement]. X, Y adjacent iff no set d-separates them (non-adjacent: the parents of the later one in a topological order separate). Given the skeleton, an unshielded $X - Z - Y$ is a collider iff Z is in *no* separating set of X, Y. Conversely, equal skeletons and v-structures give the same blocking sets on every path. ■

5.3 CPDAG

Definition: CPDAG

The completed partially directed acyclic graph of an equivalence class has a directed edge $X \to Y$ if every DAG in the class contains $X \to Y$, and an undirected edge $X - Y$ if the direction varies within the class.

CPDAG $X - Z - Y$ represents the chain, fork and reversed chain; the collider's CPDAG keeps both arrows. Health example: CPDAG $A \to B \leftarrow E$, $B \to P \to R$ ($B \to P$ forced, else a new v-structure $A \to B \leftarrow P$; likewise $P \to R$). **An undirected edge means an undistinguished direction, not "no relation".**

Summary

- Gaussian linear SEM: $\Sigma = (I - B)^{-1} D (I - B)^{-\top}$; its GGM is the moral graph.
- Conditioning closes chains and forks, opens colliders (d-separation).
- Markov + faithfulness: CIs = d-separations.
- Markov-equivalent DAGs share skeleton and v-structures; data identify only the CPDAG.

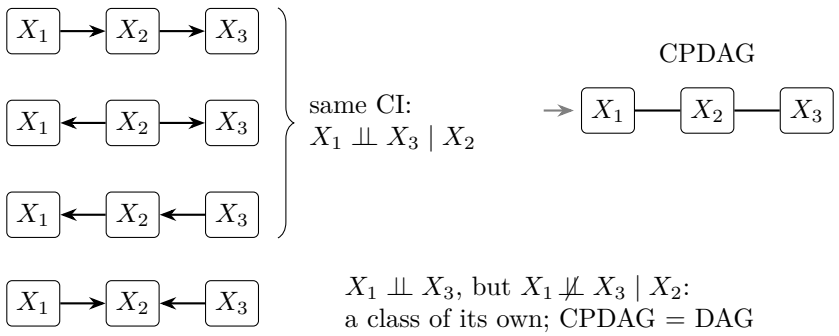


Figure 10: Markov equivalence: chain, fork and reversed chain share skeleton and CIs (undirected CPDAG); the collider is identified.

6 Selected exercises

Motivation. Structural equations as parent regressions, conditioning on a collider's descendant, and the equivalence class of a path.

Exercise 9.1 (Structural errors and non-descendants)

In a Gaussian linear SEM on a DAG G, let $\mathrm{nd}(j)$ be the set of non-descendants of j other than j.
(a) Show that ϵ_j is independent of $X_{\mathrm{nd}(j)}$.
(b) Deduce that the population least-squares regression of X_j on $X_{\mathrm{pa}(j)}$ has coefficients β_{kj} and residual variance σ_j^2, and that $X_j \perp\!\!\!\perp X_{\mathrm{nd}(j)\setminus\mathrm{pa}(j)} \mid X_{\mathrm{pa}(j)}$ (local Markov property).

Solution [Supplement].

(a)

Step 1. In topological order B is strictly lower triangular, $B^p = 0$, $(I-B)^{-1} = \sum_{m=0}^{p-1} B^m$; $(B^m)_{kl}$ sums edge-coefficient products over directed paths $l \to \cdots \to k$ of length m, so $\{(I-B)^{-1}\}_{kl} \neq 0$ only if $l = k$ or l is an ancestor of k.

Step 2. For $k \in \mathrm{nd}(j)$, $X_k = \sum_l \{(I-B)^{-1}\}_{kl}\epsilon_l$ involves only $(\epsilon_l)_{l \neq j}$, which are independent of ϵ_j; so $\epsilon_j \perp\!\!\!\perp X_{\mathrm{nd}(j)}$.

(b)

Step 3. $\mathrm{pa}(j) \subseteq \mathrm{nd}(j)$ (else a directed cycle). With $X_P = X_{\mathrm{pa}(j)}$, $\beta = (\beta_{kj})_{k \in \mathrm{pa}(j)}$: $\mathbb{E}(X_P \epsilon_j) = 0$, so β solves $\mathbb{E}\{X_P(X_j - X_P^\top \beta)\} = 0$, uniquely since $\mathrm{Var}(X_P)$ is a principal submatrix of $\Sigma \succ 0$. Residual ϵ_j, variance σ_j^2.

Step 4. With $X_N = X_{\mathrm{nd}(j)\setminus\mathrm{pa}(j)}$: given $(X_P, X_N) = (x_P, x_N)$, $X_j = x_P^\top \beta + \epsilon_j$ with $\epsilon_j \perp\!\!\!\perp (X_P, X_N)$ by (a), so the law $N(x_P^\top \beta, \sigma_j^2)$ is free of x_N: $X_j \perp\!\!\!\perp X_N \mid X_P$. ■

Exercise 9.2 (Conditioning on a descendant of a collider)

Let $X, Y, \epsilon_Z, \epsilon_W$ be independent $N(0,1)$, $Z = aX + bY + \epsilon_Z$ and $W = cZ + \epsilon_W$ with $a, b, c \neq 0$. Show that

$$\rho_{XY \cdot W} = -\frac{c^2 ab}{\sqrt{\{c^2(1+b^2)+1\}\{c^2(1+a^2)+1\}}} \neq 0,$$

and that $|\rho_{XY \cdot W}|$ is strictly smaller than the collider value $|\rho_{XY \cdot Z}| = |ab|/\sqrt{(1+a^2)(1+b^2)}$.

Solution [Supplement].

Step 1. $\mathrm{Cov}(X,W) = ca$, $\mathrm{Cov}(Y,W) = cb$, $v := \mathrm{Var}W = c^2(a^2+b^2+1)+1$.

Step 2. Residuals on W: $X - \frac{ca}{v}W$, $Y - \frac{cb}{v}W$, with covariance $-\frac{c^2ab}{v}$ and variances $1 - \frac{c^2a^2}{v} = \frac{c^2(1+b^2)+1}{v}$, $\frac{c^2(1+a^2)+1}{v}$. Divide; nonzero as $a, b, c \neq 0$.

Step 3. Dividing by c^2,

$$|\rho_{XY\cdot W}| = \frac{|ab|}{\sqrt{(1+b^2+c^{-2})(1+a^2+c^{-2})}} < \frac{|ab|}{\sqrt{(1+a^2)(1+b^2)}}.$$

W opens $X \to Z \leftarrow Y$, less than Z; the collider value is reached as $c^2 \to \infty$. ■

Exercise 9.3 (The equivalence class of a path)

Consider DAGs whose skeleton is the path $X_1 - X_2 - \cdots - X_p$, $p \geq 2$. Show that exactly p of them have no v-structure, namely those with a single source X_s,

$$X_1 \leftarrow \cdots \leftarrow X_{s-1} \leftarrow X_s \to X_{s+1} \to \cdots \to X_p, \qquad s = 1, \ldots, p,$$

that these form one Markov equivalence class, and that its CPDAG is the undirected path.

Solution [Supplement].

Step 1. Every orientation of a tree is acyclic. Let $e_i = \mathsf{R}$ if $X_i \to X_{i+1}$, $e_i = \mathsf{L}$ otherwise.

Step 2. Only consecutive edges share a node; (X_{i-1}, X_i, X_{i+1}) is unshielded and a collider iff $e_{i-1}e_i = \mathsf{RL}$. So no v-structure iff $e_1 \cdots e_{p-1}$ has no factor RL.

Step 3. If $e_i = \mathsf{R}$, $e_j = \mathsf{L}$ with $i < j$, let i' be the largest index $< j$ with $e_{i'} = \mathsf{R}$; then $e_{i'+1} = \mathsf{L}$, an RL. Hence words without RL are exactly $\mathsf{L}^{s-1}\mathsf{R}^{p-s}$, $s = 1, \ldots, p$: the single-source DAGs.

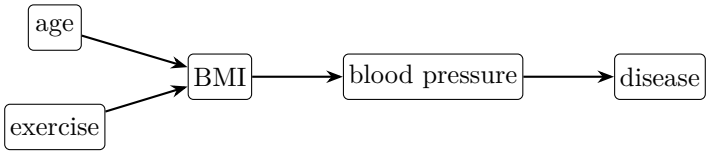
Step 4. By Verma–Pearl these p DAGs are equivalent, and any equivalent DAG has the same skeleton and no v-structure, so is one of them. e_i is R for $s \leq i$ and L for $s > i$, so every edge varies: the CPDAG is undirected. ■

Chapter 10. Structure Learning: PC, GES and MMHC

Graphical models, Lecture 3 (constraint-based and score-based discovery of the Markov equivalence class)

1 What Observational Data Can Reveal

Which relations remain given the others, which edges do CIs remove, and how far can directions be fixed?



Structure learning returns the relations the assumptions and data allow; not every arrow is fixed.

2 Assumptions Matter as Much as Data

Assumption	Practical meaning
acyclicity	relations at one time point are represented by a DAG without cycles
causal Markov condition	paths blocked in the graph appear as conditional independences in the data
faithfulness	no accidental cancellation of coefficients makes relations look independent
causal sufficiency	no unobserved common cause of two analyzed variables
same generating process	all samples come from the same system, without selection bias

PC and GES need causal sufficiency; with hidden common causes use FCI (Ch. 11).

3 The PC Algorithm

Motivation. PC removes edges by CI tests with growing conditioning sets drawn from current neighbours, then orients from colliders.

3.1 Skeleton phase

Algorithm: PC, skeleton

1. Connect all pairs of variables by undirected edges.
2. For $\ell = 0, 1, 2, \ldots$: for each adjacent pair (X_j, X_k) and each conditioning set $S \subseteq \text{adj}(X_j) \setminus \{X_k\}$ of size ℓ, test
$$H_0 : X_j \perp\!\!\!\perp X_k \mid X_S.$$
3. If H_0 is not rejected for some S, remove the edge $X_j - X_k$,
4. and store that S as the separating set $\text{Sep}(X_j, X_k)$.

Stop when no node has more than ℓ remaining neighbors.

Tests: Fisher-z partial correlation (Gaussian, Ch. 8), χ^2/G^2 (categorical), nonlinear CI tests (Ch. 11). The output is an undirected **skeleton**.

Example: skeleton of the toy example

Pair	Conditioning set	Decision	Action
age, exercise	$\varnothing$	independent	remove edge
age, blood pressure	BMI	conditionally independent	remove edge
exercise, blood pressure	BMI	conditionally independent	remove edge
BMI, disease	blood pressure	conditionally independent	remove edge

(Age–disease, exercise–disease are removed similarly.) Remaining skeleton:

$$\mathcal{E}_{\text{skel}} = \{\text{age–BMI},\ \text{exercise–BMI},\ \text{BMI–blood pressure},\ \text{blood pressure–disease}\}.$$

3.2 Orientation phase

$\text{Sep}(\text{age}, \text{exercise}) = \varnothing$, both are adjacent to BMI and not to each other, and BMI $\notin$ Sep, so

$$\text{age} \to \text{BMI} \leftarrow \text{exercise}.$$

Algorithm: PC, orientation

1. **v-structures**: for every unshielded triple $X - Z - Y$ (X, Y non-adjacent) with $Z \notin \text{Sep}(X, Y)$, orient $X \to Z \leftarrow Y$.
2. **Meek rules**, applied until nothing changes:

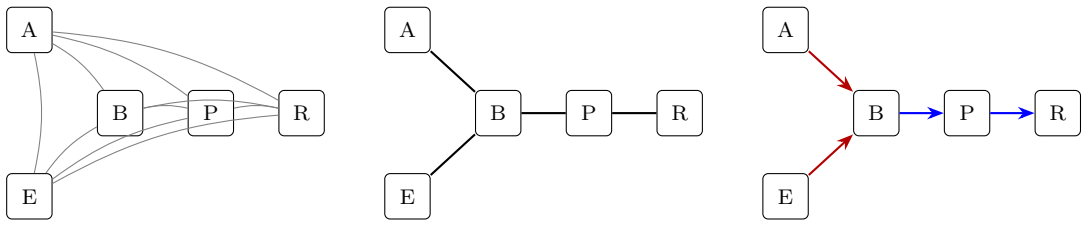
   R1 (no new collider) $X \to Z - Y$, X, Y non-adjacent $\Rightarrow Z \to Y$;

   R2 (no cycle) $X \to Z \to Y$ and $X - Y \Rightarrow X \to Y$;

   R3 $X - Z_1 \to Y$, $X - Z_2 \to Y$, $X - Y$, Z_1, Z_2 non-adjacent $\Rightarrow X \to Y$.

R1: $A \to B - C$, $A \not\sim C$; $C \to B$ would add an unsupported collider, so $B \to C$. R2: $A \to B \to C$, $A - C$; $C \to A$ would make a cycle, so $A \to C$. Example:

$$\text{age} \to \text{BMI} \leftarrow \text{exercise}, \qquad \text{BMI} \to \text{blood pressure} \to \text{disease}.$$



(1) complete graph (2) skeleton after CI tests (3) v-structure + Meek R1

Figure 11: PC on the toy example: CI tests give the skeleton; the v-structure $A \to B \leftarrow E$ (red) is oriented and R1 propagates $B \to P \to R$ (blue).

3.3 Why PC is correct

Theorem: correctness of PC with a CI oracle

Assume the Markov condition, faithfulness and causal sufficiency, and that every CI query is answered correctly. Then PC returns the CPDAG of the true DAG.

Proof [Supplement]. *Adjacencies.* By faithfulness adjacent variables are dependent given every set, so true edges stay. If X, Y are non-adjacent with Y not a descendant of X, $X \perp\!\!\!\perp Y \mid \mathrm{pa}(X)$ (local Markov); as $\mathrm{pa}(X) \subseteq \mathrm{adj}(X)$ throughout, a separating set is found by level $\ell = |\mathrm{pa}(X)|$. *v-structures.* For unshielded $X - Z - Y$: if Z is a non-collider, every separating set contains Z; if a collider, none does. So $\mathrm{Sep}(X, Y)$ decides correctly. *Remaining edges.* The Meek rules are sound (alternatives create a v-structure or cycle) and complete (Meek, 1995). ■

Remark: finite-sample issues

Early type-II errors remove edges irrecoverably and corrupt separating sets. *PC-stable* removes order dependence by updating adjacencies after each level. Conflicting v-structures signal test errors or violated assumptions.

4 Score-Based Search

4.1 BIC for Gaussian linear SEMs

$$S_{\mathrm{BIC}}(G) = \ell(\hat{\theta}_G) - \frac{d_G}{2} \log n,$$

with d_G the number of parameters.

Theorem: BIC decomposes into local scores

For a Gaussian linear SEM,

$$S_{\mathrm{BIC}}(G) = \sum_{j=1}^{p} S_j\{X_j, \mathrm{pa}_G(j)\}, \qquad S_j = -\frac{n}{2} \log \hat{\sigma}^2_{j|\mathrm{pa}(j)} - \frac{|\mathrm{pa}(j)| + 1}{2} \log n + c,$$

where $\hat{\sigma}^2_{j|\mathrm{pa}(j)}$ is the residual variance of the least-squares regression of X_j on its parents.

Proof [Supplement]. By the Markov factorization the log-likelihood is a sum over local regressions with separate parameters; each has maximum $-\frac{n}{2}\log(2\pi\hat{\sigma}^2) - \frac{n}{2}$ and $|\mathrm{pa}(j)|+1$ parameters (centered data). ■

A local move changes one or two local scores. Equivalent DAGs share maximum likelihood and d_G, so BIC is **score equivalent**: scores cannot split an equivalence class either.

4.2 GES and hill climbing

	GES	Hill climbing
search space	CPDAGs representing Markov equivalence classes	individual DAGs
moves	forward phase adds relations, backward phase removes them	add, delete or reverse an edge
typical output	a high-scoring CPDAG	one high-scoring DAG
caution	remaining undirected edges are undetermined	not every output direction is identified

Algorithm: Greedy Equivalence Search

Forward: from the empty graph, apply the best single edge insertion (in equivalence-class space) while the score increases. **Backward**: apply the best deletion while it helps. Output the CPDAG.

Theorem: consistency of GES (Chickering, 2002)

If the distribution is faithful to a DAG and the score is consistent and decomposable (e.g. BIC), then GES returns the true CPDAG with probability tending to 1 as $n \to \infty$, despite being greedy.

Hill climbing can stick in local optima (tabu lists, restarts help).

5 Computational Burden and Hybrid Methods

	Why the cost grows	Remaining problem
PC	repeats CI tests over many conditioning sets	tests multiply when the graph is dense; early test errors propagate to later edges and directions
GES	compares scores of many edge additions/deletions at each step	with many variables score computations grow; greedy results depend on sample and search path

With maximal degree k, PC needs $O(p^{k+2})$ tests: polynomial in p, exponential in k.

- **MMHC**: CI tests restrict candidate edges, then score search;
- **FLOP**: efficient search over variable orders and parent sets.

5.1 MMHC

CI tests $\longrightarrow$ candidate skeleton $\longrightarrow$ score search $\longrightarrow$ DAG candidate

Stage 1: Max-Min Parents and Children finds candidate neighbours by CI tests. **Stage 2**: hill climbing on BIC within those edges. Edges dropped in stage 1 cannot return.

6 Comparison

	PC	GES	MMHC
approach	constraint (CI) based	score based	both, in sequence
search	removes edges when independent	changes the CPDAG to increase the score	restricts candidates by tests, then score search over DAGs
typical output	CPDAG	CPDAG	DAG candidate
key settings	independence test, significance level	score, complexity penalty	tests, significance level, score
sensitive to	test errors	score and search path	edges lost in stage 1

Summary

- Under the four assumptions, data identify the Markov equivalence class.
- PC: CI tests, v-structures from separating sets, Meek rules; correct with an oracle.
- BIC is decomposable and score equivalent; GES is consistent.
- MMHC combines both; stage-1 errors are irreversible.

7 Selected exercises

Motivation. BIC on one edge, soundness of the Meek rules, and the test count of PC.

Exercise 10.1 (BIC on one edge)

Two centered variables are observed n times, with sample variances s_X^2, s_Y^2 (divisor n) and sample correlation r, $|r| < 1$.
(a) Using the local scores of the BIC theorem, show that $S_{\mathrm{BIC}}(X \to Y) = S_{\mathrm{BIC}}(Y \to X)$ and that the edge beats the empty graph iff $-n\log(1-r^2) > \log n$.
(b) For i.i.d. bivariate Gaussian data with correlation ρ, show that the probability that BIC chooses the correct adjacency tends to 1, both for $\rho = 0$ and for $\rho \neq 0$.

Solution [Supplement].

(a)

Step 1. The residual variance of Y on X is $s_Y^2 - s_{XY}^2/s_X^2 = s_Y^2(1-r^2)$, so

$$S_{\mathrm{BIC}}(X \to Y) = -\frac{n}{2}\{\log s_X^2 + \log s_Y^2 + \log(1-r^2)\} - \frac{3}{2}\log n + 2c,$$

symmetric in X, Y.

Step 2. Empty graph: $-\frac{n}{2}(\log s_X^2 + \log s_Y^2) - \log n + 2c$. Difference $-\frac{n}{2}\log(1-r^2) - \frac{1}{2}\log n > 0 \iff -n\log(1-r^2) > \log n$.

(b)

Step 3. ($\rho = 0$.) $\sqrt{n}\, r \xrightarrow{d} N(0,1)$ (Ch. 8, $q = 0$), so $nr^2 = O_p(1)$, $r^2 \xrightarrow{p} 0$. For $0 \le u \le \frac{1}{2}$, $-\log(1-u) = \sum_{k\ge 1} u^k/k \le u/(1-u) \le 2u$, hence

$$P\{-n\log(1-r^2) > \log n\} \le P\{2nr^2 > \log n\} + P\{r^2 > \tfrac{1}{2}\} \to 0.$$

Step 4. ($\rho \neq 0$.) $-\log(1-r^2) \xrightarrow{p} -\log(1-\rho^2) > 0$ and $n/\log n \to \infty$, so $-n\log(1-r^2)/\log n \xrightarrow{p} \infty$: the edge is chosen with probability $\to 1$. ■

Exercise 10.2 (Soundness of the Meek rules)

Let $\mathcal{C}$ be a Markov equivalence class and P a partially directed graph with the common skeleton such that every directed edge of P has that orientation in every DAG of $\mathcal{C}$, and every v-structure of $\mathcal{C}$ appears in P with both edges directed. Show that if the premise of R1, R2 or R3 holds in P, then every DAG of $\mathcal{C}$ contains the edge that the rule orients.

Solution [Supplement]. Fix $G' \in \mathcal{C}$; each conclusion pair is adjacent, so we exclude the wrong orientation.

Step 1. (R1.) $G' \ni X \to Z$. If $Y \to Z$, then $X \to Z \leftarrow Y$ is a v-structure of $\mathcal{C}$, so $Y \to Z$ would be directed in P; but $Z - Y$ is undirected. So $Z \to Y$.

Step 2. (R2.) $Y \to X$ would give the cycle $X \to Z \to Y \to X$. So $X \to Y$.

Step 3. (R3.) Suppose $Y \to X$. Then $X \to Z_i$ gives the cycle $X \to Z_i \to Y \to X$, so $Z_i \to X$ ($i = 1, 2$), and $Z_1 \to X \leftarrow Z_2$ is a v-structure that would be directed in P; but $X - Z_1$ is undirected. So $X \to Y$.

After PC's v-structure step the assumptions on P hold and each orientation preserves them, so all Meek orientations are correct. ■

Exercise 10.3 (How many tests PC needs)

Assume the conditions of the PC correctness theorem and let every node of the true DAG have at most k neighbours, $p \geq 3$. Show that the skeleton phase runs only the levels $\ell = 0, \ldots, k$ and performs at most $(k+1)p^{k+2}$ CI tests.

Solution [Supplement].

Step 1. A non-adjacent pair is separated by $\mathrm{pa}(X)$ and removed by level $|\mathrm{pa}(X)| \leq k$; true edges stay. So after level k the graph is the true skeleton.

Step 2. Then no node has more than k neighbours, so levels $\ell \geq k+1$ are never run.

Step 3. At level ℓ each ordered adjacent pair uses at most $\binom{|\mathrm{adj}(X_a)|-1}{\ell} \leq \binom{p-2}{\ell} \leq (p-2)^\ell$ sets, over at most $p(p-1)$ pairs:

$$p(p-1)\sum_{\ell=0}^{k}(p-2)^\ell \leq p^2(k+1)(p-2)^k \leq (k+1)p^{k+2}.$$

■

Chapter 11. Relational Inference Matched to the Data Structure

Graphical models, Lecture 4 (hidden confounders, FCI and PAGs, nonlinear CI tests, PCMCI+, feedback)

1 PC Assumes All Common Causes Are Observed

PC premises: one DAG, no missing common cause, Markov and faithfulness, correct CI tests.

Hidden common cause: $E \leftarrow U \rightarrow P$ with U unobserved makes E, P related without a direct link, and PC may output a direct edge. **With possible hidden common causes, PC edges are not direct relations.**

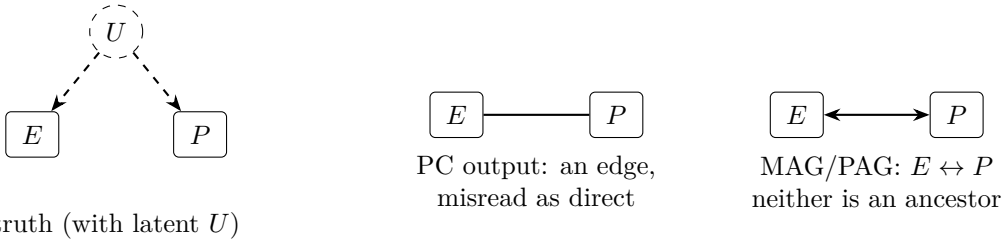


Figure 12: A hidden common cause: a DAG on the observed variables must pick a direction; FCI's ancestral graphs use a bidirected edge.

1.1 Observational data may not distinguish three explanations

Only X, Y observed and related:

Possible structure	Explanation	Observed conditional independences
$X \rightarrow Y$	X causes Y	none
$X \leftarrow Y$	Y causes X	none
$X \leftarrow U \rightarrow Y$	hidden common cause U	none

More observed variables can exclude some explanations.

2 FCI Infers a PAG Instead of a DAG

2.1 Ancestral graphs and PAGs

Marginalizing latents from a DAG gives a **maximal ancestral graph** (MAG): $X \rightarrow Y$ means X is an ancestor of Y; $X \leftrightarrow Y$ means neither is an ancestor of the other. Equivalent MAGs are summarized by a **partial ancestral graph** (PAG): marks common to all MAGs are shown, others are circles $\circ$.

	Object represented	Meaning of relations
PC's CPDAG	DAG equivalence class under the assumption of no missing common causes among analyzed variables	only directions common to all equivalent DAGs are fixed
FCI's PAG	equivalence class of observed-variable structures allowing hidden common causes	only endpoints common to all compatible structures are fixed

FCI uses CIs but does *not* assume causal sufficiency.

2.2 Reading PAG endpoints

PAG edge	How to read it
$X \to Y$	apart from selection bias, in every compatible structure X is an ancestor of Y and Y is not an ancestor of X
$X \leftrightarrow Y$	neither is an ancestor of the other; compatible with a structure with a hidden common cause
$X \circ\!\!\to Y$	it is certain that Y is not an ancestor of X; the mark at X is undetermined
$X \circ\!\!-\!\!\circ Y$	the two variables are adjacent but both endpoints are undetermined

Arrowhead = *not an ancestor*; circle = tail or arrowhead.

2.3 FCI can still find the arrowheads of colliders

Skeleton $X - Y - Z$, $X \not\sim Z$, $X \perp\!\!\!\perp Z$ but $X \not\perp\!\!\!\perp Z \mid Y$: FCI orients

$$X \circ\!\!\to Y \leftarrow\!\!\circ Z,$$

leaving open whether X causes Y or shares a hidden cause with it.

2.4 FCI checks wider separating sets, then orients conservatively

Algorithm: FCI

Edge removal.

1. Start from the complete graph on the observed variables.
2. If $X \perp\!\!\!\perp Y \mid S$ is found, remove $X - Y$ (as in PC's skeleton phase).
3. Widen the candidate separating sets beyond current neighbors to the variables on possible paths (Possible-D-Sep) and check again.

Endpoint determination.

1. Record the separating set used when an edge was removed.
2. Mark as colliders the middle variables not contained in the separating set.
3. Propagate only forced endpoints (Zhang's rules R1–R10).

Undetermined endpoints remain circles $\circ$.

Remark: why neighbors are not enough

In a MAG a separating set may need a node W reaching X only via colliders (e.g. $W \leftrightarrow V \leftrightarrow X$), not adjacent to X or Y. FCI uses the computable superset Possible-D-Sep; PC would wrongly keep the edge.

RFCI skips most of Possible-D-Sep with targeted triple tests: faster, slightly less informative, still sound.

3 When Basic Assumptions Break, Methods Change

Basic assumption	Situation where it fails	Candidate methods
all common causes observed	unobserved common causes exist	FCI, RFCI
the chosen test captures the relation	partial correlation misses nonlinear relations	KCI, distance correlation
independent observations	autocorrelation and time lags	PCMCI+
no cycles in the graph	feedback among variables	CCD, cyclic LiNGAM

3.1 Nonlinear relations: change the independence test

Example: dependence with zero correlation

$X \sim \mathrm{Unif}(-1, 1)$, $Y = X^2 + \varepsilon$, $\varepsilon \perp\!\!\!\perp X$: $\mathrm{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$ by symmetry, so Fisher-z and PC remove a real edge.

CI test	When to use
Fisher-z	linear conditional relations in continuous Gaussian data
kernel CI test (KCI)	general nonlinear conditional relations (based on cross-covariance operators in reproducing-kernel spaces)
distance correlation	nonlinear relations without strong assumptions on the distribution; $\mathrm{dCor}(X, Y) = 0$ iff $X \perp\!\!\!\perp Y$

PC/FCI stay the same; only the test changes. Nonparametric tests lose power in small samples and are slower.

3.2 Time series: PCMCI+ for lagged relations

Target relations:

$$X^j_{t-\tau} \longrightarrow X^k_t, \qquad \tau = 0, 1, \ldots, \tau_{\max}.$$

Algorithm: PCMCI+ (outline)

1. Among the past values of all variables, first reduce the candidate parents that may be related to each present value (a PC-type condition-selection step on lagged links).
2. Re-test each candidate link conditioning on the other past parents of both endpoints (momentary conditional independence, MCI), which removes autocorrelation effects.
3. Distinguish time-lagged relations from same-time (contemporaneous) relations and represent them in a time-series graph; contemporaneous links are oriented with PC-type rules.

Example output: $X^1_{t-2} \to X^2_t$, $X^2_{t-1} \to X^3_t$. Ordinary PC on autocorrelated series gives false edges (Exercise 11.2); check $\tau_{\max}$ and stationarity.

3.3 Feedback: the DAG assumption fails

E.g. $X_1 \to X_2 \to X_3 \to X_1$.

Method	Core idea
CCD	uses conditional independences to find relations common to the equivalence class of cyclic graphs
cyclic LiNGAM	uses linearity and non-Gaussian errors to further identify the cyclic structure

$X = BX + \epsilon$ is an equilibrium; it needs $I - B$ invertible (stability: $\varrho(B) < 1$). First decide whether feedback is *simultaneous* or *time-lagged* (PCMCI+).

4 Choosing a Method for the Data Structure

Feature of the data	Method	Main points to check
hidden common causes suspected	FCI or RFCI	read PAG circles and arrowheads separately
nonlinear relations expected	PC/FCI with nonparametric tests	test power and computing time
time lags and autocorrelation	PCMCI+	maximal lag and stationarity
simultaneous feedback possible	CCD or cyclic LiNGAM	distinguish cycles from time lags

Summary

- With hidden confounders FCI outputs a PAG: arrowhead = not an ancestor, circle = undetermined.
- FCI searches Possible-D-Sep; RFCI trades information for speed.
- Nonlinear: KCI/dCor tests; time series: PCMCI+; feedback: CCD, cyclic LiNGAM.

5 Selected exercises

Motivation. A hidden cause mimicking a direct effect, spurious correlation of autocorrelated series, and the equilibrium of a cyclic SEM.

Exercise 11.1 (A hidden common cause mimics a direct effect)

Let $U \sim N(0,1)$ be latent, $X = aU + \epsilon_X$, $Y = bU + \epsilon_Y$, with $U, \epsilon_X, \epsilon_Y$ independent, $\epsilon_X \sim N(0, \sigma_X^2)$, $\epsilon_Y \sim N(0, \sigma_Y^2)$, $\sigma_Y^2 > 0$.
(a) Find c and $\tau^2 > 0$ such that the SEM $X' \sim N(0, a^2 + \sigma_X^2)$, $Y' = cX' + \eta$, $\eta \sim N(0, \tau^2)$ independent of X', has the same distribution as (X, Y).
(b) Add an observed $Z \sim N(0,1)$, independent of $(U, \epsilon_X, \epsilon_Y)$, with $X = dZ + aU + \epsilon_X$, $d \neq 0$. Show that $Z \perp\!\!\!\perp Y$, whereas in any Gaussian SEM $Z \to X \to Y$ with nonzero coefficients Z and Y are dependent.

Solution [Supplement].

(a)

Step 1. (X, Y) is centered Gaussian with $\mathrm{Var}X = a^2 + \sigma_X^2$, $\mathrm{Var}Y = b^2 + \sigma_Y^2$, $\mathrm{Cov}(X, Y) = ab$, and is determined by this covariance.

Step 2. $\mathrm{Cov}(X', Y') = c(a^2 + \sigma_X^2)$, $\mathrm{Var}Y' = c^2(a^2 + \sigma_X^2) + \tau^2$; matching,

$$c = \frac{ab}{a^2 + \sigma_X^2}, \qquad \tau^2 = b^2 + \sigma_Y^2 - \frac{a^2 b^2}{a^2 + \sigma_X^2} = \sigma_Y^2 + \frac{b^2 \sigma_X^2}{a^2 + \sigma_X^2} > 0.$$

So (X, Y) alone cannot tell $X \leftarrow U \rightarrow Y$ from $X \rightarrow Y$ (or $Y \rightarrow X$).

(b)

Step 3. $Y = bU + \epsilon_Y$ is a function of $(U, \epsilon_Y) \perp\!\!\!\perp Z$, so $Z \perp\!\!\!\perp Y$.

Step 4. If $X = dZ + \epsilon_X'$, $Y = cX + \eta$ with $c, d \neq 0$, then $\mathrm{Cov}(Z, Y) = cd \neq 0$.

This CI pattern is what FCI uses to orient $Z \circ\!\!\rightarrow X \leftarrow\!\!\circ Y$, ruling out $X \rightarrow Y$. ■

Exercise 11.2 (Independent but autocorrelated series)

Let $X_t = \phi X_{t-1} + u_t$ and $Y_t = \phi Y_{t-1} + v_t$ be stationary AR(1) processes, $|\phi| < 1$, where (u_t) and (v_t) are i.i.d. $(0, 1)$ sequences independent of each other. Let $C_n = \frac{1}{n} \sum_{t=1}^{n} X_t Y_t$. Show that $\mathbb{E}C_n = 0$ and

$$n\mathrm{Var}(C_n) \longrightarrow \gamma(0)^2 \frac{1 + \phi^2}{1 - \phi^2}, \qquad \gamma(0) = \frac{1}{1 - \phi^2},$$

whereas for $\phi = 0$ the limit is $\gamma(0)^2 = 1$.

Solution [Supplement].

Step 1. $X_t = \sum_{m \geq 0} \phi^m u_{t-m}$ has mean 0 and $\gamma(h) = \phi^{|h|}/(1 - \phi^2)$, likewise Y; by independence $\mathbb{E}(X_t Y_t) = 0$.

Step 2. By independence,

$$\mathrm{Cov}(X_s Y_s, X_t Y_t) = \mathbb{E}(X_s X_t)\, \mathbb{E}(Y_s Y_t) = \gamma(t - s)^2.$$

Step 3. Hence

$$n\mathrm{Var}(C_n) = \frac{1}{n} \sum_{s=1}^{n} \sum_{t=1}^{n} \gamma(t - s)^2 = \sum_{|h| < n} \Big(1 - \frac{|h|}{n}\Big) \gamma(h)^2,$$

and since $\sum_h \gamma(h)^2 < \infty$, dominated convergence gives

$$\sum_{h \in \mathbb{Z}} \gamma(h)^2 = \gamma(0)^2 \Big(1 + 2 \sum_{h \geq 1} \phi^{2h}\Big) = \gamma(0)^2 \Big(1 + \frac{2\phi^2}{1 - \phi^2}\Big) = \gamma(0)^2 \frac{1 + \phi^2}{1 - \phi^2}.$$

So $\mathrm{Var}(r_n) \approx \frac{1}{n} \cdot \frac{1 + \phi^2}{1 - \phi^2}$ (19× the i.i.d. value at $\phi = 0.9$): naive Fisher-z declares spurious edges. ■

Exercise 11.3 (Equilibrium of a linear cyclic SEM)

Let B be a $p \times p$ matrix with spectral radius $\varrho(B) < 1$.
(a) Show that $I - B$ is invertible and $(I - B)^{-1} = \sum_{m \geq 0} B^m$.
(b) Show that for a fixed ϵ and any start $X^{(0)}$, the iteration $X^{(t+1)} = BX^{(t)} + \epsilon$ converges to $(I - B)^{-1}\epsilon$.
(c) For the feedback loop $X_1 = bX_2 + \epsilon_1$, $X_2 = aX_1 + \epsilon_2$, show that $\varrho(B) < 1$ iff $|ab| < 1$, and find the equilibrium.

Solution [Supplement].

(a)

Step 1. Gelfand: $\varrho(B) = \lim_m \|B^m\|^{1/m}$, so for $\varrho(B) < q < 1$, $\|B^m\| \leq q^m$ for $m \geq m_0$; $\sum_m B^m$ converges.

Step 2. $(I - B)\sum_{m=0}^{M} B^m = I - B^{M+1} \to I$, so $(I - B)\sum_{m\geq 0} B^m = I$, and likewise in the other order.

(b) $X^{(t)} = B^t X^{(0)} + \sum_{m=0}^{t-1} B^m \epsilon \to (I - B)^{-1}\epsilon$ by Step 1.

(c) $B = \begin{pmatrix} 0 & b \\ a & 0 \end{pmatrix}$ has characteristic polynomial $\lambda^2 - ab$, eigenvalue moduli $\sqrt{|ab|}$, so $\varrho(B) < 1 \iff |ab| < 1$, and

$$X_1 = \frac{\epsilon_1 + b\epsilon_2}{1 - ab}, \qquad X_2 = \frac{a\epsilon_1 + \epsilon_2}{1 - ab}.$$

■

Chapter 12. Identifiable Causal Discovery Models: LiNGAM

Graphical models, Lecture 5 (non-Gaussian errors, residual independence, ICA-LiNGAM, DirectLiNGAM, variants)

1 Directions That Conditional Independence Cannot Distinguish

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

All three satisfy $X \perp\!\!\!\perp Z \mid Y$ (Ch. 9). In

$$X = BX + \epsilon, \qquad \epsilon_j \text{ independent},$$

the error *distribution*, not only its covariance, can carry direction information.

2 The LiNGAM Model

Motivation. With non-Gaussian noise only the true direction leaves a residual independent of the regressor, which identifies the causal order.

Definition: LiNGAM

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad X = BX + \epsilon.$$

Assumption	Meaning
linearity	effects of the parents are linear combinations
acyclicity	with a suitable ordering of variables, B is strictly lower triangular
independent errors	$\epsilon_1, \ldots, \epsilon_p$ are mutually independent: no hidden common cause among the analyzed variables
non-Gaussianity	the structural errors are not normally distributed, which leaves information that distinguishes directions

Theorem: identifiability of LiNGAM (Shimizu et al., 2006)

Under linearity, acyclicity, mutually independent errors and non-Gaussian errors (at most one Gaussian error is allowed), the coefficient matrix B — hence the DAG and the causal order — is uniquely determined by the distribution of X.

3 In the True Direction, Cause and Regression Residual Are Independent

3.1 Two variables

True direction $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X \perp\!\!\!\perp \epsilon_Y, \quad \epsilon_X, \epsilon_Y \text{ non-Gaussian.}$$

$r_Y = Y - bX = \epsilon_Y$, so $X \perp\!\!\!\perp r_Y$.

Reverse $Y \to X$: $X = cY + r_X$; Y and r_X both contain ϵ_X, ϵ_Y, so typically $Y \not\perp\!\!\!\perp r_X$.

Theorem: Darmois–Skitovich

Let $\epsilon_1, \dots, \epsilon_m$ be independent and $U = \sum_i \alpha_i \epsilon_i$, $V = \sum_i \beta_i \epsilon_i$. If $U \perp\!\!\!\perp V$, then every ϵ_i with $\alpha_i \beta_i \neq 0$ is Gaussian.

Theorem: the reverse regression leaves a dependent residual

In the model above with $b \neq 0$ and $\sigma_X^2 = \mathrm{Var}\epsilon_X$, $\sigma_Y^2 = \mathrm{Var}\epsilon_Y$, the least-squares coefficient is $c = \mathrm{Cov}(X, Y)/\mathrm{Var}Y = b\sigma_X^2/(b^2\sigma_X^2 + \sigma_Y^2)$ and

$$r_X = X - cY = \frac{\sigma_Y^2}{b^2\sigma_X^2 + \sigma_Y^2}\,\epsilon_X - c\,\epsilon_Y, \qquad Y = b\,\epsilon_X + \epsilon_Y.$$

r_X is uncorrelated with Y but, if either error is non-Gaussian, not independent of it.

Proof [Supplement]. $r_X = (1-cb)\epsilon_X - c\epsilon_Y$ with $1 - cb = \sigma_Y^2/(b^2\sigma_X^2 + \sigma_Y^2)$; all coefficients of r_X and Y are nonzero, so $r_X \perp\!\!\!\perp Y$ would force both errors Gaussian (Darmois–Skitovich). Uncorrelatedness is the normal equation. ■

With Gaussian errors, (r_X, Y) uncorrelated $\Rightarrow$ independent in *both* directions: no identifiability. **LiNGAM picks the direction with the more independent residual.**

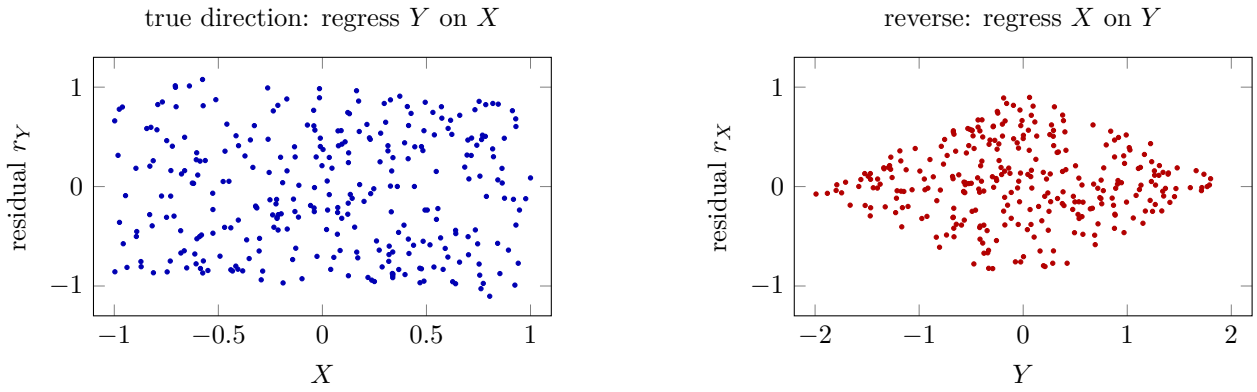


Figure 13: $X = \epsilon_X$, $Y = X + \epsilon_Y$, $\epsilon_X, \epsilon_Y \overset{\text{iid}}{\sim} \mathrm{Unif}(-1, 1)$, $n = 300$. True direction: rectangle (independent); reverse: diamond, uncorrelated but dependent (correlation of squares 0.07 vs −0.42).

Example: a simulation picture

With $\mathrm{Unif}(-1, 1)$ errors, (X, r_Y) fills a rectangle; (Y, r_X) fills a diamond whose spread of r_X shrinks at the extremes of Y.

4 Estimation

4.1 ICA-LiNGAM

$X = (I - B)^{-1}\epsilon = A\epsilon$ is a mixture of independent non-Gaussian sources (ICA).

Algorithm: ICA-LiNGAM

1. Run ICA to estimate an unmixing matrix W with WX having independent components; W equals $I - B$ up to an unknown row permutation and row scaling.
2. Find the row permutation of W giving no zeros (in practice, no small entries) on the diagonal; it is unique (Exercise 12.1).
3. Divide each row by its diagonal element to get W', and set $\hat{B} = I - W'$.
4. Find the causal order by permuting $\hat{B}$ to be as close as possible to strictly lower triangular; prune small coefficients (e.g. by bootstrap or adaptive Lasso).

ICA is non-convex: local optima, dependence on initial values.

4.2 DirectLiNGAM finds exogenous variables one by one

Theorem: characterization of an exogenous variable

In a LiNGAM, let $r_i^{(j)} = X_i - \dfrac{\mathrm{Cov}(X_i, X_j)}{\mathrm{Var}(X_j)} X_j$ for $i \neq j$. Then X_j is exogenous (has no parents) iff $X_j \perp\!\!\!\perp r_i^{(j)}$ for all $i \neq j$.

Proof ($\Rightarrow$) and idea ($\Leftarrow$) [Supplement]. ($\Rightarrow$) $X_j = \epsilon_j$, $X_i = a_{ij}\epsilon_j + \sum_{l \neq j} a_{il}\epsilon_l$ with $A = (I-B)^{-1}$, and $\mathrm{Cov}(X_i, X_j)/\mathrm{Var}(X_j) = a_{ij}$, so $r_i^{(j)} = \sum_{l \neq j} a_{il}\epsilon_l \perp\!\!\!\perp \epsilon_j$. ($\Leftarrow$) If X_j has a parent, some $r_i^{(j)}$ shares a non-Gaussian error with X_j with nonzero coefficients; Darmois–Skitovich gives dependence (Shimizu et al., 2011). ■

The residuals $(r_i^{(j)})_{i \neq j}$ again follow a LiNGAM (Exercise 12.3), so the step repeats.

Algorithm: DirectLiNGAM

For each candidate X_j, regress every other X_i on X_j to obtain residuals $r_i^{(j)}$, and add up the dependence between the candidate and all residuals:

$$T(j) = \sum_{i \neq j} \mathrm{Dep}(X_j, r_i^{(j)}).$$

1. Evaluate every variable as an exogenous candidate.
2. Choose the variable with the smallest $T(j)$ as the first in the order.
3. Replace the remaining variables by the residuals with the chosen variable's effect removed.
4. Repeat the same process on the remaining variables.

Dep: kernel independence measure or mutual information. No non-convex step; with a consistent Dep it finds the order in $p - 1$ steps.

4.3 Repeating residual independence yields the causal order

Example: three variables

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3,$$

with independent non-Gaussian errors.

Step	Independence observed	Choice
1	X_1 is independent of the residuals of X_2, X_3 after removing X_1	choose X_1 first
2	after removing X_1's effect, X_2 is independent of the residual of X_3	choose X_2
3	the last remaining variable	choose X_3

Result: $X_1 < X_2 < X_3$ (in step 1, $T(2) > T(1)$ since the residual of X_1 on X_2 shares ϵ_1, ϵ_2 with X_2).

4.4 After the order: estimate direct edges and effects

With order $X_1 < X_2 < X_3$, earlier variables are candidate parents:

$$X_2 = \beta_{21}X_1 + \epsilon_2, \qquad X_3 = \beta_{31}X_1 + \beta_{32}X_2 + \epsilon_3.$$

$\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{32} \approx 0.7$, $\hat{\beta}_{31} \approx 0$:

$$X_1 \Rightarrow X_2 \Rightarrow X_3.$$

Least squares within the order, pruning by adaptive Lasso or tests; acyclicity is guaranteed.

5 Variants for Different Data Structures

Data or problem	Method	Caveat
independent cross-sectional data	DirectLiNGAM, ICA-LiNGAM	linear, acyclic structure, no hidden confounding
repeated measurements over time	VAR-LiNGAM	distinguish lagged and contemporaneous relations
feedback cycles	cyclic LiNGAM family	the acyclicity assumption of basic LiNGAM cannot be used
hidden common causes possible	ParceLiNGAM, RCD	estimate the identifiable part of the relations rather than the whole DAG

Remark: when LiNGAM is fragile

Near-Gaussian errors need large samples; measurement error, nonlinearity, confounding or selection can favour the wrong direction. Bootstrap the order and test final residual independence.

Summary

- Non-Gaussian independent errors break ties within the Markov equivalence class.
- True direction: cause $\perp\!\!\!\perp$ residual; reverse: uncorrelated but dependent (Darmois–Skitovich); Gaussian errors: no difference.

- ICA-LiNGAM reads B off the unmixing matrix; DirectLiNGAM peels off exogenous variables, then estimates coefficients.

6 Selected exercises

Motivation. The unique permutation in ICA-LiNGAM, the exact dependence in the figure, and the DirectLiNGAM recursion.

Exercise 12.1 (The row permutation in ICA-LiNGAM is unique)

Let $M = I - B$, where the variables admit a causal order κ (a bijection onto $\{1, \dots, p\}$) with $B_{jk} \neq 0 \Rightarrow \kappa(k) < \kappa(j)$.
(a) Show that the identity is the only permutation σ with $M_{\sigma(i),i} \neq 0$ for all i.
(b) Let $W = \Lambda P M$ with Λ diagonal invertible and P a permutation matrix (the ICA indeterminacy). Show that exactly one row permutation of W has no zero on the diagonal, and that it recovers M after dividing each row by its diagonal entry.

Solution [Supplement].

(a)

Step 1. $M_{jk} \neq 0 \Rightarrow j = k$ or $B_{jk} \neq 0$, hence $\kappa(k) \leq \kappa(j)$.

Step 2. So $\kappa(i) \leq \kappa(\sigma(i))$ for all i, and $\tau = \kappa \circ \sigma \circ \kappa^{-1}$ has $\tau(m) \geq m$; as $\sum_m (\tau(m) - m) = 0$, τ, hence σ, is the identity.

(b)

Step 3. Row i of W is λ_i times row $\pi(i)$ of M. Reordering by q puts $\lambda_{q(i)} M_{\pi(q(i)),\cdot}$ in row i, with diagonal entry $\lambda_{q(i)} M_{\pi(q(i)),i}$.

Step 4. All nonzero iff $\sigma = \pi \circ q$ has $M_{\sigma(i),i} \neq 0$ for all i, i.e. $q = \pi^{-1}$ by (a). This gives $\Lambda' M$; $M_{ii} = 1$, so row normalization returns M and $\hat{B} = I - M = B$. ■

Exercise 12.2 (The dependence in the simulation figure)

Let $X = \epsilon_X$ and $Y = X + \epsilon_Y$ with $\epsilon_X, \epsilon_Y \overset{\text{iid}}{\sim} \text{Unif}(-1, 1)$.
(a) Show that in the true direction $\text{Cov}(X^2, r_Y^2) = 0$.
(b) For the reverse regression $r_X = X - cY$, show that $c = \frac{1}{2}$, $\text{Cov}(r_X, Y) = 0$ and $\text{Corr}(r_X^2, Y^2) = -\frac{3}{7}$.

Solution [Supplement]. Let $u = \epsilon_X$, $v = \epsilon_Y$: $\mathbb{E}u = \mathbb{E}u^3 = 0$, $\mathbb{E}u^2 = \frac{1}{3}$, $\mathbb{E}u^4 = \frac{1}{5}$, likewise v.

(a) $r_Y = v \perp\!\!\!\perp u = X$, so X^2, r_Y^2 are independent.

(b)

Step 1. $c = \frac{1}{3}/\frac{2}{3} = \frac{1}{2}$, $r_X = \frac{1}{2}(u - v)$, $Y = u + v$, $\text{Cov}(r_X, Y) = \frac{1}{2}(\mathbb{E}u^2 - \mathbb{E}v^2) = 0$.

Step 2. $(u - v)^2 (u + v)^2 = (u^2 - v^2)^2$, so

$$\mathbb{E}(r_X^2 Y^2) = \tfrac{1}{4}\mathbb{E}(u^4 - 2u^2v^2 + v^4) = \tfrac{1}{4}(\tfrac{2}{5} - \tfrac{2}{9}) = \tfrac{2}{45},$$
$$\mathbb{E}r_X^2 \, \mathbb{E}Y^2 = \tfrac{1}{4} \cdot \tfrac{2}{3} \cdot \tfrac{2}{3} = \tfrac{1}{9}, \qquad \text{Cov}(r_X^2, Y^2) = \tfrac{2}{45} - \tfrac{5}{45} = -\tfrac{1}{15}.$$

Step 3. $\mathbb{E}(u \pm v)^4 = \mathbb{E}u^4 + 6\mathbb{E}u^2\mathbb{E}v^2 + \mathbb{E}v^4 = \frac{16}{15}$, $\text{Var}\{(u \pm v)^2\} = \frac{16}{15} - \frac{4}{9} = \frac{28}{45}$, so $\text{Var}(Y^2) = \frac{28}{45}$, $\text{Var}(r_X^2) = \frac{7}{180}$ and

$$\text{Corr}(r_X^2, Y^2) = \frac{-1/15}{\sqrt{(7/180)(28/45)}} = \frac{-1/15}{14/90} = -\frac{3}{7} \approx -0.43,$$

matching the simulated -0.42. ■

Exercise 12.3 (The DirectLiNGAM recursion)

In a LiNGAM $X = BX + \epsilon$ with $A = (I - B)^{-1}$, let X_j be exogenous and $r_i = X_i - a_{ij}X_j$ for $i \neq j$. Show that

$$r_i = \sum_{k \neq j} b_{ik} r_k + \epsilon_i \qquad (i \neq j),$$

i.e. $(r_i)_{i \neq j}$ is a LiNGAM with coefficient matrix B with row and column j deleted, the same errors $(\epsilon_i)_{i \neq j}$, and the induced causal order.

Solution [Supplement].

Step 1. $A = I + BA$: $a_{ij} = \delta_{ij} + \sum_k b_{ik} a_{kj}$. Row j of B is zero, so $X_j = \epsilon_j$, $a_{jj} = 1$, and $a_{ij} = b_{ij} + \sum_{k \neq j} b_{ik} a_{kj}$ for $i \neq j$.

Step 2. With $X_j = \epsilon_j$, $X_k = r_k + a_{kj}\epsilon_j$:

$$X_i = b_{ij}\epsilon_j + \sum_{k \neq j} b_{ik}(r_k + a_{kj}\epsilon_j) + \epsilon_i = \sum_{k \neq j} b_{ik} r_k + a_{ij}\epsilon_j + \epsilon_i;$$

subtract $a_{ij}X_j = a_{ij}\epsilon_j$.

Step 3. The reduced B stays strictly lower triangular in the induced order and $(\epsilon_i)_{i \neq j}$ stay independent non-Gaussian, so the LiNGAM assumptions hold and the exogeneity characterization applies again. ■

Chapter 13. Identifiable Gaussian Linear SEMs

Graphical models, Lecture 6 (equal error variances, conditional-variance ordering, ANM, the US algorithm)

1 In General Gaussian SEMs, Directions Are Not Identified

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

These give the same CIs and Gaussian distribution (Ch. 9).

General Gaussian SEM: $\epsilon_j \sim N(0, \sigma_j^2)$	Equal-variance Gaussian SEM: $\epsilon_j \sim N(0, \sigma^2)$
if each variable may have its own error variance, several directions are compatible with the same distribution result: CPDAG	if all structural errors are restricted to the same variance, the residual-variance structure differs by direction result: DAG

Even for Gaussian data, an assumption on the *error variances* identifies the full DAG.

2 The Equal-Variance Gaussian SEM

Definition: equal-variance Gaussian SEM

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad \epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma^2),$$

$$X = BX + \epsilon, \qquad \Sigma = \sigma^2 (I - B)^{-1} (I - B)^{-\top}.$$

Assumption	Meaning
linearity	effects of the parents are linear combinations
acyclicity	with a suitable ordering, B is strictly lower triangular
independent errors	no hidden common cause among the analyzed variables
common error variance	all structural errors have the same σ^2
nonzero edges	coefficients of the edges in the graph are not 0

3 Two Variables: The Equal-Variance Assumption Picks the Direction

True direction $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X, \epsilon_Y \overset{\text{ind}}{\sim} N(0, \sigma^2).$$

Reverse $Y \to X$: $Y = \epsilon'_Y$, $X = cY + \eta$ with

$$\mathrm{Var}(\epsilon'_Y) = (1 + b^2)\sigma^2, \qquad \mathrm{Var}(\eta) = \frac{\sigma^2}{1 + b^2}.$$

Proof [Supplement]. $\mathrm{Var}Y = (1+b^2)\sigma^2$, $c = b/(1+b^2)$, $\mathrm{Var}(X \mid Y) = \sigma^2 - \dfrac{b^2\sigma^4}{(1+b^2)\sigma^2} = \dfrac{\sigma^2}{1+b^2}$. ■

Equal only if $(1+b^2)^2 = 1$, i.e. $b = 0$. **Only the true direction has a common error variance.**

4 Conditional Variances Reveal the Causal Order

Motivation. With equal noise variances, a source has the smallest (conditional) variance; repeating gives the causal order.

4.1 The key lemma

Theorem: conditional-variance ordering

In an equal-variance Gaussian SEM, let S be an *ancestral* set (it contains all ancestors of each of its members, e.g. the first few variables of a causal order) and $j \notin S$. Then

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \quad \text{if } \mathrm{pa}(j) \subseteq S, \qquad \mathrm{Var}(X_j \mid X_S) \geq \sigma^2(1 + b_{jk}^2) > \sigma^2 \quad \text{otherwise,}$$

for a suitable parent $k \notin S$ of j.

Proof [Supplement]. $X = A\epsilon$, $A = (I-B)^{-1}$, $a_{jj} = 1$, a_{jl} the total effect (0 unless $l \in \mathrm{an}(j) \cup \{j\}$). S ancestral $\Rightarrow$ X_S and $(\epsilon_l)_{l\in S}$ determine each other (triangular system). So in $X_j = \sum_{l\in S} a_{jl}\epsilon_l + \sum_{l\notin S} a_{jl}\epsilon_l$ the first sum is fixed by X_S and the second is independent of it:

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \sum_{l\notin S} a_{jl}^2 = \sigma^2\Big(1 + \sum_{l\notin S,\, l\neq j} a_{jl}^2\Big).$$

If $\mathrm{pa}(j) \subseteq S$, all ancestors of j lie in S and the sum is empty. Otherwise take the parent $k \notin S$ *latest* in a causal order. Another directed path $k \to \cdots \to m \to j$ would give a parent $m \notin S$ later than k, a contradiction; so $a_{jk} = b_{jk} \neq 0$. ■

Given the ordered variables, sources of the rest have conditional variance σ^2, others more:

Theorem: identifiability (Peters and Bühlmann, 2014)

The DAG and B of an equal-variance Gaussian SEM are identified from Σ. A causal order is obtained top-down: $\pi_1 = \arg\min_j \mathrm{Var}(X_j)$, and $\pi_m = \arg\min_{j\notin S} \mathrm{Var}(X_j \mid X_S)$ with $S = \{\pi_1, \ldots, \pi_{m-1}\}$.

4.2 A worked example

Example: three variables

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3, \qquad \epsilon_1, \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0,1).$$

Step	Variance comparison	Choice
1	$\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = 1.64$, $\mathrm{Var}(X_3) = 1.8036$	X_1
2	$\mathrm{Var}(X_2 \mid X_1) = 1$, $\mathrm{Var}(X_3 \mid X_1) = 1.49$	X_2
3	the last remaining variable	X_3

$\mathrm{Var}(X_3) = 0.49\times1.64+1$, $\mathrm{Var}(X_3 \mid X_1) = 0.49\,\mathrm{Var}(X_2 \mid X_1)+1 = 1.49$. Order $X_1 < X_2 < X_3$.

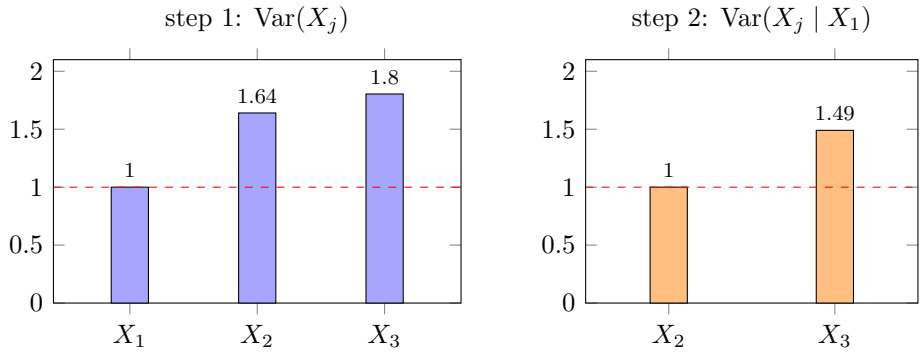


Figure 14: Top-down ordering ($\sigma^2 = 1$, red dashed): the variable at the common error variance is selected, first X_1, then X_2.

4.3 After the order: select parents and edges

With order $X_1 < X_2 < X_3$:

$$X_2 = \beta_{21}X_1 + \epsilon_2, \qquad X_3 = \beta_{31}X_1 + \beta_{32}X_2 + \epsilon_3.$$

$\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{31} \approx 0$, $\hat{\beta}_{32} \approx 0.7$:

$$X_1 \to X_2 \to X_3.$$

5 Equal Variance Does Not Mean Equal Observed Variances

Misconception	Correct meaning
all X_j must have the same variance	variation transmitted from the parents makes observed variances differ (as in the example)
standardizing all variables gives equal variance	standardization rescales the structural errors too, so it does not guarantee the equal-variance assumption — it typically destroys it
similar fitted residual variances prove the assumption	only a diagnostic compatible with the chosen DAG; it does not prove the true direction

Natural for same-kind, same-unit variables with similar measurement error (e.g. expression levels).

Remark: scale dependence and varsortability

The assumption is not scale-invariant (Exercise 13.2). Simulated benchmarks often have variances increasing along the order ("varsortability"), flattering variance-based methods; real data may lack this.

6 Additive Noise Models and the US Algorithm

6.1 The equal-variance Gaussian SEM is a special ANM

$$X_j = f_j(X_{\mathrm{pa}(j)}) + \epsilon_j, \qquad \epsilon_1, \ldots, \epsilon_p \text{ independent.}$$

With S the already selected variables,

$$U_j(S) = \mathbb{E}\{\mathrm{Var}(X_j \mid X_S)\}.$$

For ancestral S, $j \notin S$ (Exercise 13.3):

$$U_j(S) = \sigma_j^2 + \mathbb{E}[\mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}].$$

Conditioning set S	Conditional variance
all parents are in S	$U_j(S) = \sigma_j^2$
some parent not yet in S	uncertainty created by the parents is added

6.2 ANMs are identified by an ordering of conditional variances

With true order π and $S_{m-1} = \{\pi_1, \ldots, \pi_{m-1}\}$, if

$$U_{\pi_m}(S_{m-1}) < U_k(S_{m-1}), \qquad k \notin S_{m-1} \cup \{\pi_m\},$$

the variables are identified one by one.

Equal-variance Gaussian SEM	General ANM
$\sigma_1^2 = \cdots = \sigma_p^2$	$\sigma_1^2, \ldots, \sigma_p^2$ may differ
a variable whose parents are all included has no extra uncertainty, hence the smallest conditional variance (lemma above)	identified as long as the order of conditional variances, combining error variance and parent-induced uncertainty, is maintained

This needs neither linearity nor Gaussianity, only the *correct conditional-variance order* (other ANM results use residual independence, as in LiNGAM).

6.3 The US algorithm

Algorithm: US — order by conditional variances, then parents

Step 1: estimate the causal order. Let S be the selected set (initially $\varnothing$).

1. Regress each X_j, $j \notin S$, on X_S (flexibly, if f_j may be nonlinear).
2. Compute the residual variance $\widehat{U}_j(S)$.
3. Select the variable with the smallest value, $\hat{\pi}_m = \arg\min_{j \notin S} \widehat{U}_j(S)$.

Add the selected variable to S and repeat.
Step 2: select the parents. The candidate parents of each variable are the variables before it in the order. For each candidate, test conditional dependence given the other preceding variables:

$$X_{\pi_m} \not\perp\!\!\!\perp X_{\pi_j} \mid X_{\{\pi_1,\ldots,\pi_{m-1}\}\setminus\{\pi_j\}}.$$

If conditional dependence remains, add the edge $\pi_j \to \pi_m$.

In a correct order, preceding non-parents are non-descendants, so Markov makes them conditionally independent and faithfulness keeps parents dependent.

7 Summary

Model	Extra assumption	Clue for the direction
LiNGAM	linear relations with independent non-Gaussian errors	independence of the cause and the regression residual
equal-variance Gaussian SEM	common variance of independent Gaussian errors	order of conditional variances starting from the source
ANM and US	an additive noise model that keeps the conditional-variance order	conditional variances find the order; conditional independence selects the parents

All still need acyclicity, no hidden confounding and causal minimality.

Key points.

- General Gaussian SEMs identify only the equivalence class; a common error variance identifies the DAG.
- Sources have the smallest conditional variance; regression then selects edges.
- Check scales, measurement error and confounding.

8 Selected exercises

Motivation. Bottom-up ordering via Ω, unit changes that reverse the direction, and the ANM variance decomposition.

Exercise 13.1 (Bottom-up ordering from the precision matrix)

In an equal-variance Gaussian SEM with nonzero edge coefficients, let $\Omega = \Sigma^{-1}$.
(a) Show that $\Omega_{jj} = \{1 + \sum_{l \in \mathrm{ch}(j)} b_{lj}^2\}/\sigma^2$, where $\mathrm{ch}(j)$ is the set of children of j. Deduce that $\mathrm{Var}(X_j \mid X_{-j}) = \sigma^2$ if j is a sink (has no children) and $< \sigma^2$ otherwise.
(b) Show that after deleting a sink, the remaining variables again follow an equal-variance Gaussian SEM on the induced subgraph, so that repeatedly removing $\arg\min_j \Omega_{jj}$ yields a reversed causal order.

Solution [Supplement].

(a)

Step 1. $\Omega = (I - B)^\top (I - B)/\sigma^2$ (Ch. 9), so $\Omega_{jj} = \sum_l (I - B)_{lj}^2/\sigma^2$: $l = j$ gives 1, $l \neq j$ gives b_{lj}^2, nonzero iff $j \to l$.

Step 2. $\mathrm{Var}(X_j \mid X_{-j}) = 1/\Omega_{jj} = \sigma^2/\{1 + \sum_{l \in \mathrm{ch}(j)} b_{lj}^2\}$ (Ch. 8), $= \sigma^2$ iff $\mathrm{ch}(j) = \varnothing$.

(b)

Step 3. A sink appears in no other equation, so the other equations with their $N(0, \sigma^2)$ errors form an equal-variance SEM on the DAG minus j and determine the law of X_{-j}.

Step 4. Sinks have $\Omega_{jj} = 1/\sigma^2$, non-sinks more, so the argmin is a sink; repeating lists variables in reverse causal order. ■

Exercise 13.2 (A change of units can reverse the direction)

Let $X = \epsilon_X$, $Y = bX + \epsilon_Y$, $\epsilon_X, \epsilon_Y \overset{\text{ind}}{\sim} N(0, \sigma^2)$, $b \neq 0$, and $\tilde{Y} = sY$ with $s > 0$.
(a) Show that $(X, \tilde{Y})$ follows the Gaussian SEM $X \to \tilde{Y}$ with error variances σ^2 and $s^2\sigma^2$, which is equal-variance only for $s = 1$.
(b) Show that the top-down rule applied to $(X, \tilde{Y})$ selects $\tilde{Y}$ first iff $s^2 < 1/(1 + b^2)$.
(c) Show that for $s = 1/(1 + b^2)$ the pair $(X, \tilde{Y})$ is *exactly* an equal-variance Gaussian SEM with $\tilde{Y} \to X$.

Solution [Supplement].

(a) $\tilde{Y} = sbX + s\epsilon_Y$, $s\epsilon_Y \sim N(0, s^2\sigma^2) \perp\!\!\!\perp X$.

(b) $\mathrm{Var}X = \sigma^2$, $\mathrm{Var}\tilde{Y} = s^2(1 + b^2)\sigma^2$; $\tilde{Y}$ first iff $s^2(1 + b^2) < 1$.

(c)

Step 1. The reverse factorization is $Y = \epsilon'_Y$, $X = cY + \eta$, $\mathrm{Var}\eta = \sigma^2/(1 + b^2)$, $\eta \perp\!\!\!\perp Y$; so $X = (c/s)\tilde{Y} + \eta$ with $\eta \perp\!\!\!\perp \tilde{Y}$.

Step 2. Error variances $s^2(1 + b^2)\sigma^2$ and $\sigma^2/(1 + b^2)$ coincide at $s = 1/(1 + b^2)$; with Gaussianity this is an equal-variance SEM $\tilde{Y} \to X$.

The method then confidently outputs the wrong direction. ■

Exercise 13.3 (Conditional variances in an additive noise model)

In an ANM $X_j = f_j(X_{\mathrm{pa}(j)}) + \epsilon_j$ with independent errors, $\mathbb{E}\epsilon_j = 0$, $\mathrm{Var}\epsilon_j = \sigma_j^2$ and $\mathbb{E}f_j(X_{\mathrm{pa}(j)})^2 < \infty$, let S be an ancestral set and $j \notin S$.
(a) Show that ϵ_j is independent of $(X_S, X_{\mathrm{pa}(j)})$.
(b) Show that $\mathrm{Var}(X_j \mid X_S) = \sigma_j^2 + \mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}$ almost surely; hence $U_j(S) = \sigma_j^2 + \mathbb{E}[\mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}]$, which equals σ_j^2 when $\mathrm{pa}(j) \subseteq S$.

Solution [Supplement].

(a)

Step 1. By induction along a causal order, X_k is a function of $(\epsilon_l)_{l \in \mathrm{an}(k) \cup \{k\}}$.

Step 2. For $k \in S$, $\mathrm{an}(k) \cup \{k\} \subseteq S \not\ni j$; for $k \in \mathrm{pa}(j)$, $\mathrm{an}(k) \cup \{k\} \subseteq \mathrm{an}(j) \not\ni j$. So $(X_S, X_{\mathrm{pa}(j)})$ is a function of $(\epsilon_l)_{l \neq j} \perp\!\!\!\perp \epsilon_j$.

(b)

Step 3. With $F = f_j(X_{\mathrm{pa}(j)})$, $\bar{F} = \mathbb{E}(F \mid X_S)$, (a) gives $\mathbb{E}(X_j \mid X_S) = \bar{F}$, so

$$\mathrm{Var}(X_j \mid X_S) = \mathbb{E}\{(F - \bar{F})^2 \mid X_S\} + 2\,\mathbb{E}\{(F - \bar{F})\epsilon_j \mid X_S\} + \mathbb{E}(\epsilon_j^2 \mid X_S).$$

Step 4. $\epsilon_j \perp\!\!\!\perp (F, X_S)$: the cross term is $\mathbb{E}\epsilon_j \cdot \mathbb{E}(F - \bar{F} \mid X_S) = 0$ and the last is σ_j^2; the first is $\mathrm{Var}(F \mid X_S)$. Take expectations for $U_j(S)$.

Step 5. If $\mathrm{pa}(j) \subseteq S$, F is X_S-measurable, so $\mathrm{Var}(F \mid X_S) = 0$ and $U_j(S) = \sigma_j^2$. ■